THE WORLD ECONOMY, MAPPED

Global Industry & Business Atlas

Software and AI adoption map

A hierarchical reference to economic activities, recognizable businesses, the software they use, and the depth of AI adoption.

22	87	258	463
sectors	divisions	groups	activity classes

1,663 business examples across all classes, plus **72 focused subtype profiles**. Includes modern technology ecosystems, traditional trades, informal operators, public services and non-commercial activities.

Coverage promise

Complete at the United Nations ISIC Revision 5 four-digit activity-class level. Business examples are a curated expansion, not a claim to enumerate every possible niche or business-model combination. [S01] [S02]

Research cut-off: 11 October 2026
Evidence: official classifications, public statistics, sector research and primary operator examples.
Important: class-level maturity bands are explicitly labeled analyst estimates, not invented worldwide adoption percentages.

Contents

Use the clickable entries below or the PDF bookmarks. Search for a familiar business name or a four-digit activity code. A short legend appears in the footer; the full rating definitions follow.

How the hierarchy works . . . 3
Reading software and AI maturity . . . 4
Evidence, uncertainty and interpretation . . . 5
What measured evidence actually says . . . 6
Sector map at a glance . . . 8
Software functions and common AI layers . . . 10
A | Agriculture, forestry and fishing . . . 11
B | Mining and quarrying . . . 20
C | Manufacturing . . . 24
D | Electricity, gas, steam and cooling . . . 56
E | Water, sewerage, waste and remediation . . . 58
F | Construction . . . 61
G | Wholesale and retail . . . 64
H | Transportation and storage . . . 73
I | Accommodation and food services . . . 79
J | Publishing, broadcasting and content . . . 82
K | Telecom, computing and information services . . . 86
L | Finance and insurance . . . 89
M | Real estate . . . 95
N | Professional, scientific and technical services . . . 96
O | Administrative and support services . . . 101
P | Public administration, defence and social security . . . 108
Q | Education . . . 110
R | Health and social work . . . 113
S | Arts, sports and recreation . . . 116
T | Membership, repair and personal services . . . 121
U | Household employment and own-use production . . . 126
V | Extraterritorial organizations and bodies . . . 128
Modern and cross-cutting ecosystem finder . . . 129
Focused business subtype profiles . . . 132
Business models: another axis, not another industry . . . 143
Assessing a specific business . . . 144
Going deeper: countries, products and occupations . . . 145
Coverage audit . . . 146
Sources and evidence notes . . . 147

01 / SCOPE AND NAVIGATION

How the hierarchy works

Your idea is right: broad sectors contain increasingly specific activities. The important correction is that an **industry, a business model, a product and a company are different things**. A single company may operate several businesses.

Level	Example	Meaning
Global economy	All economic activity	The umbrella for this atlas.
Section / sector	A - Agriculture, forestry and fishing	22 broad official categories.
Division	01 - Crop and animal production	87 two-digit categories.
Group	012 - Perennial crop growing	258 three-digit categories.
Class	0121 - Grape growing	463 four-digit categories; the complete backbone.
Business subtype	Wine-grape vineyard / table-grape farm	Editorial examples within a class.
Operating model	Family-owned / cooperative / enterprise; local / export	Cross-cutting dimensions, not extra official levels.

ISIC classifies economic activities rather than brands, products or job titles. Its principal-activity rules and national implementations matter when an establishment performs more than one activity. The atlas uses short, plain-English labels rather than reproducing the official explanatory notes. [S01] [S02]

What is and is not exhaustive

Included: every ISIC Rev. 5 class, with its section, division and group; business examples; software categories; software and AI maturity bands; an AI workflow; scale context; and a practical watchpoint. Residual classes such as "other" activities remain included, rather than silently dropping difficult-to-name businesses.

Not claimed: a census of every company, every product sold, every informal occupation, every geography-specific licence, or all combinations of franchise, subscription, online, wholesale and ownership models. Public administration, household activity and international bodies are included for economic coverage even though they are not ordinary commercial businesses. [S01]

How to find a business

Start with the sector chapter, then follow division > group > class. The ecosystem finder maps labels such as fintech, climate technology, robotics and tourism to several relevant classes. The focused subtype appendix goes deeper into broad categories such as software, allied health and personal services.

02 / TWO SEPARATE SCALES

Reading software and AI maturity

Digital software maturity (D) describes how software supports the operating model. **AI maturity (A)** describes the depth of intentional AI use. A business can be D4 and A0: it may depend completely on conventional software and use no purposeful AI.

Band	Meaning	Practical interpretation
D0	Mostly manual	Paper, face-to-face work and minimal dedicated software.
D1	Basic digital administration	Messaging, payments, spreadsheets, basic accounts and web presence.
D2	Dedicated workflow software	Systems for orders, appointments, inventory, cases, design or jobs.
D3	Integrated operations	Connected workflows, shared data and operational monitoring; sensors where relevant.
D4	Software-critical operation	The core service or operating model cannot realistically function without software.

Band	Meaning	Practical interpretation
A0	No purposeful AI	No intentional AI deployment established.
A1	Assistance or experiments	Occasional assistants, trials or pilots; not an embedded production process.
A2	Bounded production use	One or a few live AI workflows with defined scope and human review.
A3	Integrated routine use	Several operational workflows routinely use AI and share data or controls.
A4	AI-central product or service	AI is an essential part of what the business sells or delivers.

These scales are editorial diagnostic tools. They are **not percentages, statistical confidence intervals or a league table**. A range such as D1-D3 / A0-A2 describes a plausible operating span across business sizes and contexts; it does not measure how many firms occupy each level. A0 can exist in any industry, including an industry whose more connected operators are usually described at higher levels.

03 / RESEARCH METHOD

Evidence, uncertainty and interpretation

Three layers of evidence

1. Classification backbone: official UN codes and hierarchy, with selected boundary checks against the explanatory notes and August 2026 errata. Structural coverage is auditable: 22 sections, 87 divisions, 258 groups and 463 classes. [S01] [S02] [S03]

2. Measured anchors: surveys and statistics with a stated year, geography, denominator and definition. These appear on the next pages. They are not extrapolated into fabricated worldwide class-by-class adoption rates.

3. Analyst profiles: sector evidence and workflow analysis generate the class-level software and AI bands. The 463 profiles use 191 reusable workflow families, with class-specific business examples and AI applications. This is not 463 independent field surveys.

What the confidence label means

Label	What it supports	What it does not mean
Medium	A directional workflow estimate with a relevant sector evidence base and plausible operating systems.	A measured adoption rate for that class, or proof that every listed subtype uses AI.
Low	A workflow-based inference with sparse direct evidence, broad class boundaries or especially variable microbusiness practice.	That the business is unimportant or that it has no technology opportunity.

There are no high-confidence class-level global maturity ratings. Shared ratings are inherited from the workflow family; the 72 additional subtype profiles are low-confidence refinements. Source links at the start of each sector explain the evidence scope. They do not turn every AI example below them into a documented deployment.

Read the AI line carefully

"AI applications / candidates" mixes established applications in the related sector with plausible, bounded workflows for the exact business. It is a map of where AI fits, not a claim that all examples are already operating it. The adoption-context line explains whether basic administration, selective deployments or software-critical operations are more relevant.

A barcode, a calendar reminder, a rules engine, a normal report or ordinary factory automation is not automatically AI. Forecasting and anomaly detection only count here when implemented using an AI method. A shop receiving orders through an AI-powered marketplace is also not necessarily operating its own AI system.

Compare like with like: firm size, country, connectivity, language, data quality, regulation and business model all matter. Power, compute, usable data and skills are important enabling conditions; smaller firms may have very different implementation capacity from large enterprises. [S06] [S07] [S37]

04 / EMPIRICAL ANCHORS

What measured evidence actually says

These findings establish that adoption differs sharply by sector, firm size and measurement method. They should not be averaged into a worldwide percentage or treated as direct measurements for all 463 classes.

EU enterprises: any AI

19.95% overall; 17% small, 30.36% medium, 55.03% large. [S04]

2025; covered EU enterprises with at least 10 employees/self-employed persons. Small = 10-49; medium = 50-249; large = 250+. Excludes several sectors and microbusinesses.

EU sector comparison

62.52% information/communication; 40.43% professional/scientific/technical; 10.79% construction. [S04]

2025; same survey. Uses NACE Rev. 2, not this atlas's ISIC Rev. 5 section letters. A sector average is not a class-specific rate.

EU selected software

46.45% ERP; 28.51% CRM; 16.28% business intelligence; 79.02% have a website. [S05]

2025; covered EU enterprises with 10+ persons. These selected tools are not a measure of all software use.

US farm connectivity

85% of farms have internet access. [S08]

2025 USDA farm-technology survey. Connectivity does not establish use of farm-management software, sensors or AI.

UK financial firms

75% of respondent firms use AI. [S21]

2024 survey of 118 regulated financial firms. UK respondent evidence, not all global financial companies.

Measured evidence, continued

US hospital predictive AI

71% use the defined predictive AI. [S25]

2024 informative responses from 2,080 non-federal acute-care hospitals. Models embedded in or linked to electronic health records; missing/do-not-know answers excluded.

Construction professionals

19% report regular AI use in specific processes; less than 1% report full organization-wide integration. [S14]

RICS Q1 2026 construction sample: 1,883 respondents. A professional sample with regional weighting, not a census of builders or trades.

Selected professional services

40% report organizational generative-AI use. [S22]

2026 Thomson Reuters study of more than 1,500 professionals in selected legal, tax, accounting, risk, fraud and government fields.

Developers using AI tools

About 83% report using at least one AI tool or agent at work. [S20]

2026 Stack Overflow AI-tools question: 17,464 respondents. Measures voluntary individual respondents, not software-company adoption.

Why the numbers differ

"Used a chatbot once," "runs one production model," "uses predictive AI through an EHR," and "fully integrates AI across operations" answer different questions. The class profiles retain that distinction instead of attaching an unjustified single adoption percentage to each business.

05 / BROWSE THE WHOLE ECONOMY

Sector map at a glance

The summaries below are an editorial reading guide, not a statistical ranking. The chapter for each sector contains every underlying class and its own software/AI profile.

Sec.	Activity family	Classes	Software base / AI pattern
A	Agriculture, forestry and fishing	39	Basic apps to precision operations Selective field, crop and animal monitoring.
B	Mining and quarrying	14	Basic sites to integrated industrial mines Condition monitoring and geological analysis at advanced operators.
C	Manufacturing	141	Craft production to connected factories Quality inspection, maintenance and planning; large variation.
D	Electricity, gas, steam and cooling	6	Usually operationally software-dependent Forecasting and network/plant anomaly detection.
E	Water, sewerage, waste and remediation	8	Small collectors to connected treatment plants Monitoring and sorting; uneven deployment.
F	Construction	12	Trade administration to integrated project systems Pilots and selective estimating/document workflows.
G	Wholesale and retail	41	Micro-retail to integrated commerce networks Recommendations, forecasting, fraud and service assistance.
H	Transportation and storage	23	Owner-operators to digital logistics networks Routing, document processing, maintenance and demand.
I	Accommodation and food services	10	Basic bookings to integrated hotel/food operations Guest support, purchasing, pricing and demand.
J	Publishing, broadcasting and content	15	Production tools to digital-native publishing Content assistance, discovery and software development.
K	Telecom, computing and information services	9	Highly software-dependent technical services Coding, networks, operations and AI product infrastructure.
L	Finance and insurance	24	Regulated digital core; outsourced fund exceptions Fraud, document analysis and supervised decision support.
M	Real estate	3	Single landlords to integrated portfolios Documents, maintenance and valuation assistance.
N	Professional, scientific and technical services	17	Specialist knowledge and project systems Drafting, search, analysis and professional review.
O	Administrative and support services	25	Manual field services to digital intermediaries Scheduling, service, documents and workforce assistance.
P	Public administration, defence and social security	8	Public records and critical administrative systems Bounded case/document assistance with accountability.

Sec.	Activity family	Classes	Software base / AI pattern
Q	Education	12	Low-resource schools to digital learning systems Teaching assistance; institutional adoption varies.
R	Health and social work	11	Basic clinics to connected hospitals/labs Administrative and specialist clinical applications differ.
S	Arts, sports and recreation	19	Independent practitioners to large institutions Creative assistance, archives, sport analysis and service.
T	Membership, repair and personal services	22	Often microbusiness administration Mostly communications, bookings and repair assistance.
U	Household employment and own-use production	3	Optional household tools; not enterprises Personal assistance, not business adoption.
V	Extraterritorial organizations and bodies	1	Institutional systems plus variable field access Translation, documents and supervised programme support.

The class counts sum to 463. Software and AI descriptions are original synthesis informed by the sources cited in each sector chapter. The official hierarchy is the coverage backbone. [S01] [S02]

06 / WHAT "TECH USE" MEANS

Software functions and common AI layers

Most businesses combine a small number of common software functions with systems specific to their activity. The class profiles emphasize the activity-specific layer instead of repeating email and accounting in every line. The table is an analytical organizing framework.

Function	Conventional software	Possible AI layer
Customer and revenue	Customer records, quotes, checkout, subscriptions, marketing, bookings.	Draft replies, classify enquiries, recommend products, forecast demand.
Operations and delivery	Orders, cases, jobs, projects, routes, production, clinical or care workflows.	Extract documents, detect exceptions, propose plans, assist skilled staff.
Assets and materials	Stock, warehouses, procurement, machines, fleets, buildings.	Visual checks, failure alerts, demand estimation and maintenance triage.
People and administration	Payroll, staffing, training, recruitment, expenses, documents.	Summaries, scheduling assistance and searchable internal knowledge.
Finance and control	Accounting, billing, risk, compliance, audit and regulated records.	Reconciliation support, anomaly review and evidence retrieval.
Research, design and content	Design tools, scientific data, engineering models, source code, media.	Coding and drafting, image analysis, experiments and design assistance.
Data, infrastructure and security	Databases, networks, cloud services, identities, logs and access controls.	Incident triage, anomaly analysis and governed AI deployment.

Useful abbreviations: ERP = enterprise resource planning; CRM = customer-relationship management; BI = business intelligence; EHR = electronic health record; POS = point of sale; GIS = geographic information system; CAD = computer-aided design; BIM = building information modelling; SaaS = software supplied as a hosted service. The presence of any one of these does not prove AI adoption.

Common deployment caution: decisions affecting health, safety, employment, benefits, money or rights need appropriate human accountability, reliable evaluation and secure data handling. Each profile adds a more specific watchpoint; these are screening considerations, not jurisdiction-specific legal advice. [S30] [S26] [S24]

COMPLETE ACTIVITY CATALOGUE / SECTION A

A | Agriculture, forestry and fishing

3 divisions / 14 groups / 39 classes / 144 business examples

Physical production dominates, but farm records, connected machinery and remote monitoring create very different operating models. Smallholders and industrial farms should never receive one unquestioned adoption score.

Evidence scope: Farm connectivity is measurable in some countries; class-level global AI use is not. FAO provides application context, not a census of adoption. [S08] [S09] [S06] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 01 | Crop and animal production; hunting and related services

Group 011 | Non-perennial crops

0111 | Cereal, pulse and oilseed growing

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: wheat farms; maize farms; soybean farms; lentil and chickpea farms.

Software: Farm records, accounting, field maps, machinery guidance, weather and irrigation tools.

AI applications / candidates: Crop scouting and yield estimates.

Adoption context: Small farms may rely on phones; larger farms add sensors and imagery. AI is selective.

Watchpoint: Local crop data and reliable connectivity matter.

0112 | Rice growing

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: irrigated rice farms; rain-fed rice farms; rice-seed growers.

Software: Farm records, accounting, field maps, machinery guidance, weather and irrigation tools.

AI applications / candidates: Paddy monitoring and irrigation alerts.

Adoption context: Small farms may rely on phones; larger farms add sensors and imagery. AI is selective.

Watchpoint: Local crop data and reliable connectivity matter.

0113 | Vegetables, melons, roots and tubers

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: vegetable market gardens; potato farms; melon farms; greenhouse vegetable growers.

Software: Crop planning, nursery inventory, greenhouse controls, irrigation records and sales software.

AI applications / candidates: Disease spotting and harvest timing.

Adoption context: Controlled environments have more usable data; open-field microbusinesses are less connected.

Watchpoint: Validate recommendations for the actual crop and climate.

0114 | Sugar-cane growing

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: sugar-cane estates; independent cane growers; cooperative cane farms.

Software: Farm records, accounting, field maps, machinery guidance, weather and irrigation tools.

AI applications / candidates: Harvest planning and field-yield estimates.

Adoption context: Small farms may rely on phones; larger farms add sensors and imagery. AI is selective.

Watchpoint: Local crop data and reliable connectivity matter.

0115 | Tobacco growing

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: leaf-tobacco farms; contract tobacco growers; tobacco-seed growers.

Software: Farm records, accounting, field maps, machinery guidance, weather and irrigation tools.

AI applications / candidates: Leaf grading and field monitoring.

Adoption context: Small farms may rely on phones; larger farms add sensors and imagery. AI is selective.

Watchpoint: Local crop data and reliable connectivity matter.

0116 | Fibre-crop growing

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: cotton farms; flax growers; hemp-fibre farms; jute farms.

Software: Farm records, accounting, field maps, machinery guidance, weather and irrigation tools.

AI applications / candidates: Plant-health monitoring and fibre-yield forecasts.

Adoption context: Small farms may rely on phones; larger farms add sensors and imagery. AI is selective.

Watchpoint: Local crop data and reliable connectivity matter.

0119 | Other non-perennial crops

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: cut-flower farms; fodder growers; annual ornamental crops; other annual crop farms.

Software: Crop planning, nursery inventory, greenhouse controls, irrigation records and sales software.

AI applications / candidates: Quality grading and harvest forecasts.

Adoption context: Controlled environments have more usable data; open-field microbusinesses are less connected.

Watchpoint: Validate recommendations for the actual crop and climate.

Group 012 | Perennial crops

0121 | Grape growing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: table-grape farms; wine-grape vineyards; raisin-grape growers.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Vine disease detection and ripeness estimates.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

0122 | Tropical and subtropical fruit growing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: banana farms; mango orchards; pineapple farms; avocado orchards.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Fruit counting and crop-health monitoring.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

0123 | Citrus growing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: orange groves; lemon orchards; mandarin farms; grapefruit growers.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Disease detection and fruit-size grading.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

0124 | Pome and stone-fruit growing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: apple orchards; pear orchards; peach farms; cherry orchards.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Blossom monitoring and harvest forecasts.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

0125 | Other tree and bush fruits and nuts

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: berry farms; walnut orchards; almond orchards; date-palm farms.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Nut and berry yield prediction.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

0126 | Oil-bearing fruit growing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: oil-palm plantations; olive groves; coconut plantations.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Plantation inspection and harvest scheduling.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

0127 | Beverage-crop growing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: coffee farms; tea estates; cocoa farms; mate growers.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Leaf or bean quality assessment.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

0128 | Spice, aromatic and medicinal crops

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: pepper farms; vanilla growers; lavender farms; medicinal-herb growers.

Software: Crop planning, nursery inventory, greenhouse controls, irrigation records and sales software.

AI applications / candidates: Plant identification and harvest-quality checks.

Adoption context: Controlled environments have more usable data; open-field microbusinesses are less connected.

Watchpoint: Validate recommendations for the actual crop and climate.

0129 | Other perennial crops

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: rubber plantations; Christmas-tree growers; sap-tree growers; energy-coppice growers.

Software: Orchard maps, harvest records, irrigation tools, traceability and farm accounting.

AI applications / candidates: Tree-health and productivity monitoring.

Adoption context: Larger plantations can support image analysis; many small growers use basic apps only.

Watchpoint: Image accuracy changes with season and cultivar.

Group 013 | Plant propagation

0130 | Plant propagation

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: seedling nurseries; ornamental plant nurseries; tissue-culture nurseries; fruit-tree propagators.

Software: Crop planning, nursery inventory, greenhouse controls, irrigation records and sales software.

AI applications / candidates: Seedling grading and environmental alerts.

Adoption context: Controlled environments have more usable data; open-field microbusinesses are less connected.

Watchpoint: Validate recommendations for the actual crop and climate.

Group 014 | Animal production

0141 | Cattle and buffalo raising

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: dairy-cattle farms; beef-cattle ranches; buffalo dairies; cattle breeding farms.

Software: Herd or flock records, breeding tools, feed management, milking controls and accounting.

AI applications / candidates: Herd-health alerts and feeding forecasts.

Adoption context: Commercial units can deploy sensor-based alerts; small herds often remain partly manual.

Watchpoint: Keep veterinarians and animal-welfare checks involved.

0142 | Horses and other equines

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: horse-breeding farms; donkey breeders; mule breeders; working-equine farms.

Software: Herd or flock records, breeding tools, feed management, milking controls and accounting.

AI applications / candidates: Animal-health records and observation alerts.

Adoption context: Commercial units can deploy sensor-based alerts; small herds often remain partly manual.

Watchpoint: Keep veterinarians and animal-welfare checks involved.

0143 | Camels and camelids

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: camel dairy farms; camel breeding farms; alpaca farms; llama farms.

Software: Herd or flock records, breeding tools, feed management, milking controls and accounting.

AI applications / candidates: Herd-health and feed monitoring.

Adoption context: Commercial units can deploy sensor-based alerts; small herds often remain partly manual.

Watchpoint: Keep veterinarians and animal-welfare checks involved.

0144 | Sheep and goat raising

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: sheep ranches; goat dairies; meat-goat farms; wool-producing farms.

Software: Herd or flock records, breeding tools, feed management, milking controls and accounting.

AI applications / candidates: Grazing and animal-health monitoring.

Adoption context: Commercial units can deploy sensor-based alerts; small herds often remain partly manual.

Watchpoint: Keep veterinarians and animal-welfare checks involved.

0145 | Pig raising

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: pig breeding units; pig finishing farms; integrated pig farms.

Software: Herd or flock records, breeding tools, feed management, milking controls and accounting.

AI applications / candidates: Feed conversion and welfare alerts.

Adoption context: Commercial units can deploy sensor-based alerts; small herds often remain partly manual.

Watchpoint: Keep veterinarians and animal-welfare checks involved.

0146 | Poultry raising

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: broiler farms; egg farms; hatcheries; duck and turkey farms.

Software: Herd or flock records, breeding tools, feed management, milking controls and accounting.

AI applications / candidates: Flock-health and barn-condition alerts.

Adoption context: Commercial units can deploy sensor-based alerts; small herds often remain partly manual.

Watchpoint: Keep veterinarians and animal-welfare checks involved.

0149 | Other animal raising

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: beekeepers; rabbit farms; silkworm farms; farmed-insect producers.

Software: Herd or flock records, breeding tools, feed management, milking controls and accounting.

AI applications / candidates: Colony or enclosure health monitoring.

Adoption context: Commercial units can deploy sensor-based alerts; small herds often remain partly manual.

Watchpoint: Keep veterinarians and animal-welfare checks involved.

Group 015 | Mixed crop and livestock farming

0150 | Mixed crop and livestock farming

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: mixed dairy-crop farms; livestock-and-grain farms; diversified smallholdings.

Software: Crop and livestock records, farm accounting, feed inventories and weather tools.

AI applications / candidates: Joint crop, feed and herd planning.

Adoption context: Integration is harder across several small enterprises; advanced farms connect datasets.

Watchpoint: Avoid treating missing field or herd records as facts.

Group 016 | Agricultural support and post-harvest activities

0161 | Crop-production support services

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: agricultural machinery contractors; crop-spraying services; irrigation contractors; harvesting contractors.

Software: Job scheduling, equipment tracking, customer records, invoicing and field maps.

AI applications / candidates: Field-job planning and crop-image interpretation.

Adoption context: AI usually enters through equipment or service apps rather than an in-house team.

Watchpoint: Check local conditions before acting on automated advice.

0162 | Animal-production support services

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: livestock breeding services; shearing contractors; farm animal boarding; herd-management contractors.

Software: Job scheduling, equipment tracking, customer records, invoicing and field maps.

AI applications / candidates: Service scheduling and animal-record summaries.

Adoption context: AI usually enters through equipment or service apps rather than an in-house team.

Watchpoint: Check local conditions before acting on automated advice.

0163 | Post-harvest crop activities

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: crop cleaning businesses; grain drying services; fruit preparation services; cotton ginning services.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Produce grading and spoilage alerts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

0164 | Seed processing for propagation

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: seed cleaning plants; seed grading businesses; seed treatment processors.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Seed defect detection and lot-quality forecasts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Group 017 | Hunting, trapping and related services

0170 | Hunting, trapping and related services

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: commercial game trapping; hunting-support services; wildlife-control trapping.

Software: Mobile messaging, maps, work orders, permits, basic accounting and payment tools.

AI applications / candidates: Field-record organization and permitted-activity reporting.

Adoption context: Small field operators often use little dedicated software; AI is mainly occasional assistance.

Watchpoint: Permissions, safety and local knowledge remain essential.

Division 02 | Forestry and logging

Group 021 | Silviculture and other forestry

0210 | Silviculture and other forestry

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: timber plantations; forest nurseries; forest regeneration contractors; managed woodland operators.

Software: Forest maps, inventories, harvest planning, fleet tracking and certification records.

AI applications / candidates: Tree inventory and forest-health mapping.

Adoption context: Large operators can combine satellite, drone and machine data; small contractors may not.

Watchpoint: Remote estimates need ground checks.

Group 022 | Logging

0220 | Logging

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: timber harvesting contractors; logging businesses; fuelwood logging; forest extraction contractors.

Software: Forest maps, inventories, harvest planning, fleet tracking and certification records.

AI applications / candidates: Harvest planning and machine-maintenance alerts.

Adoption context: Large operators can combine satellite, drone and machine data; small contractors may not.

Watchpoint: Remote estimates need ground checks.

Group 023 | Gathering non-wood forest products

0230 | Gathering non-wood forest products

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: wild mushroom gatherers; cork harvesters; natural resin collectors; wild berry collectors.

Software: Mobile messaging, maps, work orders, permits, basic accounting and payment tools.

AI applications / candidates: Product identification support and collection records.

Adoption context: Small field operators often use little dedicated software; AI is mainly occasional assistance.

Watchpoint: Permissions, safety and local knowledge remain essential.

Group 024 | Forestry support services

0240 | Forestry support services

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: forest survey contractors; forest fire-protection services; forest management contractors.

Software: Forest maps, inventories, harvest planning, fleet tracking and certification records.

AI applications / candidates: Remote forest monitoring and risk mapping.

Adoption context: Large operators can combine satellite, drone and machine data; small contractors may not.

Watchpoint: Remote estimates need ground checks.

Division 03 | Fishing and aquaculture

Group 031 | Fishing

0311 | Marine fishing

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: offshore fishing fleets; coastal fishers; shellfish fishing operators; commercial seaweed gatherers.

Software: Navigation, electronic logbooks, vessel monitoring, catch traceability and accounting.

AI applications / candidates: Catch documentation and environmental forecasting.

Adoption context: Fleet operators are more digital than small-scale fishers; AI remains uneven.

Watchpoint: Respect fishing limits; forecasts do not guarantee catches.

0312 | Freshwater fishing

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: lake fisheries; river fishing businesses; inland shellfish fisheries.

Software: Navigation, electronic logbooks, vessel monitoring, catch traceability and accounting.

AI applications / candidates: Catch records and fishing-condition analysis.

Adoption context: Fleet operators are more digital than small-scale fishers; AI remains uneven.

Watchpoint: Respect fishing limits; forecasts do not guarantee catches.

Group 032 | Aquaculture

0321 | Marine aquaculture

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: salmon sea farms; oyster farms; mussel farms; seaweed farms.

Software: Stock records, feeding systems, water sensors, health logs and traceability.

AI applications / candidates: Feeding control and stock-health monitoring.

Adoption context: Intensive farms can use automated monitoring; small ponds may use manual sampling.

Watchpoint: Poor sensor calibration can create unsafe advice.

0322 | Freshwater aquaculture

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: carp farms; tilapia farms; trout farms; freshwater hatcheries.

Software: Stock records, feeding systems, water sensors, health logs and traceability.

AI applications / candidates: Water-quality alerts and growth prediction.

Adoption context: Intensive farms can use automated monitoring; small ponds may use manual sampling.

Watchpoint: Poor sensor calibration can create unsafe advice.

Group 033 | Fishing and aquaculture support

0330 | Fishing and aquaculture support

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: fish-farm monitoring contractors; fishing support services; aquaculture technical services.

Software: Stock records, feeding systems, water sensors, health logs and traceability.

AI applications / candidates: Water-quality interpretation and service planning.

Adoption context: Intensive farms can use automated monitoring; small ponds may use manual sampling.

Watchpoint: Poor sensor calibration can create unsafe advice.

COMPLETE ACTIVITY CATALOGUE / SECTION B

B | Mining and quarrying

5 divisions / 10 groups / 14 classes / 47 business examples

The main split is between artisanal or small extraction sites and industrial operators with geological models, fleets and continuous sensor data. Expensive equipment makes selected predictive workflows commercially plausible.

Evidence scope: Large-operator cases demonstrate feasibility, not prevalence across all mines. Oil, gas and mineral subtypes inherit cautious analyst estimates. [S10] [S06] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 05 | Coal and lignite mining

Group 051 | Hard-coal mining

0510 | Hard-coal mining

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: underground coal mines; open-cut coal mines; hard-coal preparation operations.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Equipment-failure prediction and condition monitoring.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

Group 052 | Lignite mining

0520 | Lignite mining

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: lignite pits; brown-coal mining operators; lignite preparation operations.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Mine planning and equipment monitoring.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

Division 06 | Crude petroleum and natural gas extraction

Group 061 | Crude-petroleum extraction

0610 | Crude-petroleum extraction

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: onshore oil producers; offshore oil operators; oil-sands extraction businesses.

Software: Asset models, production controls, process historians, maintenance and enterprise systems.

AI applications / candidates: Equipment anomaly detection and production forecasting.

Adoption context: Industrial operations provide substantial sensor data; AI is deployed in selected monitored workflows.

Watchpoint: Protect operational networks and independent safety systems.

Group 062 | Natural-gas extraction

0620 | Natural-gas extraction

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: onshore gas producers; offshore gas producers; coal-seam gas operators.

Software: Asset models, production controls, process historians, maintenance and enterprise systems.

AI applications / candidates: Leak detection and asset-condition monitoring.

Adoption context: Industrial operations provide substantial sensor data; AI is deployed in selected monitored workflows.

Watchpoint: Protect operational networks and independent safety systems.

Division 07 | Metal-ore mining

Group 071 | Iron-ore mining

0710 | Iron-ore mining

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: iron-ore open pits; underground iron mines; iron-ore beneficiation operations.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Ore characterization and maintenance forecasts.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

Group 072 | Non-ferrous metal ores

0721 | Uranium and thorium ore mining

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: licensed uranium mines; licensed thorium ore operations.

Software: Production planning, quality records, traceability, permits, maintenance and controlled document systems.

AI applications / candidates: Compliance-document review and asset maintenance.

Adoption context: Digital records may be essential even where AI remains limited to administrative or inspection assistance.

Watchpoint: Do not delegate regulated release or safety decisions.

0729 | Other non-ferrous metal-ore mining

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: copper mines; gold mines; bauxite mines; nickel and rare-earth mines.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Ore sorting and plant-condition analysis.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

Division 08 | Other mining and quarrying

Group 081 | Stone, sand and clay quarrying

0810 | Stone, sand and clay quarrying

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: dimension-stone quarries; sand pits; gravel quarries; clay pits.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Stockpile measurement and maintenance scheduling.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

Group 089 | Other mining and quarrying not elsewhere classified

0891 | Chemical and fertilizer mineral mining

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: phosphate mines; potash mines; sulphur extraction; chemical-mineral quarries.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Material-quality and equipment monitoring.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

0892 | Peat extraction

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: peat harvesting businesses; horticultural peat suppliers; peat preparation operators.

Software: Mobile messaging, maps, work orders, permits, basic accounting and payment tools.

AI applications / candidates: Site mapping and production planning.

Adoption context: Small field operators often use little dedicated software; AI is mainly occasional assistance.

Watchpoint: Permissions, safety and local knowledge remain essential.

0893 | Salt extraction

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: salt mines; sea-salt works; salt-brine extraction businesses.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Evaporation-condition and equipment monitoring.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

0899 | Other mining and quarrying

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: gemstone mines; natural graphite mines; industrial-mineral mines; other mineral extraction.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Mineral grading and site monitoring.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

Division 09 | Mining support services

Group 091 | Oil and gas extraction support

0910 | Oil and gas extraction support

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: drilling contractors; well-service companies; exploration support providers; offshore support specialists.

Software: Asset models, production controls, process historians, maintenance and enterprise systems.

AI applications / candidates: Maintenance planning and operational-document search.

Adoption context: Industrial operations provide substantial sensor data; AI is deployed in selected monitored workflows.

Watchpoint: Protect operational networks and independent safety systems.

Group 099 | Other mining support services

0990 | Other mining support services

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: mineral exploration contractors; test-drilling contractors; mine dewatering services; mining technical contractors.

Software: Geological models, mine planning, fleet dispatch, maintenance, safety and enterprise systems.

AI applications / candidates: Survey interpretation and equipment monitoring.

Adoption context: Large mines have connected operations; artisanal sites may be largely manual. AI concentrates in larger operators.

Watchpoint: Safety-critical decisions need qualified review.

COMPLETE ACTIVITY CATALOGUE / SECTION C

C | Manufacturing

24 divisions / 71 groups / 141 classes / 494 business examples

The catalogue covers every manufacturing division, from food and textiles to semiconductors, aircraft and repair. Software used to design and run a factory is different from software or AI embedded in its products.

Evidence scope: NIST supplies manufacturing research context; Eurostat supplies selected regional software and AI benchmarks. Individual product-class ratings are estimates, not survey results. [S11] [S04] [S05] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 10 | Food products

Group 101 | Meat processing and preservation

1010 | Meat processing and preservation

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: abattoirs; meat cutting plants; sausage manufacturers; meat preservation plants.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Hygiene checks, grading and production forecasts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Group 102 | Fish and shellfish processing

1020 | Fish and shellfish processing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: fish filleting plants; seafood freezing plants; canned-seafood factories; shellfish processors.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Species or defect checks and cold-chain alerts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Group 103 | Fruit and vegetable processing

1030 | Fruit and vegetable processing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: canning plants; frozen-vegetable factories; dried-fruit processors; juice processors.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Produce sorting and spoilage prediction.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Group 104 | Vegetable and animal oils and fats

1040 | Vegetable and animal oils and fats

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: edible-oil mills; oil refineries; margarine factories; animal-fat rendering plants.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Process-quality monitoring and maintenance.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

Group 105 | Dairy-product manufacturing

1050 | Dairy-product manufacturing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: milk processors; cheese factories; yoghurt producers; ice-cream manufacturers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Batch-quality monitoring and demand forecasts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Group 106 | Grain milling and starch products

1061 | Grain-mill products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: flour mills; rice mills; breakfast-cereal plants; grain milling businesses.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Grain grading and mill-condition monitoring.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1062 | Starches and starch products

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: corn-starch plants; potato-starch plants; glucose-syrup factories.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Process-yield and quality monitoring.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

Group 107 | Other food products

1071 | Bakery products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial bread factories; biscuit manufacturers; cake bakeries; packaged pastry producers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Demand forecasting and visual quality checks.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1072 | Sugar manufacturing

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: cane-sugar mills; beet-sugar factories; sugar refineries.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Process-quality and equipment monitoring.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

1073 | Cocoa, chocolate and confectionery

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: chocolate makers; candy manufacturers; chewing-gum producers; cocoa processors.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Defect inspection and seasonal demand forecasts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1074 | Pasta, noodles and related products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: pasta factories; noodle manufacturers; couscous producers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Shape or drying defects and demand forecasts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1075 | Prepared meals and dishes

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: frozen-meal manufacturers; packaged-meal factories; shelf-stable meal producers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Production forecasting and label checks.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1076 | Coffee and tea processing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: coffee roasters; instant-coffee manufacturers; tea processing factories; tea blending businesses.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Roast or leaf grading and batch consistency.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1079 | Other food products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: sauce makers; spice processors; infant-food manufacturers; specialty food-ingredient producers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Ingredient-quality checks and demand forecasts.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Group 108 | Prepared animal feeds

1080 | Prepared animal feeds

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: livestock feed mills; aquaculture-feed factories; pet-food manufacturers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Ingredient-quality and feed-demand prediction.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Division 11 | Beverages

Group 110 | Beverages

1101 | Distilled alcoholic beverages

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: distilleries; spirit blending businesses; liqueur producers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Batch consistency and production planning.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1102 | Wine manufacturing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: wineries; fruit-wine producers; sparkling-wine producers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Fermentation monitoring and quality grading.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1103 | Beer manufacturing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: large breweries; craft breweries; alcohol-free beer producers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Fermentation anomalies and demand forecasting.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1104 | Malt manufacturing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: barley maltsters; specialty-malt producers; industrial malting plants.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Germination and kiln-quality monitoring.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

1105 | Soft drinks and bottled waters

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: soft-drink bottlers; mineral-water bottlers; flavored-water producers; energy-drink manufacturers.

Software: Batch production, recipes, inventory, quality checks, expiry tracking and food traceability.

AI applications / candidates: Bottle defects and demand forecasting.

Adoption context: Factories adopt more connected systems than workshops; AI often arrives through inspection or planning tools.

Watchpoint: Human food-safety responsibility is unchanged.

Division 12 | Tobacco products

Group 120 | Tobacco-product manufacturing

1200 | Tobacco-product manufacturing

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: cigarette factories; cigar makers; smoking-tobacco processors.

Software: Production planning, quality records, traceability, permits, maintenance and controlled document systems.

AI applications / candidates: Packaging checks and compliance documentation.

Adoption context: Digital records may be essential even where AI remains limited to administrative or inspection assistance.

Watchpoint: Do not delegate regulated release or safety decisions.

Division 13 | Textiles

Group 131 | Textile spinning, weaving and finishing

1311 | Textile-fibre preparation and spinning

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: cotton spinning mills; wool preparation mills; yarn manufacturers; textile-fibre processors.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Yarn-defect detection and machine monitoring.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

1312 | Textile weaving

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: cotton weaving mills; silk weaving mills; industrial fabric weavers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Fabric-defect detection and loom monitoring.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

1313 | Textile finishing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: dye houses; fabric finishing plants; textile bleaching businesses; textile printing finishers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Shade consistency and process monitoring.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

Group 139 | Other textiles

1391 | Knitted and crocheted fabrics

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: knitted fabric mills; hosiery-fabric producers; technical knit manufacturers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Knit-defect detection and production planning.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

1392 | Made-up textiles other than apparel

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: bed-linen factories; curtain makers; tent manufacturers; tarpaulin producers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Cutting-layout assistance and seam inspection.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

1393 | Carpets and rugs

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: carpet mills; handmade rug workshops; mat manufacturers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Pattern checks and defect inspection.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

1394 | Cordage, rope, twine and netting

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: rope factories; fishing-net makers; cordage manufacturers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Strength-test analytics and visual inspection.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

1399 | Other textiles

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: nonwoven manufacturers; felt makers; technical-textile producers; narrow-fabric makers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Material inspection and process anomalies.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

Division 14 | Wearing apparel

Group 141 | Wearing apparel other than fur

1410 | Wearing apparel other than fur

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: garment factories; tailoring workshops; uniform manufacturers; leather-clothing makers.

Software: Pattern design, cutting plans, orders, inventory, production tracking and supplier records.

AI applications / candidates: Sizing support, pattern assistance and visual checks.

Adoption context: Large factories connect design to production; tailoring shops may use only orders and payments.

Watchpoint: Check fit, material use and labor implications.

Group 142 | Fur articles

1420 | Fur articles

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: fur garment makers; fur-accessory manufacturers; fur article workshops.

Software: Pattern design, cutting plans, orders, inventory, production tracking and supplier records.

AI applications / candidates: Material grading and cutting-layout assistance.

Adoption context: Large factories connect design to production; tailoring shops may use only orders and payments.

Watchpoint: Check fit, material use and labor implications.

Group 143 | Knitted and crocheted apparel

1430 | Knitted and crocheted apparel

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: sweater manufacturers; hosiery factories; knitted-underwear makers.

Software: Pattern design, cutting plans, orders, inventory, production tracking and supplier records.

AI applications / candidates: Pattern assistance and stitch inspection.

Adoption context: Large factories connect design to production; tailoring shops may use only orders and payments.

Watchpoint: Check fit, material use and labor implications.

Division 15 | Leather and related products

Group 151 | Leather preparation, luggage and saddlery

1511 | Leather and fur tanning and dressing

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: tanneries; leather finishing plants; fur dressing businesses.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Surface grading and process consistency.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

1512 | Luggage, handbags, saddlery and harness

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: handbag manufacturers; suitcase makers; saddlery workshops; harness manufacturers.

Software: Pattern design, cutting plans, orders, inventory, production tracking and supplier records.

AI applications / candidates: Material inspection and cutting-layout support.

Adoption context: Large factories connect design to production; tailoring shops may use only orders and payments.

Watchpoint: Check fit, material use and labor implications.

Group 152 | Footwear manufacturing

1520 | Footwear manufacturing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: shoe factories; boot makers; sandal manufacturers; specialist footwear workshops.

Software: Pattern design, cutting plans, orders, inventory, production tracking and supplier records.

AI applications / candidates: Fit modeling and visual quality inspection.

Adoption context: Large factories connect design to production; tailoring shops may use only orders and payments.

Watchpoint: Check fit, material use and labor implications.

Division 16 | Wood, cork and plaited products, excluding furniture

Group 161 | Sawmilling and wood planing

1610 | Sawmilling and wood planing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: sawmills; timber planing mills; wood drying businesses.

Software: Design and cutting software, timber inventories, machine controls and production planning.

AI applications / candidates: Timber grading and cutting optimization.

Adoption context: Industrial mills and factories can use vision; artisan shops usually use lighter tools.

Watchpoint: Keep machine safety independent of model suggestions.

Group 162 | Wood, cork and plaited products

1621 | Veneer sheets and wood panels

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: plywood mills; particleboard plants; veneer manufacturers; fibreboard plants.

Software: Design and cutting software, timber inventories, machine controls and production planning.

AI applications / candidates: Panel-defect checks and yield planning.

Adoption context: Industrial mills and factories can use vision; artisan shops usually use lighter tools.

Watchpoint: Keep machine safety independent of model suggestions.

1622 | Builders' carpentry and joinery

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: door factories; window-frame makers; prefabricated timber component makers; stair workshops.

Software: Design and cutting software, timber inventories, machine controls and production planning.

AI applications / candidates: Design assistance and cutting schedules.

Adoption context: Industrial mills and factories can use vision; artisan shops usually use lighter tools.

Watchpoint: Keep machine safety independent of model suggestions.

1623 | Wooden containers

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: pallet manufacturers; wooden-crate makers; cooperages; wooden packaging manufacturers.

Software: Design and cutting software, timber inventories, machine controls and production planning.

AI applications / candidates: Damage inspection and production forecasting.

Adoption context: Industrial mills and factories can use vision; artisan shops usually use lighter tools.

Watchpoint: Keep machine safety independent of model suggestions.

1629 | Other wood, cork and plaited products

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: cork-product factories; basket workshops; wooden utensil makers; straw-product makers.

Software: Design tools, order books, online selling, stock records, accounting and payments.

AI applications / candidates: Design drafting and order forecasting.

Adoption context: Handwork remains central; AI is generally a design or marketing aid rather than autonomous production.

Watchpoint: Protect original designs and customer information.

Division 17 | Paper and paper products

Group 170 | Paper and paper products

1701 | Pulp, paper and paperboard

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: pulp mills; paper mills; paperboard factories.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Process control support and defect detection.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

1702 | Corrugated paper and paper containers

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: corrugated-box plants; paper-bag factories; carton manufacturers.

Software: Packaging design, print specifications, production planning, quality and inventory software.

AI applications / candidates: Print or fold inspection and demand planning.

Adoption context: Larger converters can apply vision and scheduling models; smaller firms adopt embedded tools.

Watchpoint: Packaging and labeling errors still need accountable checks.

1709 | Other paper and paperboard articles

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: tissue factories; stationery manufacturers; paper-label producers; sanitary-paper manufacturers.

Software: Packaging design, print specifications, production planning, quality and inventory software.

AI applications / candidates: Product inspection and line monitoring.

Adoption context: Larger converters can apply vision and scheduling models; smaller firms adopt embedded tools.

Watchpoint: Packaging and labeling errors still need accountable checks.

Division 18 | Printing and recorded-media reproduction

Group 181 | Printing and printing-related services

1811 | Printing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: commercial printers; book printers; label printers; packaging printers.

Software: Prepress, layout, print job management, color control, estimating and billing software.

AI applications / candidates: Preflight checks and print-defect detection.

Adoption context: Core work is digital in many firms; AI adoption is more limited than basic print automation.

Watchpoint: Verify rights, spelling and print-ready specifications.

1812 | Printing-related services

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: prepress studios; bookbinders; plate-making businesses; print-finishing businesses.

Software: Prepress, layout, print job management, color control, estimating and billing software.

AI applications / candidates: File checking and job estimation.

Adoption context: Core work is digital in many firms; AI adoption is more limited than basic print automation.

Watchpoint: Verify rights, spelling and print-ready specifications.

Group 182 | Reproduction of recorded media

1820 | Reproduction of recorded media

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: disc replication plants; recorded-media duplication services; physical software-media duplicators.
Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.
AI applications / candidates: Production inspection and scheduling.
Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.
Watchpoint: Automation is not automatically AI.

Division 19 | Coke and refined petroleum products

Group 191 | Coke-oven products

1910 | Coke-oven products

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: coke works; coke processing plants; coke by-product recovery plants.
Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.
AI applications / candidates: Condition monitoring and process anomalies.
Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.
Watchpoint: Optimization must stay inside verified operating limits.

Group 192 | Petroleum and other fossil-fuel products

1920 | Petroleum and other fossil-fuel products

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: oil refineries; lubricant blenders; petroleum-wax producers; bituminous-product manufacturers.
Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.
AI applications / candidates: Process anomalies and maintenance forecasting.
Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.
Watchpoint: Optimization must stay inside verified operating limits.

Division 20 | Chemicals and chemical products

Group 201 | Basic chemicals, fertilizers and primary polymers

2011 | Basic chemicals

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: industrial-gas producers; basic organic-chemical plants; inorganic-chemical plants; pigment manufacturers.
Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.
AI applications / candidates: Process-quality monitoring and asset reliability.
Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.
Watchpoint: Optimization must stay inside verified operating limits.

2012 | Fertilizers and nitrogen compounds

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: fertilizer factories; nitrogen-compound plants; compound-fertilizer blenders.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Batch consistency and maintenance forecasting.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

2013 | Primary plastics and synthetic rubber

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: polymer resin plants; synthetic-rubber plants; primary-plastic producers.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Product-quality prediction and asset monitoring.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

Group 202 | Other chemical products

2021 | Pesticides and agrochemicals

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: licensed agrochemical manufacturers; biopesticide producers; plant-protection product factories.

Software: Production planning, quality records, traceability, permits, maintenance and controlled document systems.

AI applications / candidates: Label review, quality records and compliance checks.

Adoption context: Digital records may be essential even where AI remains limited to administrative or inspection assistance.

Watchpoint: Do not delegate regulated release or safety decisions.

2022 | Paints, coatings, inks and mastics

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: paint factories; industrial-coating makers; printing-ink manufacturers; sealant producers.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Color consistency and quality prediction.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

2023 | Soap, detergents, cleaners and cosmetics

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: soap factories; detergent makers; perfume houses; skincare manufacturers.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Batch-quality checks and demand forecasts.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

2029 | Other chemical products

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: adhesive manufacturers; specialty-chemical producers; photographic-chemical makers; chemical-preparation plants.

Software: Production planning, quality records, traceability, permits, maintenance and controlled document systems.

AI applications / candidates: Quality-document review and maintenance planning.

Adoption context: Digital records may be essential even where AI remains limited to administrative or inspection assistance.

Watchpoint: Do not delegate regulated release or safety decisions.

Group 203 | Man-made fibres

2030 | Man-made fibres

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: synthetic-fibre plants; regenerated-cellulose fibre producers; industrial-filament manufacturers.

Software: Production planning, loom or knitting controls, inventory, quality and order-management systems.

AI applications / candidates: Fibre consistency and machine monitoring.

Adoption context: Export mills are more integrated than small workshops; visual inspection is a practical AI entry point.

Watchpoint: Training images must reflect local materials and defects.

Division 21 | Pharmaceutical and medicinal products

Group 210 | Pharmaceutical and medicinal products

2100 | Pharmaceutical and medicinal products

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: generic-drug factories; biologics manufacturers; medicinal-botanical processors; vaccine manufacturers.

Software: Validated manufacturing, laboratory, quality, clinical or regulatory records and supply-chain systems.

AI applications / candidates: Quality-event review and production anomaly detection.

Adoption context: Some firms have advanced analytics; regulated deployment is slower than experimentation.

Watchpoint: Validate systems; protect health data and quality release.

Division 22 | Rubber and plastic products

Group 221 | Rubber products

2211 | Tyres, tubes and tyre retreading

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: tyre factories; inner-tube manufacturers; tyre retreaders.

Software: Mould and product design, machine controls, production records, quality and maintenance.

AI applications / candidates: Surface inspection and process consistency.

Adoption context: Connected molding or extrusion lines support predictive models; small plants may not.

Watchpoint: Validate models against actual materials and process windows.

2219 | Other rubber products

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: rubber-seal makers; industrial-hose manufacturers; rubber-sheet factories.

Software: Mould and product design, machine controls, production records, quality and maintenance.

AI applications / candidates: Defect inspection and maintenance forecasting.

Adoption context: Connected molding or extrusion lines support predictive models; small plants may not.

Watchpoint: Validate models against actual materials and process windows.

Group 222 | Plastic products

2220 | Plastic products

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: injection-moulding businesses; plastic-pipe factories; plastic packaging makers; plastic household-goods makers.

Software: Mould and product design, machine controls, production records, quality and maintenance.

AI applications / candidates: Moulding-defect detection and machine monitoring.

Adoption context: Connected molding or extrusion lines support predictive models; small plants may not.

Watchpoint: Validate models against actual materials and process windows.

Division 23 | Other non-metallic mineral products

Group 231 | Glass and glass products

2310 | Glass and glass products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: flat-glass plants; bottle manufacturers; glass-fibre producers; technical-glass manufacturers.

Software: Production recipes, kiln or furnace controls, quality, dispatch and maintenance systems.

AI applications / candidates: Surface-defect inspection and furnace monitoring.

Adoption context: Industrial plants have stronger data foundations than small workshops.

Watchpoint: Independent product testing remains necessary.

Group 239 | Other non-metallic mineral products

2391 | Refractory products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: refractory-brick factories; furnace-lining manufacturers; refractory castable producers.

Software: Production recipes, kiln or furnace controls, quality, dispatch and maintenance systems.

AI applications / candidates: Batch-quality monitoring and product inspection.

Adoption context: Industrial plants have stronger data foundations than small workshops.

Watchpoint: Independent product testing remains necessary.

2392 | Clay building materials

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: brickworks; roof-tile factories; clay-pipe makers.

Software: Production recipes, kiln or furnace controls, quality, dispatch and maintenance systems.

AI applications / candidates: Kiln monitoring and visual grading.

Adoption context: Industrial plants have stronger data foundations than small workshops.

Watchpoint: Independent product testing remains necessary.

2393 | Other porcelain and ceramics

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: tableware factories; sanitaryware manufacturers; technical-ceramic producers; ceramic workshops.

Software: Production recipes, kiln or furnace controls, quality, dispatch and maintenance systems.

AI applications / candidates: Glaze or shape defect detection.

Adoption context: Industrial plants have stronger data foundations than small workshops.

Watchpoint: Independent product testing remains necessary.

2394 | Cement, lime and plaster

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: cement plants; lime kilns; plaster manufacturers.

Software: Enterprise planning, process controls, batch records, laboratory systems and maintenance software.

AI applications / candidates: Kiln anomalies and energy-use forecasting.

Adoption context: Connected plants can use predictive models; smaller processors have less integrated data.

Watchpoint: Optimization must stay inside verified operating limits.

2395 | Concrete, cement and plaster articles

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: precast-concrete factories; ready-mix concrete producers; concrete-block makers; plasterboard factories.

Software: Production recipes, kiln or furnace controls, quality, dispatch and maintenance systems.

AI applications / candidates: Mix-quality prediction and production scheduling.

Adoption context: Industrial plants have stronger data foundations than small workshops.

Watchpoint: Independent product testing remains necessary.

2396 | Stone cutting, shaping and finishing

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: stone-cutting workshops; countertop fabricators; monument-stone workshops.

Software: Design tools, order books, online selling, stock records, accounting and payments.

AI applications / candidates: Slab-layout assistance and surface inspection.

Adoption context: Handwork remains central; AI is generally a design or marketing aid rather than autonomous production.

Watchpoint: Protect original designs and customer information.

2399 | Other non-metallic mineral products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: abrasive makers; mineral insulation factories; specialty mineral-product manufacturers.

Software: Production recipes, kiln or furnace controls, quality, dispatch and maintenance systems.

AI applications / candidates: Material-defect and process monitoring.

Adoption context: Industrial plants have stronger data foundations than small workshops.

Watchpoint: Independent product testing remains necessary.

Division 24 | Basic metals

Group 241 | Basic iron and steel

2410 | Basic iron and steel

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: integrated steelworks; electric-arc steel mills; steel rolling mills.

Software: Process controls, production planning, laboratory quality, traceability and asset maintenance.

AI applications / candidates: Surface inspection and asset-condition prediction.

Adoption context: Large mills use extensive sensors; foundries and workshops vary widely.

Watchpoint: Maintain independently validated quality and safety controls.

Group 242 | Basic precious and non-ferrous metals

2420 | Basic precious and non-ferrous metals

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: aluminium smelters; copper refineries; precious-metal refiners; non-ferrous rolling mills.

Software: Process controls, production planning, laboratory quality, traceability and asset maintenance.

AI applications / candidates: Product grading and energy/process anomalies.

Adoption context: Large mills use extensive sensors; foundries and workshops vary widely.

Watchpoint: Maintain independently validated quality and safety controls.

Group 243 | Metal casting

2431 | Iron and steel casting

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: iron foundries; steel foundries; cast-component suppliers.

Software: Process controls, production planning, laboratory quality, traceability and asset maintenance.

AI applications / candidates: Casting-defect inspection and process monitoring.

Adoption context: Large mills use extensive sensors; foundries and workshops vary widely.

Watchpoint: Maintain independently validated quality and safety controls.

2432 | Non-ferrous metal casting

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: aluminium foundries; copper-alloy foundries; non-ferrous die casters.

Software: Process controls, production planning, laboratory quality, traceability and asset maintenance.

AI applications / candidates: Casting inspection and scrap reduction support.

Adoption context: Large mills use extensive sensors; foundries and workshops vary widely.

Watchpoint: Maintain independently validated quality and safety controls.

Division 25 | Fabricated metal products, excluding machinery

Group 251 | Structural metal products, tanks and steam generators

2511 | Structural metal products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: steel fabricators; metal door and window makers; prefabricated metal-building makers.

Software: Computer-aided design, machining programs, quoting, production planning and inspection records.

AI applications / candidates: Drawing interpretation and weld-inspection support.

Adoption context: Modern shops are software-dependent but AI use is usually task-specific.

Watchpoint: A generated drawing or toolpath is not automatically safe.

2512 | Metal tanks, reservoirs and containers

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: storage-tank fabricators; pressure-container makers; metal-reservoir manufacturers.

Software: Computer-aided design, machining programs, quoting, production planning and inspection records.

AI applications / candidates: Inspection-record review and defect detection.

Adoption context: Modern shops are software-dependent but AI use is usually task-specific.

Watchpoint: A generated drawing or toolpath is not automatically safe.

2513 | Steam generators other than heating boilers

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial boiler makers; steam-generator fabricators; boiler component makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Inspection documentation and condition monitoring.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

Group 252 | Weapons and ammunition manufacturing

2520 | Weapons and ammunition manufacturing

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: licensed firearms manufacturers; licensed ammunition manufacturers; defence ordnance contractors.

Software: Production planning, quality records, traceability, permits, maintenance and controlled document systems.

AI applications / candidates: Inventory reconciliation and compliance-document review only.

Adoption context: Digital records may be essential even where AI remains limited to administrative or inspection assistance.

Watchpoint: Do not delegate regulated release or safety decisions.

Group 259 | Other fabricated metal products and metalworking

2591 | Metal forging, pressing and stamping

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: forging shops; metal stampers; roll-forming businesses; powder-metallurgy manufacturers.

Software: Computer-aided design, machining programs, quoting, production planning and inspection records.

AI applications / candidates: Tool-wear forecasting and defect inspection.

Adoption context: Modern shops are software-dependent but AI use is usually task-specific.

Watchpoint: A generated drawing or toolpath is not automatically safe.

2592 | Metal treatment, coating and machining

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: machine shops; electroplating businesses; heat-treatment shops; metal-coating contractors.

Software: Computer-aided design, machining programs, quoting, production planning and inspection records.

AI applications / candidates: Tool-condition monitoring and surface inspection.

Adoption context: Modern shops are software-dependent but AI use is usually task-specific.

Watchpoint: A generated drawing or toolpath is not automatically safe.

2593 | Cutlery, hand tools and general hardware

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: cutlery makers; hand-tool manufacturers; lock makers; hardware factories.

Software: Computer-aided design, machining programs, quoting, production planning and inspection records.

AI applications / candidates: Visual inspection and production scheduling.

Adoption context: Modern shops are software-dependent but AI use is usually task-specific.

Watchpoint: A generated drawing or toolpath is not automatically safe.

2599 | Other fabricated metal products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: wire-product makers; fastener factories; metal-can manufacturers; metal household-product makers.

Software: Computer-aided design, machining programs, quoting, production planning and inspection records.

AI applications / candidates: Product inspection and maintenance prediction.

Adoption context: Modern shops are software-dependent but AI use is usually task-specific.

Watchpoint: A generated drawing or toolpath is not automatically safe.

Division 26 | Computer, electronic and optical products

Group 261 | Electronic components and boards

2611 | Photovoltaic cells, panels and inverters

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: solar-cell plants; photovoltaic module assemblers; solar-inverter manufacturers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Cell-defect inspection and test-data analysis.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

2619 | Other electronic components and boards

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: semiconductor manufacturers; circuit-board makers; electronic component factories; board assemblers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Wafer or board inspection and yield analysis.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

Group 262 | Computers and peripheral equipment

2620 | Computers and peripheral equipment

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: computer assemblers; server manufacturers; storage-device makers; printer manufacturers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Test-log analysis and component inspection.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

Group 263 | Communication equipment

2630 | Communication equipment

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: network-equipment manufacturers; telecom radio makers; smartphone factories; communications-device makers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Hardware test analysis and quality inspection.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

Group 264 | Consumer electronics

2640 | Consumer electronics

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: television manufacturers; audio-equipment makers; consumer video-device makers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Assembly inspection and return-failure analysis.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

Group 265 | Measuring, testing, navigation and control equipment; clocks

2651 | Measuring, testing, navigation and control equipment

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: sensor manufacturers; laboratory instrument makers; industrial-control equipment makers; navigation-instrument makers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Calibration anomaly detection and test analysis.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

2652 | Watches and clocks

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: watch factories; clock makers; watch-movement manufacturers.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: Component inspection and test-data analysis.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

Group 266 | Irradiation and electromedical equipment

2660 | Irradiation and electromedical equipment

D3-D4 / A0-A3 / Estimate confidence: medium

Businesses: medical imaging equipment makers; radiotherapy equipment manufacturers; electrotherapeutic-device makers.

Software: Design controls, quality systems, test records, traceability and regulated manufacturing software.

AI applications / candidates: Manufacturing quality and service-log analysis.

Adoption context: AI in the device itself is separate from AI used by its manufacturer.

Watchpoint: Clinical and regulatory evidence cannot be replaced by generated text.

Group 267 | Optical instruments and photographic equipment

2670 | Optical instruments and photographic equipment

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: lens manufacturers; camera makers; microscope manufacturers; optical-instrument makers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Optical-defect inspection and calibration checks.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

Group 268 | Magnetic and optical media

2680 | Magnetic and optical media

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: magnetic-media manufacturers; blank optical-disc producers; recording-media makers.

Software: Product design, manufacturing execution, automated testing, quality and component traceability.

AI applications / candidates: Surface inspection and quality-test analysis.

Adoption context: Complex plants support mature analytics and inspection; smaller assemblers may use vendor features only.

Watchpoint: False rejects, missed defects and confidential designs matter.

Division 27 | Electrical equipment

Group 271 | Motors, generators and power-control equipment

2710 | Motors, generators and power-control equipment

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: electric-motor factories; transformer makers; generator manufacturers; switchgear producers.

Software: Electrical design, production planning, test systems, quality and serial-number traceability.

AI applications / candidates: Test-data analytics and predictive maintenance.

Adoption context: Scale and test-data quality determine how far AI is embedded.

Watchpoint: Safety certification requires validated evidence.

Group 272 | Batteries and accumulators

2720 | Batteries and accumulators

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: battery-cell manufacturers; battery-pack assemblers; industrial battery makers.

Software: Electrical design, production planning, test systems, quality and serial-number traceability.

AI applications / candidates: Cell-quality inspection and test anomaly detection.

Adoption context: Scale and test-data quality determine how far AI is embedded.

Watchpoint: Safety certification requires validated evidence.

Group 273 | Wiring and wiring devices

2731 | Fibre-optic cables

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: fibre-optic cable factories; optical-fibre cable assemblers; specialty optical-cable makers.

Software: Electrical design, production planning, test systems, quality and serial-number traceability.

AI applications / candidates: Cable-defect detection and line monitoring.

Adoption context: Scale and test-data quality determine how far AI is embedded.

Watchpoint: Safety certification requires validated evidence.

2732 | Other electrical and electronic wires and cables

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: power-cable factories; data-cable manufacturers; insulated-wire producers.

Software: Electrical design, production planning, test systems, quality and serial-number traceability.

AI applications / candidates: Insulation inspection and process consistency.

Adoption context: Scale and test-data quality determine how far AI is embedded.

Watchpoint: Safety certification requires validated evidence.

2733 | Wiring devices

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: switch and socket makers; electrical connector manufacturers; conduit-fitting makers.

Software: Electrical design, production planning, test systems, quality and serial-number traceability.

AI applications / candidates: Assembly inspection and test-log analysis.

Adoption context: Scale and test-data quality determine how far AI is embedded.

Watchpoint: Safety certification requires validated evidence.

Group 274 | Electric lighting equipment

2740 | Electric lighting equipment

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: LED luminaire factories; lamp manufacturers; outdoor-lighting equipment makers.

Software: Electrical design, production planning, test systems, quality and serial-number traceability.

AI applications / candidates: Light-quality testing and assembly inspection.

Adoption context: Scale and test-data quality determine how far AI is embedded.

Watchpoint: Safety certification requires validated evidence.

Group 275 | Domestic appliances

2750 | Domestic appliances

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: refrigerator factories; washing-machine makers; cooking-appliance manufacturers; small-appliance makers.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: End-of-line inspection and service-failure prediction.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

Group 279 | Other electrical equipment

2790 | Other electrical equipment

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: uninterruptible power-supply makers; electrical signaling equipment makers; specialty electrical-device manufacturers.

Software: Electrical design, production planning, test systems, quality and serial-number traceability.

AI applications / candidates: Electrical-test analysis and condition monitoring.

Adoption context: Scale and test-data quality determine how far AI is embedded.

Watchpoint: Safety certification requires validated evidence.

Division 28 | Machinery and equipment not elsewhere classified

Group 281 | General-purpose machinery

2811 | Engines and turbines other than vehicle engines

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial-engine makers; wind-turbine manufacturers; steam-turbine producers; stationary turbine makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Maintenance forecasts and production-quality analysis.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2812 | Fluid-power equipment

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: hydraulic-cylinder makers; pneumatic equipment manufacturers; hydraulic motor makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Leak-test analysis and condition monitoring.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2813 | Other pumps, compressors, taps and valves

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: pump manufacturers; compressor makers; industrial-valve manufacturers; tap factories.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Test-bench anomalies and defect detection.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2814 | Bearings, gears and driving elements

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: bearing factories; gearbox makers; transmission-element manufacturers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Wear detection and machining-quality analysis.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2815 | Ovens, furnaces and permanent heating equipment

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial-furnace makers; boiler manufacturers; permanent heating-equipment makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Service-log analysis and thermal-condition monitoring.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2816 | Lifting and handling equipment

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: crane manufacturers; forklift makers; elevator manufacturers; conveyor-system makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Condition monitoring and inspection assistance.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2817 | Office machinery other than computers

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: office equipment manufacturers; document-processing machine makers; office-machine assemblers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Test-log analysis and maintenance scheduling.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2818 | Power-driven hand tools

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: electric drill makers; portable saw manufacturers; pneumatic hand-tool makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Assembly inspection and tool-test analysis.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2819 | Other general-purpose machinery

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial cooling-equipment makers; packaging-machine makers; filtration-equipment manufacturers; general machine builders.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Machine-condition and production-quality monitoring.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

Group 282 | Special-purpose machinery

2821 | Agricultural and forestry machinery

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: tractor makers; harvester manufacturers; forestry-machine makers; farm implement manufacturers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Test-data analysis and maintenance forecasting.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2822 | Metal-forming machinery and machine tools

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: press manufacturers; CNC machine-tool builders; metal-forming equipment makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Machine calibration and quality-data analysis.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2823 | Metallurgy machinery

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: rolling-mill equipment makers; metal-processing plant equipment makers; metallurgy machine builders.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Asset-condition and project-document analysis.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2824 | Mining, quarrying and construction machinery

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: excavator manufacturers; rock-crusher makers; mining-machine manufacturers; road-equipment makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Maintenance forecasting and factory inspection.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2825 | Food, beverage and tobacco processing machinery

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: food-processing equipment makers; brewery-equipment manufacturers; dairy-machine builders.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Hygienic equipment inspection and service analysis.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2826 | Textile, apparel and leather machinery

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: loom makers; textile-spinning equipment makers; industrial sewing-machine makers; tanning-machine builders.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Machine-test analysis and service diagnostics.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

2829 | Other special-purpose machinery

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: semiconductor equipment makers; printing-machine builders; industrial-robot manufacturers; specialty production-equipment makers.

Software: Engineering design, product lifecycle records, enterprise planning, service and maintenance systems.

AI applications / candidates: Test anomaly detection and service-log analysis.

Adoption context: AI commonly begins in engineering assistance or asset analytics, not autonomous end-to-end manufacture.

Watchpoint: Engineers must approve designs and maintenance actions.

Division 29 | Motor vehicles, trailers and semi-trailers

Group 291 | Motor vehicles

2910 | Motor vehicles

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: car manufacturers; commercial truck makers; bus manufacturers; electric-vehicle assemblers.

Software: Product lifecycle management, factory execution, supplier planning, diagnostics and quality systems.

AI applications / candidates: Factory inspection and warranty-data analysis.

Adoption context: Large manufacturers are highly connected; component suppliers vary substantially.

Watchpoint: Safety-critical software and hardware need formal validation.

Group 292 | Vehicle bodies, trailers and semi-trailers

2920 | Vehicle bodies, trailers and semi-trailers

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: coachbuilders; trailer factories; truck-body makers; caravan manufacturers.

Software: Product lifecycle management, factory execution, supplier planning, diagnostics and quality systems.

AI applications / candidates: Design-document checking and assembly inspection.

Adoption context: Large manufacturers are highly connected; component suppliers vary substantially.

Watchpoint: Safety-critical software and hardware need formal validation.

Group 293 | Motor-vehicle parts and accessories

2930 | Motor-vehicle parts and accessories

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: brake suppliers; automotive electrical-component makers; seating manufacturers; powertrain-component suppliers.

Software: Product lifecycle management, factory execution, supplier planning, diagnostics and quality systems.

AI applications / candidates: Component inspection and failure forecasting.

Adoption context: Large manufacturers are highly connected; component suppliers vary substantially.

Watchpoint: Safety-critical software and hardware need formal validation.

Division 30 | Other transport equipment

Group 301 | Ship and boat building

3011 | Ships and floating structures

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: commercial shipyards; offshore platform builders; barge builders; floating-structure manufacturers.

Software: Marine design, project controls, materials, procurement and inspection records.

AI applications / candidates: Inspection-record review and project scheduling.

Adoption context: Digital engineering can be mature while shop-floor integration and AI remain patchy.

Watchpoint: Class and engineering approvals remain human responsibilities.

3012 | Pleasure and sporting boats

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: yacht builders; sailboat manufacturers; leisure-boat yards; rowing-boat makers.

Software: Marine design, project controls, materials, procurement and inspection records.

AI applications / candidates: Design-document assistance and material inspection.

Adoption context: Digital engineering can be mature while shop-floor integration and AI remain patchy.

Watchpoint: Class and engineering approvals remain human responsibilities.

Group 302 | Railway locomotives and rolling stock

3020 | Railway locomotives and rolling stock

D3-D4 / A0-A3 / Estimate confidence: medium

Businesses: locomotive manufacturers; railcar builders; tram manufacturers; railway component producers.

Software: Engineering design, product lifecycle, manufacturing, maintenance and safety-document systems.

AI applications / candidates: Inspection and maintenance-document analysis.

Adoption context: Long asset lives and regulated change slow deployment of new AI models.

Watchpoint: Use approved configurations and validated inspection procedures.

Group 303 | Air and spacecraft and related machinery

3030 | Air and spacecraft and related machinery

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: aircraft manufacturers; spacecraft manufacturers; civil drone manufacturers; aerospace component suppliers.

Software: Engineering models, configuration control, production traceability, simulation and maintenance systems.

AI applications / candidates: Manufacturing inspection and approved maintenance support.

Adoption context: Digital design is central; AI deployment is selective and tightly reviewed.

Watchpoint: Airworthiness and mission safety need independent verification.

Group 304 | Military fighting vehicles

3040 | Military fighting vehicles

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: licensed military-vehicle manufacturers; armored-vehicle assemblers; defence vehicle contractors.

Software: Production planning, quality records, traceability, permits, maintenance and controlled document systems.

AI applications / candidates: Inventory, quality records and compliance documentation only.

Adoption context: Digital records may be essential even where AI remains limited to administrative or inspection assistance.

Watchpoint: Do not delegate regulated release or safety decisions.

Group 309 | Other transport equipment not elsewhere classified

3091 | Motorcycles

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: motorcycle factories; motor-scooter manufacturers; moped assemblers.

Software: Product lifecycle management, factory execution, supplier planning, diagnostics and quality systems.

AI applications / candidates: Assembly inspection and warranty-data analysis.

Adoption context: Large manufacturers are highly connected; component suppliers vary substantially.

Watchpoint: Safety-critical software and hardware need formal validation.

3092 | Bicycles and mobility carriages

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: bicycle manufacturers; e-bike makers; wheelchair manufacturers; mobility-carriage makers.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: Component inspection and demand forecasting.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

3099 | Other transport equipment

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: handcart makers; specialized non-motorized vehicle makers; other transport-equipment workshops.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: Design assistance and order scheduling.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

Division 31 | Furniture

Group 310 | Furniture

3101 | Wooden furniture

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: wood furniture factories; cabinet makers; office wood-furniture makers; custom furniture workshops.

Software: Design and cutting software, timber inventories, machine controls and production planning.

AI applications / candidates: Design drafting and cutting-layout assistance.

Adoption context: Industrial mills and factories can use vision; artisan shops usually use lighter tools.

Watchpoint: Keep machine safety independent of model suggestions.

3102 | Other furniture

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: metal furniture factories; plastic furniture makers; mattress manufacturers; upholstered furniture makers.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: Design support and visual inspection.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

Division 32 | Other manufacturing

Group 321 | Jewellery and related articles

3211 | Jewellery and related articles

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: fine-jewellery workshops; precious-metal jewellery factories; gemstone-setting businesses.

Software: Design tools, order books, online selling, stock records, accounting and payments.

AI applications / candidates: Design assistance and visual grading.

Adoption context: Handwork remains central; AI is generally a design or marketing aid rather than autonomous production.

Watchpoint: Protect original designs and customer information.

3212 | Imitation jewellery

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: costume-jewellery factories; fashion accessory makers; imitation-jewellery workshops.

Software: Design tools, order books, online selling, stock records, accounting and payments.

AI applications / candidates: Design variations and demand forecasting.

Adoption context: Handwork remains central; AI is generally a design or marketing aid rather than autonomous production.

Watchpoint: Protect original designs and customer information.

Group 322 | Musical instruments

3220 | Musical instruments

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: piano makers; guitar manufacturers; wind-instrument makers; percussion workshops.

Software: Design tools, order books, online selling, stock records, accounting and payments.

AI applications / candidates: Acoustic-test analysis and design assistance.

Adoption context: Handwork remains central; AI is generally a design or marketing aid rather than autonomous production.

Watchpoint: Protect original designs and customer information.

Group 323 | Sporting goods

3230 | Sporting goods

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: fitness-equipment makers; ball manufacturers; outdoor sports-gear makers; racket manufacturers.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: Product inspection and demand forecasting.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

Group 324 | Games and toys

3240 | Games and toys

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: board-game manufacturers; toy factories; puzzle makers; model-kit manufacturers.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: Artwork checking and seasonal demand forecasts.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

Group 325 | Medical and dental instruments and supplies

3250 | Medical and dental instruments and supplies

D3-D4 / A0-A3 / Estimate confidence: medium

Businesses: surgical-instrument makers; dental-laboratory manufacturers; orthotic makers; medical consumable factories.

Software: Design controls, quality systems, test records, traceability and regulated manufacturing software.

AI applications / candidates: Manufacturing inspection and traceability checks.

Adoption context: AI in the device itself is separate from AI used by its manufacturer.

Watchpoint: Clinical and regulatory evidence cannot be replaced by generated text.

Group 329 | Other manufacturing

3290 | Other manufacturing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: broom and brush factories; pen manufacturers; protective-equipment makers; candle factories.

Software: Design, production planning, inventory, quality, maintenance and enterprise accounting.

AI applications / candidates: Product inspection and order forecasting.

Adoption context: Larger plants integrate machines and records; smaller producers may work from spreadsheets.

Watchpoint: Automation is not automatically AI.

Division 33 | Machinery and equipment repair and installation

Group 331 | Repair of fabricated metal, machinery and equipment

3311 | Repair of fabricated metal products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: tank repair contractors; metal-container repairers; boiler metalwork repair shops.

Software: Service orders, maintenance histories, spare-parts inventory, field-service and invoicing systems.

AI applications / candidates: Inspection-record summaries and fault prioritization.

Adoption context: Connected customers enable predictive maintenance; local shops rely more on technician experience.

Watchpoint: Do not substitute generated advice for safe isolation procedures.

3312 | Repair of machinery

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial machinery repair shops; pump repairers; farm machinery repairers; machine-tool service firms.

Software: Service orders, maintenance histories, spare-parts inventory, field-service and invoicing systems.

AI applications / candidates: Fault diagnosis support and failure prediction.

Adoption context: Connected customers enable predictive maintenance; local shops rely more on technician experience.

Watchpoint: Do not substitute generated advice for safe isolation procedures.

3313 | Repair of electronic and optical equipment

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: instrument repair laboratories; optical-equipment repairers; medical-electronics service firms.

Software: Service orders, maintenance histories, spare-parts inventory, field-service and invoicing systems.

AI applications / candidates: Service-log analysis and fault lookup.

Adoption context: Connected customers enable predictive maintenance; local shops rely more on technician experience.

Watchpoint: Do not substitute generated advice for safe isolation procedures.

3314 | Repair of electrical equipment

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: electric-motor rewinders; transformer service firms; industrial electrical repair shops.

Software: Service orders, maintenance histories, spare-parts inventory, field-service and invoicing systems.

AI applications / candidates: Condition analysis and maintenance prioritization.

Adoption context: Connected customers enable predictive maintenance; local shops rely more on technician experience.

Watchpoint: Do not substitute generated advice for safe isolation procedures.

3315 | Repair of transport equipment other than motor vehicles

D2-D4 / A0-A2 / Estimate confidence: medium

Businesses: ship repair yards; aircraft maintenance providers; railway repair workshops.

Software: Approved maintenance manuals, work orders, configuration records, parts traceability and inspection systems.

AI applications / candidates: Approved-maintenance search and inspection support.

Adoption context: AI can assist retrieval and documentation, but release-to-service remains controlled.

Watchpoint: Use approved technical information and authorized sign-off.

3319 | Repair of other equipment

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial equipment repairers; specialist equipment restoration shops; other commercial equipment service firms.

Software: Service orders, maintenance histories, spare-parts inventory, field-service and invoicing systems.

AI applications / candidates: Fault-record search and job scheduling.

Adoption context: Connected customers enable predictive maintenance; local shops rely more on technician experience.

Watchpoint: Do not substitute generated advice for safe isolation procedures.

Group 332 | Installation of industrial machinery and equipment

3320 | Installation of industrial machinery and equipment

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: factory machinery installers; production-line commissioning firms; industrial equipment riggers.

Software: Service orders, maintenance histories, spare-parts inventory, field-service and invoicing systems.

AI applications / candidates: Commissioning-document checks and schedule assistance.

Adoption context: Connected customers enable predictive maintenance; local shops rely more on technician experience.

Watchpoint: Do not substitute generated advice for safe isolation procedures.

COMPLETE ACTIVITY CATALOGUE / SECTION D

D | Electricity, gas, steam and cooling

1 divisions / 4 groups / 6 classes / 21 business examples

Utilities can be software-critical without being AI-native. Plant controls, network operations, billing and maintenance form the base; forecasting and anomaly detection sit above it.

Evidence scope: IEA evidence supports energy use cases. Local grid size, asset type, regulation and operating data determine actual deployment. [S12] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 35 | Electricity, gas, steam and air-conditioning supply

Group 351 | Electricity generation, transmission and distribution

3511 | Non-renewable electricity generation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: gas-fired power stations; coal power stations; nuclear power operators; oil-fired generators.

Software: Plant controls, maintenance, asset histories, dispatch, market and compliance systems.

AI applications / candidates: Asset anomalies and generation forecasting.

Adoption context: Sensor-rich assets support analytics; level of integration differs by operator.

Watchpoint: Protect safety systems and critical infrastructure.

3512 | Renewable electricity generation

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: solar-farm operators; wind-farm operators; hydropower operators; geothermal power operators.

Software: Plant monitoring, weather forecasts, asset maintenance, dispatch and energy trading software.

AI applications / candidates: Weather-linked generation forecasts and inspection.

Adoption context: Utility-scale assets often have remote monitoring; small installations use simpler vendor dashboards.

Watchpoint: Forecasts need local weather and outage information.

3513 | Electricity transmission and distribution

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: grid operators; electricity distributors; electricity trading operators; EV charging-station operators.

Software: Network controls, outage management, grid maps, metering, billing and asset systems.

AI applications / candidates: Load forecasts and network-condition alerts.

Adoption context: AI can augment load and condition analysis; safe operation requires validated controls.

Watchpoint: Cybersecurity and grid reliability are non-negotiable.

Group 352 | Gas manufacture and distribution through mains

3520 | Gas manufacture and distribution through mains

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: gas distribution utilities; manufactured-gas suppliers; district gas network operators.

Software: Network maps, supervisory controls, metering, billing, maintenance and customer-service systems.

AI applications / candidates: Demand forecasting and leak-alert analysis.

Adoption context: Larger utilities have richer operational data; legacy infrastructure limits integration.

Watchpoint: Keep verified manual and safety fallbacks.

Group 353 | Steam and air-conditioning supply

3530 | Steam and air-conditioning supply

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: district heating operators; district cooling operators; industrial steam suppliers.

Software: Network maps, supervisory controls, metering, billing, maintenance and customer-service systems.

AI applications / candidates: Heat-demand forecasts and equipment monitoring.

Adoption context: Larger utilities have richer operational data; legacy infrastructure limits integration.

Watchpoint: Keep verified manual and safety fallbacks.

Group 354 | Electricity and natural-gas brokers and agents

3540 | Electricity and natural-gas brokers and agents

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: energy brokers; electricity procurement agents; natural-gas brokers.

Software: Customer records, energy contracts, tariffs, billing analysis and procurement tools.

AI applications / candidates: Contract comparison and demand analysis.

Adoption context: Digital comparison is common in sophisticated firms; AI adoption is task-specific.

Watchpoint: Check tariff terms and disclose uncertainty.

COMPLETE ACTIVITY CATALOGUE / SECTION E

E | Water, sewerage, waste and remediation

4 divisions / 6 groups / 8 classes / 27 business examples

This sector spans sensor-rich treatment facilities and very small waste collectors. Keep waste collection, disposal, recovery and environmental remediation separate.

Evidence scope: EPA provides connected-sewer examples; manufacturing and infrastructure research inform related operational analogies. Waste-class AI estimates are especially uncertain. [S13] [S11] [S06]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 36 | Water collection, treatment and supply

Group 360 | Water collection, treatment and supply

3600 | Water collection, treatment and supply

D1-D4 / A0-A2 / Estimate confidence: medium

Businesses: municipal water suppliers; private water utilities; desalination operators; bulk water suppliers.

Software: Network maps, treatment controls, sensors, work orders, billing and water-quality records.

AI applications / candidates: Leak alerts and treatment-process anomalies.

Adoption context: Large networks can use smart monitoring; smaller operators may lack reliable instrumentation.

Watchpoint: Protect public health and verify sensor data.

Division 37 | Sewerage

Group 370 | Sewerage

3700 | Sewerage

D1-D4 / A0-A2 / Estimate confidence: medium

Businesses: sewer network operators; wastewater treatment plants; septic tank service businesses.

Software: Network maps, treatment controls, sensors, work orders, billing and water-quality records.

AI applications / candidates: Overflow prediction and treatment monitoring.

Adoption context: Large networks can use smart monitoring; smaller operators may lack reliable instrumentation.

Watchpoint: Protect public health and verify sensor data.

Division 38 | Waste collection, treatment, disposal and recovery

Group 381 | Waste collection

3811 | Non-hazardous waste collection

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: municipal waste collectors; commercial refuse firms; recycling collection businesses.

Software: Fleet tracking, route plans, customer accounts, bin records and weighing systems.

AI applications / candidates: Route planning and contamination-image checks.

Adoption context: AI is more feasible in fleet-scale operations than informal collection.

Watchpoint: Routes and image tools must respect privacy and safety.

3812 | Hazardous waste collection

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: licensed medical-waste collectors; hazardous industrial-waste collectors; hazardous-material collection firms.

Software: Waste manifests, permits, chain-of-custody records, transport tracking and asset systems.

AI applications / candidates: Manifest checking and collection planning.

Adoption context: AI is mainly supervised document and inspection support, not autonomous handling.

Watchpoint: Licensed procedures and traceability remain essential.

Group 382 | Waste treatment and disposal

3821 | Non-hazardous waste treatment and disposal

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: landfill operators; non-hazardous incinerators; organic-waste treatment firms.

Software: Weighbridge systems, plant controls, maintenance, environmental monitoring and compliance records.

AI applications / candidates: Waste-stream monitoring and asset reliability.

Adoption context: Connected facilities can add analytics; deployment is uneven across sites.

Watchpoint: Maintain independent environmental and safety monitoring.

3822 | Hazardous waste treatment and disposal

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: licensed hazardous-waste treatment plants; medical-waste disposal firms; hazardous-waste landfill operators.

Software: Waste manifests, permits, chain-of-custody records, transport tracking and asset systems.

AI applications / candidates: Compliance-record review and asset monitoring.

Adoption context: AI is mainly supervised document and inspection support, not autonomous handling.

Watchpoint: Licensed procedures and traceability remain essential.

Group 383 | Materials and other waste recovery

3830 | Materials and other waste recovery

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: scrap-metal recyclers; plastics recyclers; electronic-waste recovery plants; materials recovery facilities.

Software: Material inventories, grading, weighing, production controls, traceability and sales systems.

AI applications / candidates: Material recognition and automated sorting support.

Adoption context: Industrial sorting lines can use vision; informal collectors often use basic mobile tools.

Watchpoint: Distinguish model confidence from actual material purity.

Division 39 | Remediation and other waste services

Group 390 | Remediation and other waste services

3900 | Remediation and other waste services

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: contaminated-land remediators; spill clean-up contractors; asbestos-remediation firms; environmental cleanup businesses.

Software: Site maps, laboratory results, sampling records, projects and compliance documentation.

AI applications / candidates: Site-data interpretation and remediation reporting.

Adoption context: AI supports interpretation and reporting where consistent site data exist.

Watchpoint: Professionals must validate site-specific risks.

COMPLETE ACTIVITY CATALOGUE / SECTION F

F | Construction

3 divisions / 9 groups / 12 classes / 46 business examples

General builders, infrastructure contractors, specialist trades and referral platforms have different software needs. Digital drawings or building models do not imply AI is routinely integrated.

Evidence scope: RICS gives current professional-sample evidence; Eurostat provides a separate EU enterprise measure. Neither establishes worldwide adoption by each trade. [S14] [S04] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 41 | Residential and non-residential building construction

Group 410 | Residential and non-residential construction

4100 | Residential and non-residential construction

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: homebuilders; apartment builders; commercial building contractors; industrial-building contractors.

Software: Estimating, project schedules, building models, drawings, procurement and site records.

AI applications / candidates: Document search, quantity checks and progress monitoring.

Adoption context: Major projects are more digital than small contractors; AI is often pilot or limited workflow use.

Watchpoint: Check designs, quantities and safety claims independently.

Division 42 | Civil engineering

Group 421 | Road and railway construction

4210 | Road and railway construction

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: road contractors; railway builders; bridge contractors; tunnel builders.

Software: Civil design, mapping, scheduling, cost control, machine tracking and inspection records.

AI applications / candidates: Site-progress comparison and maintenance-risk analysis.

Adoption context: Large infrastructure programmes support more analytics than small one-off projects.

Watchpoint: Survey accuracy and engineering sign-off are critical.

Group 422 | Utility-project construction

4220 | Utility-project construction

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: water-network contractors; power-line builders; telecom-network contractors; sewer construction firms.

Software: Civil design, mapping, scheduling, cost control, machine tracking and inspection records.

AI applications / candidates: Survey interpretation and project-document checking.

Adoption context: Large infrastructure programmes support more analytics than small one-off projects.

Watchpoint: Survey accuracy and engineering sign-off are critical.

Group 429 | Other civil engineering

4290 | Other civil engineering

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: port builders; dam contractors; civil land-development firms; other infrastructure contractors.
Software: Civil design, mapping, scheduling, cost control, machine tracking and inspection records.
AI applications / candidates: Site-progress and project-risk analysis.
Adoption context: Large infrastructure programmes support more analytics than small one-off projects.
Watchpoint: Survey accuracy and engineering sign-off are critical.

Division 43 | Specialized construction

Group 431 | Demolition and site preparation

4311 | Demolition

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: building demolition contractors; selective demolition firms; industrial dismantling contractors.
Software: Estimating, project schedules, building models, drawings, procurement and site records.
AI applications / candidates: Survey-image review and documentation support.
Adoption context: Major projects are more digital than small contractors; AI is often pilot or limited workflow use.
Watchpoint: Check designs, quantities and safety claims independently.

4312 | Site preparation

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: earthworks contractors; excavation firms; land-clearing contractors; site-drilling businesses.
Software: Estimating, project schedules, building models, drawings, procurement and site records.
AI applications / candidates: Site measurement and equipment maintenance.
Adoption context: Major projects are more digital than small contractors; AI is often pilot or limited workflow use.
Watchpoint: Check designs, quantities and safety claims independently.

Group 432 | Electrical, plumbing and other construction installation

4321 | Electrical installation

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: electrical contractors; building wiring firms; solar installation contractors; alarm installation firms.
Software: Quoting, job scheduling, invoices, drawings, customer records and field-service apps.
AI applications / candidates: Quote drafting and service fault lookup.
Adoption context: Many microbusinesses use messaging first; AI mainly supports office tasks or troubleshooting.
Watchpoint: Only qualified people should approve safety-critical work.

4322 | Plumbing, heating and air-conditioning installation

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: plumbers; HVAC installers; heating contractors; ventilation contractors.
Software: Quoting, job scheduling, invoices, drawings, customer records and field-service apps.
AI applications / candidates: Job triage and equipment fault lookup.
Adoption context: Many microbusinesses use messaging first; AI mainly supports office tasks or troubleshooting.
Watchpoint: Only qualified people should approve safety-critical work.

4329 | Other construction installation

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: lift installers; insulation contractors; building automation installers; other installation specialists.

Software: Quoting, job scheduling, invoices, drawings, customer records and field-service apps.

AI applications / candidates: Drawing search and job-planning assistance.

Adoption context: Many microbusinesses use messaging first; AI mainly supports office tasks or troubleshooting.

Watchpoint: Only qualified people should approve safety-critical work.

Group 433 | Building completion and finishing

4330 | Building completion and finishing

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: painters; plasterers; tilers; flooring and glazing contractors.

Software: Quoting, job scheduling, invoices, drawings, customer records and field-service apps.

AI applications / candidates: Quote drafting and material take-off support.

Adoption context: Many microbusinesses use messaging first; AI mainly supports office tasks or troubleshooting.

Watchpoint: Only qualified people should approve safety-critical work.

Group 434 | Specialized construction intermediation

4340 | Specialized construction intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: trade-matching services; contractor referral agencies; home-improvement marketplaces.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Provider matching and enquiry qualification.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

Group 439 | Other specialized construction

4390 | Other specialized construction

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: roofing contractors; scaffolding firms; foundation contractors; masonry contractors.

Software: Quoting, job scheduling, invoices, drawings, customer records and field-service apps.

AI applications / candidates: Estimate assistance and site-image review.

Adoption context: Many microbusinesses use messaging first; AI mainly supports office tasks or troubleshooting.

Watchpoint: Only qualified people should approve safety-critical work.

COMPLETE ACTIVITY CATALOGUE / SECTION G

G | Wholesale and retail

2 divisions / 17 groups / 41 classes / 154 business examples

Merchants own or trade goods; agents and marketplaces intermediate. Online versus offline selling is a channel distinction, not a complete industry hierarchy. Micro-retail and large chains vary sharply.

Evidence scope: Retail industry cases illustrate workflows. Regional software surveys and SME research anchor only broad directional assessments. [S15] [S04] [S05] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 46 | Wholesale trade

Group 461 | Wholesale on a fee or contract basis

4610 | Wholesale on a fee or contract basis

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: commission merchants; import-export agents; industrial sales agents; wholesale brokers.

Software: Customer records, quotations, product catalogues, commissions and trade documents.

AI applications / candidates: Lead matching and quotation-document extraction.

Adoption context: AI commonly enters as document assistance or search rather than autonomous negotiation.

Watchpoint: Confirm product specifications and counterparties.

Group 462 | Wholesale of farm raw materials and live animals

4620 | Wholesale of farm raw materials and live animals

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: grain merchants; seed wholesalers; livestock dealers; raw-cotton merchants.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Demand forecasting and product grading.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

Group 463 | Wholesale of food, beverages and tobacco

4630 | Wholesale of food, beverages and tobacco

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: foodservice distributors; produce wholesalers; beverage distributors; tobacco wholesalers.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Demand forecasting and expiry-risk analysis.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

Group 464 | Household-goods wholesale

4641 | Wholesale of textiles, clothing and footwear

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: fabric wholesalers; apparel distributors; footwear wholesalers; workwear distributors.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Assortment forecasts and catalogue enrichment.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4642 | Wholesale of furniture, carpets and lighting

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: furniture wholesalers; lighting distributors; carpet wholesalers; shop-fixture distributors.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Catalogue search and demand forecasting.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4649 | Wholesale of other household goods

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: cosmetics distributors; pharmaceutical wholesalers; appliance wholesalers; household-goods distributors.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Inventory forecasts and document extraction.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

Group 465 | Machinery, equipment and supply wholesale

4651 | Wholesale of computers, peripherals and software

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: computer distributors; enterprise hardware wholesalers; software distributors; peripheral wholesalers.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Product matching and quote generation support.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4652 | Wholesale of electronic and telecom equipment

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: electronic component distributors; telecom equipment wholesalers; network hardware distributors.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Part matching and supply-risk analysis.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4653 | Wholesale of farm machinery and supplies

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: tractor distributors; farm implement wholesalers; agricultural equipment dealers.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Equipment matching and spare-parts forecasting.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4659 | Wholesale of other machinery and equipment

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: industrial machinery distributors; medical equipment wholesalers; construction machinery wholesalers; office equipment distributors.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Technical product search and demand forecasting.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

Group 466 | Vehicle, motorcycle and related-parts wholesale

4661 | Wholesale of motor vehicles

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: vehicle importers; fleet vehicle wholesalers; auto distributors; used-vehicle wholesalers.

Software: Dealer management, inventory, leads, finance paperwork, valuations and service records.

AI applications / candidates: Stock valuation support and demand forecasting.

Adoption context: Dealership groups use more analytics; smaller traders often use listing portals and spreadsheets.

Watchpoint: Verify vehicle condition, prices and finance disclosures.

4662 | Wholesale of vehicle parts

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: automotive parts distributors; tyre wholesalers; vehicle-accessory wholesalers.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Part compatibility matching and inventory forecasts.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4663 | Wholesale of motorcycles and parts

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: motorcycle importers; motorcycle distributors; motorcycle-parts wholesalers.

Software: Dealer management, inventory, leads, finance paperwork, valuations and service records.

AI applications / candidates: Catalogue matching and demand forecasting.

Adoption context: Dealership groups use more analytics; smaller traders often use listing portals and spreadsheets.

Watchpoint: Verify vehicle condition, prices and finance disclosures.

Group 467 | Other specialized wholesale

4671 | Wholesale of fuels and related products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: petroleum distributors; coal merchants; bulk gas distributors; lubricant wholesalers.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Demand forecasts and delivery planning.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4672 | Wholesale of metals and ores

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: metal stockholders; ore traders; precious-metal wholesalers; industrial metal distributors.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Price-risk summaries and inventory planning.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4673 | Wholesale of construction and plumbing materials

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: building-material merchants; plumbing wholesalers; hardware distributors; heating-supply distributors.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Quotation extraction and stock forecasting.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

4679 | Wholesale of chemicals, scrap and other products

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: chemical distributors; scrap merchants; industrial raw-material traders; specialist commodity wholesalers.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Document extraction and material matching.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

Group 469 | Non-specialized wholesale

4690 | Non-specialized wholesale

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: general trading companies; cash-and-carry wholesalers; mixed-product import-export merchants.

Software: Enterprise planning, warehouse stock, purchasing, sales orders, catalogues and credit control.

AI applications / candidates: Cross-category demand and catalogue analysis.

Adoption context: Large distributors integrate demand and logistics; small merchants rely on accounts and messaging.

Watchpoint: Forecasts need accurate stock, lead times and margins.

Division 47 | Retail trade

Group 471 | Non-specialized retail

4711 | General retail led by food, beverages or tobacco

D0-D4 / A0-A3 / Estimate confidence: medium

Businesses: supermarkets; grocery convenience stores; village grocery shops; food-led general stores.

Software: Checkout, stock, purchasing, loyalty, expiry and delivery systems.

AI applications / candidates: Replenishment forecasts and checkout anomaly review.

Adoption context: Chains can use forecasting and vision; informal grocery shops may use no dedicated software.

Watchpoint: Check waste, availability and privacy rather than trusting forecasts alone.

4719 | Other non-specialized retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: department stores; general merchandise stores; variety stores; mixed-goods online retailers.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Assortment forecasting and product recommendations.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

Group 472 | Food, beverage and tobacco retail

4721 | Specialized food retail

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: butchers; fishmongers; greengrocers; specialist bakery-product shops.

Software: Checkout, payments, basic stock, messaging, online listings and bookkeeping.

AI applications / candidates: Demand forecasts and expiry-risk alerts.

Adoption context: Small shops often use a few disconnected tools; AI is mainly embedded or occasional.

Watchpoint: Do not confuse a platform's AI with the shop's own capability.

4722 | Beverage retail

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: bottle shops; specialist tea retailers; coffee-bean shops; non-alcoholic drink stores.

Software: Checkout, payments, basic stock, messaging, online listings and bookkeeping.

AI applications / candidates: Inventory suggestions and customer enquiry assistance.

Adoption context: Small shops often use a few disconnected tools; AI is mainly embedded or occasional.

Watchpoint: Do not confuse a platform's AI with the shop's own capability.

4723 | Tobacco retail

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: tobacconists; cigar shops; specialist tobacco-product retailers.

Software: Checkout, payments, basic stock, messaging, online listings and bookkeeping.

AI applications / candidates: Stock planning and compliance-document support.

Adoption context: Small shops often use a few disconnected tools; AI is mainly embedded or occasional.

Watchpoint: Do not confuse a platform's AI with the shop's own capability.

Group 473 | Automotive fuel retail

4730 | Automotive fuel retail

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: petrol stations; diesel filling stations; automotive gas stations; hydrogen vehicle-fueling stations.

Software: Forecourt or charging controls, payments, inventory or metering, pricing and maintenance.

AI applications / candidates: Demand forecasting and equipment anomaly alerts.

Adoption context: Larger networks integrate demand data; independent sites may only use transaction systems.

Watchpoint: Protect payment systems and equipment safety.

Group 474 | ICT equipment retail

4740 | ICT equipment retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: computer shops; phone retailers; software retailers; telecom equipment stores.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Product matching and customer-support assistance.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

Group 475 | Household-equipment retail

4751 | Textile retail

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: fabric shops; yarn retailers; haberdashery shops; household textile retailers.

Software: Checkout, payments, basic stock, messaging, online listings and bookkeeping.

AI applications / candidates: Catalogue tagging and stock suggestions.

Adoption context: Small shops often use a few disconnected tools; AI is mainly embedded or occasional.

Watchpoint: Do not confuse a platform's AI with the shop's own capability.

4752 | Hardware, building-material, paint and glass retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: hardware stores; paint shops; building-material stores; glass retailers.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Project-product matching and stock forecasts.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

4753 | Carpet, rug, wall and floor-covering retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: carpet shops; flooring stores; wallpaper shops; rug retailers.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Room visualization and quotation support.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

4759 | Appliance, furniture, lighting and household retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: furniture stores; appliance retailers; lighting shops; kitchenware retailers.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Product recommendations and visual search.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

Group 476 | Cultural and recreational goods retail

4761 | Books, newspapers and stationery retail

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: bookshops; newsagents; stationery stores; office-supply retailers.

Software: Checkout, payments, basic stock, messaging, online listings and bookkeeping.

AI applications / candidates: Product recommendations and stock forecasting.

Adoption context: Small shops often use a few disconnected tools; AI is mainly embedded or occasional.

Watchpoint: Do not confuse a platform's AI with the shop's own capability.

4762 | Sporting-equipment retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: sports shops; bicycle retailers; fishing-tackle shops; camping-equipment stores.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Equipment matching and seasonal stock forecasts.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

4763 | Game and toy retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: toy shops; board-game retailers; video-game stores; hobby-game shops.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Product discovery and seasonal demand forecasts.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

4769 | Other cultural and recreational retail

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: musical-instrument stores; art-supply shops; specialist recreation retailers.

Software: Checkout, payments, basic stock, messaging, online listings and bookkeeping.

AI applications / candidates: Product matching and customer-support drafting.

Adoption context: Small shops often use a few disconnected tools; AI is mainly embedded or occasional.

Watchpoint: Do not confuse a platform's AI with the shop's own capability.

Group 477 | Other-goods retail, excluding vehicles

4771 | Clothing, footwear and leather retail

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: fashion boutiques; shoe stores; leather-goods shops; online apparel retailers.

Software: Checkout, e-commerce, inventory, merchandising, loyalty and returns systems.

AI applications / candidates: Size assistance, visual search and demand forecasts.

Adoption context: Online and chain retailers use richer recommendation tools than local boutiques.

Watchpoint: Check fit advice, bias and customer-data consent.

4772 | Pharmaceutical, medical and cosmetic retail

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: pharmacies; medical-supply stores; cosmetics shops; personal-care retailers.

Software: Dispensing or checkout, stock, expiry, prescription records and purchasing systems.

AI applications / candidates: Inventory and expiry alerts; reviewed customer support.

Adoption context: Clinical pharmacy workflows need stronger controls than ordinary cosmetic retail.

Watchpoint: Never let a generic assistant replace pharmacist judgment.

4773 | Other new-goods retail

D0-D3 / A0-A1 / Estimate confidence: medium

Businesses: jewellery stores; flower shops; pet-supply stores; specialist gift and equipment shops.

Software: Checkout, payments, basic stock, messaging, online listings and bookkeeping.

AI applications / candidates: Catalogue enrichment and stock suggestions.

Adoption context: Small shops often use a few disconnected tools; AI is mainly embedded or occasional.

Watchpoint: Do not confuse a platform's AI with the shop's own capability.

4774 | Second-hand goods retail

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: thrift stores; antique dealers; used-book shops; used-furniture retailers.

Software: Listings, intake records, pricing, payments, condition notes and inventory tools.

AI applications / candidates: Item identification, listing drafts and valuation support.

Adoption context: Online resellers can use image or text tools; local informal sellers may not.

Watchpoint: Verify authenticity, ownership and condition.

Group 478 | Vehicle, motorcycle and related-parts retail

4781 | Motor-vehicle retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: new-car dealerships; used-car dealers; commercial vehicle dealers; online vehicle retailers.

Software: Dealer management, inventory, leads, finance paperwork, valuations and service records.

AI applications / candidates: Lead qualification and inventory valuation support.

Adoption context: Dealership groups use more analytics; smaller traders often use listing portals and spreadsheets.

Watchpoint: Verify vehicle condition, prices and finance disclosures.

4782 | Vehicle-parts retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: auto-parts shops; tyre retailers; vehicle-accessory stores.

Software: Checkout, inventory, e-commerce, customer records, merchandising and accounting.

AI applications / candidates: Part compatibility search and stock forecasts.

Adoption context: Chains and online sellers are more integrated than independent stores.

Watchpoint: Product and price claims must remain accurate.

4783 | Motorcycle and related retail

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: motorcycle dealerships; scooter retailers; motorcycle-parts shops.

Software: Dealer management, inventory, leads, finance paperwork, valuations and service records.

AI applications / candidates: Product matching and lead assistance.

Adoption context: Dealership groups use more analytics; smaller traders often use listing portals and spreadsheets.

Watchpoint: Verify vehicle condition, prices and finance disclosures.

Group 479 | Retail intermediation

4790 | Retail intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: multi-seller marketplaces; retail commission platforms; product-matching agents.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Search ranking, fraud detection and seller support.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

COMPLETE ACTIVITY CATALOGUE / SECTION H

H | Transportation and storage

5 divisions / 13 groups / 23 classes / 82 business examples

Separate the carrier, terminal, warehouse, forwarder and booking intermediary. A large logistics network can run extensive AI while an owner-driver relies on navigation and a phone.

Evidence scope: DHL supplies concrete operator examples. Estimates for rail, aviation, ports and small carriers are workflow-based transfers, not direct class-level measurements. [S16] [S12] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 49 | Land and pipeline transport

Group 491 | Rail transport

4911 | Interurban passenger rail

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: long-distance train operators; high-speed passenger rail operators; intercity rail services.

Software: Train planning, signaling interfaces, asset maintenance, ticketing or freight and control systems.

AI applications / candidates: Demand forecasts and asset-condition monitoring.

Adoption context: AI is selectively added to established operational software and sensor streams.

Watchpoint: Safety-critical railway decisions require certified controls.

4912 | Freight rail

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: rail freight carriers; bulk freight rail operators; intermodal rail operators.

Software: Train planning, signaling interfaces, asset maintenance, ticketing or freight and control systems.

AI applications / candidates: Wagon maintenance and train planning support.

Adoption context: AI is selectively added to established operational software and sensor streams.

Watchpoint: Safety-critical railway decisions require certified controls.

Group 492 | Other land transport

4921 | Urban and suburban passenger land transport

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: city bus operators; metro operators; tram operators; suburban transit services.

Software: Scheduling, ticketing, vehicle tracking, passenger information and maintenance systems.

AI applications / candidates: Ridership forecasts and service disruption detection.

Adoption context: Large networks are more integrated than small operators; AI focuses on prediction.

Watchpoint: Preserve accessibility and service reliability.

4922 | Other passenger land transport

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: taxi businesses; coach operators; school bus contractors; chauffeur services.

Software: Booking, dispatch, vehicle tracking, payments, driver schedules and maintenance.

AI applications / candidates: Dispatch assistance and demand forecasting.

Adoption context: Platforms and fleets are more digital than independent drivers or boat operators.

Watchpoint: Operational automation is not proof of safe autonomy.

4923 | Road freight

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: trucking companies; owner-driver hauliers; refrigerated transport firms; household moving companies.

Software: Transport management, dispatch, telematics, proof of delivery, fuel and maintenance tools.

AI applications / candidates: Route planning, load matching and maintenance alerts.

Adoption context: Large fleets combine datasets; owner-drivers may mainly use broker and navigation apps.

Watchpoint: Respect driving limits, loading constraints and human oversight.

Group 493 | Pipeline transport

4930 | Pipeline transport

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: oil pipeline operators; gas transmission pipelines; other bulk pipeline carriers.

Software: Pipeline controls, network maps, integrity inspection, leak monitoring and maintenance systems.

AI applications / candidates: Leak detection and asset-condition analysis.

Adoption context: Connected networks can add anomaly detection to existing controls.

Watchpoint: Maintain independent safety and incident-response systems.

Division 50 | Water transport

Group 501 | Sea and coastal water transport

5011 | Sea and coastal passenger transport

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: ferry operators; cruise lines; coastal passenger boat services.

Software: Vessel management, navigation, schedules, cargo records, maintenance and commercial systems.

AI applications / candidates: Demand forecasts and vessel-condition monitoring.

Adoption context: Large carriers use prediction and optimization; smaller vessels use lighter tools.

Watchpoint: Marine safety and routing need qualified supervision.

5012 | Sea and coastal freight transport

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: container shipping lines; bulk shipping operators; tanker operators; coastal cargo carriers.

Software: Vessel management, navigation, schedules, cargo records, maintenance and commercial systems.

AI applications / candidates: Arrival-time forecasts and vessel maintenance.

Adoption context: Large carriers use prediction and optimization; smaller vessels use lighter tools.

Watchpoint: Marine safety and routing need qualified supervision.

Group 502 | Inland water transport

5021 | Inland passenger water transport

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: river ferry operators; lake passenger services; inland excursion-boat operators.

Software: Booking, dispatch, vehicle tracking, payments, driver schedules and maintenance.

AI applications / candidates: Booking forecasts and schedule assistance.

Adoption context: Platforms and fleets are more digital than independent drivers or boat operators.

Watchpoint: Operational automation is not proof of safe autonomy.

5022 | Inland freight water transport

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: river barge carriers; inland bulk shipping businesses; canal freight operators.

Software: Vessel management, navigation, schedules, cargo records, maintenance and commercial systems.

AI applications / candidates: Arrival-time and maintenance forecasting.

Adoption context: Large carriers use prediction and optimization; smaller vessels use lighter tools.

Watchpoint: Marine safety and routing need qualified supervision.

Division 51 | Air transport

Group 511 | Passenger air transport

5110 | Passenger air transport

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: scheduled airlines; charter airlines; air-taxi operators; passenger helicopter services.

Software: Reservations or cargo systems, flight operations, crew planning, maintenance and revenue tools.

AI applications / candidates: Maintenance analytics and disruption recovery support.

Adoption context: Software is operationally essential; AI augments selected planning and monitoring workflows.

Watchpoint: No unvalidated model should control safety-critical decisions.

Group 512 | Freight air transport

5120 | Freight air transport

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: cargo airlines; air-cargo charter businesses; freight helicopter operators; commercial cargo space-transport operators.

Software: Reservations or cargo systems, flight operations, crew planning, maintenance and revenue tools.

AI applications / candidates: Capacity forecasting and maintenance analysis.

Adoption context: Software is operationally essential; AI augments selected planning and monitoring workflows.

Watchpoint: No unvalidated model should control safety-critical decisions.

Division 52 | Warehousing and transport support

Group 521 | Warehousing and storage

5210 | Warehousing and storage

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: general warehouses; cold stores; grain storage businesses; self-storage facilities.

Software: Warehouse management, inventory, barcodes, labor plans, automation and customer billing.

AI applications / candidates: Slotting, demand forecasts and visual inventory checks.

Adoption context: Large fulfillment centres can use vision and robotics; small stores may use spreadsheets.

Watchpoint: Check inventory accuracy and worker safety.

Group 522 | Transport support services

5221 | Land-transport support services

D2-D4 / A0-A2 / Estimate confidence: medium

Businesses: parking operators; rail infrastructure services; bus-terminal operators; road toll operators.

Software: Asset maintenance, access control, ticketing or tolls, scheduling and incident systems.

AI applications / candidates: Demand forecasts and asset-condition monitoring.

Adoption context: Connected infrastructure has more scope for analytics than small terminals or parking lots.

Watchpoint: Keep accessibility and security controls auditable.

5222 | Water-transport support services

D3-D4 / A0-A3 / Estimate confidence: medium

Businesses: port operators; marine terminal services; pilotage services; harbor towage businesses.

Software: Terminal operating systems, berth plans, cargo records, customs interfaces and equipment maintenance.

AI applications / candidates: Vessel arrival forecasts and berth-planning support.

Adoption context: Large ports can coordinate predictive systems across several operators.

Watchpoint: Bad partner data can propagate across the network.

5223 | Air-transport support services

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: airport operators; ground handlers; air traffic support services; airport terminal operators.

Software: Airport operations, flight information, baggage, ground handling, access and asset systems.

AI applications / candidates: Passenger-flow forecasts and baggage anomaly checks.

Adoption context: Large airports support predictive flow analysis; deployment must fit aviation assurance.

Watchpoint: Protect passenger data and operational safety.

5224 | Cargo handling

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: stevedores; terminal cargo handlers; freight loading contractors; intermodal handling businesses.

Software: Warehouse management, inventory, barcodes, labor plans, automation and customer billing.

AI applications / candidates: Cargo recognition and handling-plan assistance.

Adoption context: Large fulfillment centres can use vision and robotics; small stores may use spreadsheets.

Watchpoint: Check inventory accuracy and worker safety.

5229 | Other transport support services

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: freight forwarders; customs clearance agents; shipping documentation firms; logistics coordinators.

Software: Freight management, shipment tracking, customs documents, customer records and billing.

AI applications / candidates: Document extraction and shipment exception detection.

Adoption context: Document-heavy workflows are practical AI targets, but many exceptions require specialists.

Watchpoint: Confirm customs and shipment information against source documents.

Group 523 | Transport intermediation

5231 | Freight-transport intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: freight brokers; digital freight exchanges; cargo-booking intermediaries.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Carrier matching and shipment-risk alerts.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

5232 | Passenger-transport intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: ride-booking platforms; transport booking agents; passenger journey intermediaries.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Demand matching and dispatch support.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

Division 53 | Postal and courier activities

Group 531 | Postal activities

5310 | Postal activities

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: national postal operators; universal-service postal businesses; postal delivery networks.

Software: Dispatch, parcel scanning, sorting, delivery tracking, customer support and billing.

AI applications / candidates: Address recognition and delivery forecasting.

Adoption context: Networks can embed routing and vision models; local couriers use simpler apps.

Watchpoint: Protect addresses and avoid unsafe delivery instructions.

Group 532 | Courier activities

5320 | Courier activities

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: express delivery firms; local couriers; parcel networks; bicycle courier businesses.

Software: Dispatch, parcel scanning, sorting, delivery tracking, customer support and billing.

AI applications / candidates: Parcel sorting, routing and arrival-time estimates.

Adoption context: Networks can embed routing and vision models; local couriers use simpler apps.

Watchpoint: Protect addresses and avoid unsafe delivery instructions.

Group 533 | Postal and courier intermediation

5330 | Postal and courier intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: parcel-booking aggregators; courier comparison services; delivery-broker platforms.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Carrier selection and delivery exception triage.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

COMPLETE ACTIVITY CATALOGUE / SECTION I

I | Accommodation and food services

2 divisions / 9 groups / 10 classes / 37 business examples

Hotels, holiday properties, restaurants, caterers and booking platforms differ in scale and data quality. Reservation systems and checkout tools are the foundation, not evidence of autonomous operation.

Evidence scope: Tourism sources provide digital context. Most detailed maturity bands here are low- or medium-confidence analytical estimates. [S17] [S05] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 55 | Accommodation

Group 551 | Hotels and similar accommodation

5510 | Hotels and similar accommodation

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: hotels; motels; resort hotels; hotel-style guesthouses.

Software: Property management, reservations, channel management, housekeeping, billing and guest records.

AI applications / candidates: Demand forecasts, pricing support and guest assistance.

Adoption context: Chains use integrated revenue tools; small independent properties are less connected.

Watchpoint: Guest privacy and accurate availability are essential.

Group 552 | Other short-term accommodation

5520 | Other short-term accommodation

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: holiday homes; vacation apartments; hostels; short-stay guest accommodation.

Software: Booking calendars, listing channels, payments, guest messages and basic property records.

AI applications / candidates: Guest-message drafting and booking forecasts.

Adoption context: Platforms supply much of the technology; operator-owned AI capability is often limited.

Watchpoint: Confirm booking terms and protect guest access information.

Group 553 | Campgrounds and recreational vehicle parks

5530 | Campgrounds and recreational vehicle parks

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: campgrounds; caravan parks; motorhome parks; trailer parks.

Software: Booking calendars, listing channels, payments, guest messages and basic property records.

AI applications / candidates: Occupancy forecasts and guest enquiry assistance.

Adoption context: Platforms supply much of the technology; operator-owned AI capability is often limited.

Watchpoint: Confirm booking terms and protect guest access information.

Group 554 | Accommodation intermediation

5540 | Accommodation intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: online accommodation marketplaces; lodging reservation agents; accommodation booking platforms.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Property matching, search ranking and fraud detection.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

Group 559 | Other accommodation

5590 | Other accommodation

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: student residences; workers' hostels; boarding houses; other longer-stay lodging.

Software: Lease or tenancy records, property accounting, work orders, portfolios and document management.

AI applications / candidates: Occupancy planning and maintenance triage.

Adoption context: Large portfolios integrate building and tenant data; small landlords use basic tools.

Watchpoint: Review legal terms and do not automate discriminatory decisions.

Division 56 | Food and beverage services

Group 561 | Restaurants and mobile food services

5610 | Restaurants and mobile food services

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: restaurants; takeaway shops; street-food vendors; food trucks.

Software: Checkout, ordering, reservations, kitchen displays, purchasing, inventory and staff schedules.

AI applications / candidates: Demand forecasts, order assistance and waste reduction.

Adoption context: Chains have richer data; stalls and small cafes may rely on messaging and payments.

Watchpoint: Food safety and allergen information need verified sources.

Group 562 | Event catering and other food services

5621 | Event catering

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: wedding caterers; corporate event caterers; private-party caterers.

Software: Event orders, menu costing, purchasing, production plans, staffing and invoicing.

AI applications / candidates: Headcount-based purchasing and quote assistance.

Adoption context: Larger contract caterers can forecast repeat demand; one-off events remain harder.

Watchpoint: Check allergies, quantities and event-specific requirements.

5629 | Other food services

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: school meal contractors; workplace caterers; hospital foodservice contractors; transport catering firms.

Software: Event orders, menu costing, purchasing, production plans, staffing and invoicing.

AI applications / candidates: Meal-demand forecasts and production planning.

Adoption context: Larger contract caterers can forecast repeat demand; one-off events remain harder.

Watchpoint: Check allergies, quantities and event-specific requirements.

Group 563 | Beverage serving

5630 | Beverage serving

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: cafes; bars; pubs; juice bars.

Software: Checkout, ordering, reservations, kitchen displays, purchasing, inventory and staff schedules.

AI applications / candidates: Demand forecasts and staff scheduling support.

Adoption context: Chains have richer data; stalls and small cafes may rely on messaging and payments.

Watchpoint: Food safety and allergen information need verified sources.

Group 564 | Food and beverage service intermediation

5640 | Food and beverage service intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: meal delivery-order platforms; restaurant booking marketplaces; catering referral services.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Menu search, order matching and demand forecasts.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

COMPLETE ACTIVITY CATALOGUE / SECTION J

J | Publishing, broadcasting and content

3 divisions / 7 groups / 15 classes / 56 business examples

This includes software publishing as well as books, games, music, film and distribution. Digital content production can be highly software-dependent while AI policies differ by creator, rights holder and audience.

Evidence scope: Creative-sector research informs rights and workflow issues. Developer survey evidence is relevant to software work, not all media businesses. [S18] [S20] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 58 | Publishing

Group 581 | Books, newspapers, periodicals and other publishing

5811 | Book publishing

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: trade publishers; educational publishers; academic book publishers; self-publishing businesses.

Software: Editorial workflow, content management, page layout, rights, distribution and subscription systems.

AI applications / candidates: Metadata, editing assistance and accessible formats.

Adoption context: AI can assist production, but maturity varies greatly by publisher and genre.

Watchpoint: Verify facts, licenses and author agreements.

5812 | Newspaper publishing

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: daily newspaper publishers; local news publishers; digital newspaper businesses.

Software: Newsroom production, content management, archives, subscriptions and distribution tools.

AI applications / candidates: Transcription, archive search and copy assistance.

Adoption context: Text-heavy work supports AI assistance, but trustworthy reporting needs editorial control.

Watchpoint: Never publish unverified generated facts or fabricated sources.

5813 | Journal and periodical publishing

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: magazine publishers; scholarly journal publishers; professional periodical publishers.

Software: Editorial workflow, content management, page layout, rights, distribution and subscription systems.

AI applications / candidates: Metadata extraction and editorial workflow assistance.

Adoption context: AI can assist production, but maturity varies greatly by publisher and genre.

Watchpoint: Verify facts, licenses and author agreements.

5819 | Other publishing

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: directory publishers; calendar publishers; greeting-card publishers; other information publishers.

Software: Editorial workflow, content management, page layout, rights, distribution and subscription systems.

AI applications / candidates: Content classification and layout assistance.

Adoption context: AI can assist production, but maturity varies greatly by publisher and genre.

Watchpoint: Verify facts, licenses and author agreements.

Group 582 | Software publishing

5821 | Video-game publishing

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: console game publishers; mobile game publishers; PC game publishers; independent game publishers.

Software: Release management, storefronts, analytics, customer support, rights and marketing tools.

AI applications / candidates: Localization, support and player-behavior analysis.

Adoption context: AI can enter support and content workflows; core game production may be handled separately.

Watchpoint: Respect creator rights, age protections and player privacy.

5829 | Other software publishing

D4 / A1-A4 / Estimate confidence: medium

Businesses: business SaaS publishers; cybersecurity software publishers; operating-system vendors; consumer app publishers.

Software: Source control, testing, cloud infrastructure, product analytics, subscriptions and support tools.

AI applications / candidates: Coding, support, analytics and embedded AI features.

Adoption context: AI is central for some vendors and optional for others; ordinary software is not inherently AI.

Watchpoint: Evaluate generated code, data access and model behavior.

Division 59 | Film, video, television and sound production

Group 591 | Film, video and television activities

5911 | Film, video and television production

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: film studios; TV production companies; video production agencies; independent video producers.

Software: Production planning, scripts, editing, asset storage, rights and collaboration systems.

AI applications / candidates: Transcription, previsualization and production planning.

Adoption context: AI-assisted production is growing but uneven across budgets and content types.

Watchpoint: Obtain likeness, voice, performer and copyright permissions.

5912 | Film, video and television post-production

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: editing studios; visual-effects houses; dubbing studios; color-grading businesses.

Software: Editing, visual effects, compositing, audio, asset management and rendering systems.

AI applications / candidates: Rotoscoping, captioning, cleanup and localization.

Adoption context: AI features can be embedded in specialist tools without replacing the whole workflow.

Watchpoint: Check artifacts, provenance and contractual permissions.

5913 | Film, video and television distribution

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: film distributors; video licensing businesses; television content distributors.

Software: Rights catalogues, licensing contracts, royalties, sales and digital delivery systems.

AI applications / candidates: Catalogue tagging and rights-document search.

Adoption context: Search and document tools are easier to adopt than fully automated rights decisions.

Watchpoint: Rights conflicts require authoritative contract review.

5914 | Film projection

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: cinema chains; independent cinemas; mobile cinemas; film-screening venues.

Software: Ticketing or reservations, payments, membership, staffing and event schedules.

AI applications / candidates: Programming forecasts and customer enquiry assistance.

Adoption context: Chains and online-first venues integrate more data than independent sites.

Watchpoint: Keep customer and event information accurate.

Group 592 | Sound recording and music publishing

5920 | Sound recording and music publishing

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: recording studios; record labels; music publishers; audio production businesses.

Software: Digital audio production, editing, rights, royalties and catalogue management.

AI applications / candidates: Audio restoration, transcription and catalogue tagging.

Adoption context: AI features vary from restoration to generation; studios retain substantial human craft.

Watchpoint: Obtain voice, music and recording permissions.

Division 60 | Broadcasting, news agencies and content distribution

Group 601 | Radio broadcasting and audio distribution

6010 | Radio broadcasting and audio distribution

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: radio stations; internet radio networks; audio distribution services; podcast distribution businesses.

Software: Playout, content management, schedules, ad sales, streaming and audience analytics.

AI applications / candidates: Transcription, scheduling and content discovery.

Adoption context: Large broadcasters and streamers embed more AI than small stations.

Watchpoint: Editorial review and rights clearance remain necessary.

Group 602 | Television broadcasting and video distribution

6020 | Television broadcasting and video distribution

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: television networks; cable programme services; streaming video services; television channel operators.

Software: Playout, content management, schedules, ad sales, streaming and audience analytics.

AI applications / candidates: Content tagging, recommendations and captioning.

Adoption context: Large broadcasters and streamers embed more AI than small stations.

Watchpoint: Editorial review and rights clearance remain necessary.

Group 603 | News agencies and other content distribution

6031 | News agencies

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: wire services; photo news agencies; news syndication businesses.

Software: Newsroom production, content management, archives, subscriptions and distribution tools.

AI applications / candidates: Archive search, translation and verification support.

Adoption context: Text-heavy work supports AI assistance, but trustworthy reporting needs editorial control.

Watchpoint: Never publish unverified generated facts or fabricated sources.

6039 | Social networks and other content distribution

D4 / A2-A4 / Estimate confidence: medium

Businesses: social networks; creator-content distribution platforms; media sharing platforms; other content distribution services.

Software: Content delivery, social graphs, recommendations, advertising, moderation and identity systems.

AI applications / candidates: Recommendations, spam detection and moderation triage.

Adoption context: AI can be part of the core service, but smaller communities may use simpler tools.

Watchpoint: Address bias, manipulation, illegal content and user appeals.

COMPLETE ACTIVITY CATALOGUE / SECTION K

K | Telecom, computing and information services

3 divisions / 8 groups / 9 classes / 33 business examples

This is the technical service and infrastructure layer: telecom, custom development, IT services, hosting and search. A provider selling AI capacity does not necessarily use advanced AI in its own operations.

Evidence scope: GSMA and developer evidence support selected workflows. Respondent and operator-level findings are not interchangeable with firm adoption rates. [S19] [S20] [S06] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 61 | Telecommunications

Group 611 | Wired, wireless and satellite telecom

6110 | Wired, wireless and satellite telecom

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: fixed-line operators; mobile network operators; internet access providers; satellite telecom operators.

Software: Network operations, customer billing, service provisioning, support and security systems.

AI applications / candidates: Network anomalies, demand prediction and customer assistance.

Adoption context: AI builds on extensive network data; deployment differs by operator and legacy estate.

Watchpoint: Protect critical communications and subscriber privacy.

Group 612 | Telecom reselling and intermediation

6120 | Telecom reselling and intermediation

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: virtual mobile operators; telecom resellers; telecom service brokers.

Software: Subscriber billing, provisioning, customer service, partner integration and sales systems.

AI applications / candidates: Churn analysis and customer enquiry support.

Adoption context: AI is usually embedded in customer or network-support products rather than owned models.

Watchpoint: Confirm service entitlement and customer consent.

Group 619 | Other telecommunications

6190 | Other telecommunications

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: specialist telecom service providers; telecom access service businesses; other telecom operators.

Software: Subscriber billing, provisioning, customer service, partner integration and sales systems.

AI applications / candidates: Service triage and network-log interpretation.

Adoption context: AI is usually embedded in customer or network-support products rather than owned models.

Watchpoint: Confirm service entitlement and customer consent.

Division 62 | Programming, IT consultancy and related services

Group 621 | Computer programming

6211 | Video-game and game-tool development

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: game development studios; contract game developers; game-engine tool developers.

Software: Game engines, source control, asset pipelines, testing, build systems and analytics.

AI applications / candidates: Code assistance, asset workflows and automated testing.

Adoption context: Coding and art tools offer AI assistance; adoption depends on studio policies and rights.

Watchpoint: Generated assets and code need quality and license review.

6219 | Other computer programming

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: custom software developers; website developers; mobile-app developers; AI application engineering firms.

Software: Source control, code editors, testing, issue tracking, cloud deployment and documentation.

AI applications / candidates: Code generation, tests, debugging and documentation.

Adoption context: Developer AI use can be high without fully autonomous project delivery.

Watchpoint: Test outputs and protect source code and credentials.

Group 622 | IT consultancy and computer facilities management

6220 | IT consultancy and computer facilities management

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: IT consultancies; managed IT providers; enterprise systems integrators; computer facilities managers.

Software: Service desks, monitoring, remote management, configuration records, projects and security tools.

AI applications / candidates: Incident triage, knowledge search and code assistance.

Adoption context: AI assists repetitive support and analysis; privileged actions need controls.

Watchpoint: Use least privilege, audit logs and approval gates.

Group 629 | Other IT and computer services

6290 | Other IT and computer services

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: disaster-recovery services; computer setup services; data-recovery firms; other IT support providers.

Software: Service desks, monitoring, remote management, configuration records, projects and security tools.

AI applications / candidates: Service-ticket triage and technical knowledge search.

Adoption context: AI assists repetitive support and analysis; privileged actions need controls.

Watchpoint: Use least privilege, audit logs and approval gates.

Division 63 | Computing infrastructure and information services

Group 631 | Computing infrastructure, data processing and hosting

6310 | Computing infrastructure, data processing and hosting

D4 / A1-A4 / Estimate confidence: medium

Businesses: cloud infrastructure providers; data centres; web hosting firms; data-processing bureaus.

Software: Compute orchestration, monitoring, billing, security, data platforms and service operations.

AI applications / candidates: Capacity planning, security alerts and workload operations.

Adoption context: AI infrastructure is core for some providers; ordinary hosting may use little operational AI.

Watchpoint: Track cost, tenant isolation and data access.

Group 639 | Web search portals and other information services

6390 | Web search portals and other information services

D3-D4 / A1-A4 / Estimate confidence: medium

Businesses: search engines; information retrieval services; business information providers; information portals.

Software: Search indexes, databases, content pipelines, APIs, analytics and subscriptions.

AI applications / candidates: Information retrieval, summarization and entity matching.

Adoption context: AI can be central to retrieval and synthesis; traditional information services vary.

Watchpoint: Show provenance and avoid unsupported summaries.

COMPLETE ACTIVITY CATALOGUE / SECTION L

L | Finance and insurance

3 divisions / 10 groups / 24 classes / 81 business examples

Banks, lenders, funds, brokers, insurers and administrators are separate activities. A fund vehicle can outsource technology to its manager; regulated software dependence is not the same as autonomous decision-making.

Evidence scope: The UK financial survey provides a strong regional anchor, but individual class bands and global extrapolations remain estimates. [S21] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 64 | Financial services other than insurance and pensions

Group 641 | Monetary intermediation

6411 | Central banking

D3-D4 / A1-A2 / Estimate confidence: low

Businesses: national central banks; regional monetary authorities; reserve banks.

Software: Payments infrastructure, statistical systems, supervisory data, economic models and secure records.

AI applications / candidates: Economic-text analysis and supervised data-quality review.

Adoption context: AI use is typically supervised and specialized rather than autonomous policy-making.

Watchpoint: Maintain confidentiality and accountable monetary decisions.

6419 | Other monetary intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: commercial banks; savings banks; credit unions; cooperative banks.

Software: Core banking, payments, customer records, risk, compliance and financial reporting.

AI applications / candidates: Fraud alerts, service assistance and credit-risk support.

Adoption context: Larger banks have established fraud and risk models; smaller institutions may use vendor modules.

Watchpoint: Fairness, explainability, security and human escalation matter.

Group 642 | Holding companies and financing conduits

6421 | Holding companies

D1-D3 / A0-A1 / Estimate confidence: low

Businesses: financial holding companies; non-financial holding companies; ownership holding vehicles.

Software: Ownership registers, consolidation, treasury, governance and company reporting.

AI applications / candidates: Portfolio-document search and reporting assistance.

Adoption context: An ownership vehicle may have few staff and outsource most operations.

Watchpoint: Separate the vehicle from its operating subsidiaries.

6422 | Financing conduits

D1-D3 / A0-A1 / Estimate confidence: low

Businesses: group financing conduits; financial special-purpose conduits; funding pass-through entities.

Software: Ownership registers, consolidation, treasury, governance and company reporting.

AI applications / candidates: Transaction-document review and compliance reporting.

Adoption context: An ownership vehicle may have few staff and outsource most operations.

Watchpoint: Separate the vehicle from its operating subsidiaries.

Group 643 | Trusts, funds and similar financial entities

6431 | Money-market funds

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: money-market mutual funds; institutional cash funds; short-term investment funds.

Software: Portfolio accounting, custody interfaces, investor records, risk and regulatory reporting.

AI applications / candidates: Risk-report summaries and compliance-document checks.

Adoption context: Technology is often supplied by managers or administrators, not owned by the fund vehicle.

Watchpoint: Do not confuse fund management with the fund itself.

6432 | Other investment funds

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: equity funds; bond funds; private investment funds; real-estate investment funds.

Software: Portfolio accounting, custody interfaces, investor records, risk and regulatory reporting.

AI applications / candidates: Portfolio monitoring and reporting support.

Adoption context: Technology is often supplied by managers or administrators, not owned by the fund vehicle.

Watchpoint: Do not confuse fund management with the fund itself.

6433 | Trust, estate and agency accounts

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: trust vehicles; estate accounts; fiduciary agency accounts.

Software: Portfolio accounting, custody interfaces, investor records, risk and regulatory reporting.

AI applications / candidates: Document classification and reporting assistance.

Adoption context: Technology is often supplied by managers or administrators, not owned by the fund vehicle.

Watchpoint: Do not confuse fund management with the fund itself.

Group 649 | Other financial services

6491 | Financial leasing

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: equipment finance lessors; vehicle finance lessors; asset-financing businesses.

Software: Loan origination, servicing, identity checks, collections, accounting and risk systems.

AI applications / candidates: Asset-document checks and credit-risk support.

Adoption context: Lenders can use scoring and document models; small lenders are often less integrated.

Watchpoint: Review bias, affordability and adverse decisions.

6492 | International trade financing

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: trade-finance companies; export-finance providers; import-finance specialists.

Software: Loan origination, servicing, identity checks, collections, accounting and risk systems.

AI applications / candidates: Trade-document checking and transaction-risk alerts.

Adoption context: Lenders can use scoring and document models; small lenders are often less integrated.

Watchpoint: Review bias, affordability and adverse decisions.

6493 | Factoring

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: invoice factoring firms; receivables finance providers; export factoring businesses.

Software: Loan origination, servicing, identity checks, collections, accounting and risk systems.

AI applications / candidates: Invoice verification and collection-risk forecasting.

Adoption context: Lenders can use scoring and document models; small lenders are often less integrated.

Watchpoint: Review bias, affordability and adverse decisions.

6494 | Securitisation

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: securitisation vehicles; asset-backed finance entities; mortgage securitisation vehicles.

Software: Portfolio accounting, custody interfaces, investor records, risk and regulatory reporting.

AI applications / candidates: Document checks and collateral-report analysis.

Adoption context: Technology is often supplied by managers or administrators, not owned by the fund vehicle.

Watchpoint: Do not confuse fund management with the fund itself.

6495 | Other credit granting

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: consumer lenders; microfinance lenders; business lenders; licensed pawn lenders.

Software: Loan origination, servicing, identity checks, collections, accounting and risk systems.

AI applications / candidates: Fraud checks, affordability analysis and credit support.

Adoption context: Lenders can use scoring and document models; small lenders are often less integrated.

Watchpoint: Review bias, affordability and adverse decisions.

6499 | Other financial services

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: own-account financial traders; other investment financial businesses; specialist financial service entities.

Software: Market data, order management, trading, risk, reconciliation and compliance tools.

AI applications / candidates: Market-data analysis and risk monitoring.

Adoption context: Some firms use advanced models; many strategies and controls are conventional software.

Watchpoint: Backtests do not establish future performance.

Division 65 | Insurance, reinsurance and pension funding

Group 651 | Insurance

6511 | Life insurance

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: life insurers; annuity providers; mutual life insurers; endowment insurers.

Software: Policy administration, underwriting, claims, customer records, billing and risk systems.

AI applications / candidates: Underwriting-document extraction and claims assistance.

Adoption context: Document and fraud models can be embedded; legacy systems constrain wider integration.

Watchpoint: Review coverage, fairness and claims decisions.

6512 | Non-life insurance

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: property insurers; motor insurers; health insurers; commercial liability insurers.

Software: Policy administration, underwriting, claims, customer records, billing and risk systems.

AI applications / candidates: Claims triage, fraud alerts and damage assessment support.

Adoption context: Document and fraud models can be embedded; legacy systems constrain wider integration.

Watchpoint: Review coverage, fairness and claims decisions.

Group 652 | Reinsurance

6520 | Reinsurance

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: life reinsurers; property catastrophe reinsurers; specialty reinsurers.

Software: Exposure data, catastrophe models, treaty administration, pricing and portfolio systems.

AI applications / candidates: Exposure-document analysis and portfolio-risk support.

Adoption context: Advanced analytics coexist with specialist human underwriting and model uncertainty.

Watchpoint: Distinguish modeled risk from reliable prediction.

Group 653 | Pension funding

6530 | Pension funding

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: occupational pension funds; retirement pension funds; pension funding entities.

Software: Portfolio accounting, custody interfaces, investor records, risk and regulatory reporting.

AI applications / candidates: Member-service assistance and reporting checks.

Adoption context: Technology is often supplied by managers or administrators, not owned by the fund vehicle.

Watchpoint: Do not confuse fund management with the fund itself.

Division 66 | Auxiliary financial and insurance services

Group 661 | Auxiliary financial services other than insurance and pensions

6611 | Financial market administration

D4 / A1-A3 / Estimate confidence: medium

Businesses: stock exchanges; derivatives exchanges; securities market infrastructure operators.

Software: Market operations, clearing or settlement interfaces, surveillance, risk and resilience systems.

AI applications / candidates: Market-abuse alerts and operational monitoring.

Adoption context: Operational software is core; AI usually augments monitoring and analysis.

Watchpoint: Preserve market integrity and tested fallback operations.

6612 | Securities and commodity brokerage

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: stockbrokers; commodity brokers; derivatives brokers; online securities brokers.

Software: Client accounts, orders, market data, compliance, suitability and reporting tools.

AI applications / candidates: Service assistance, trade surveillance and suitability support.

Adoption context: Large and digital brokers embed more analytics than small advisory practices.

Watchpoint: Do not let generated advice bypass suitability controls.

6619 | Other auxiliary financial activities

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: payment processors; financial transaction support providers; financial advisers; mortgage advisory services.

Software: Transaction processing, identity checks, customer records, risk and document systems.

AI applications / candidates: Fraud alerts and financial-document extraction.

Adoption context: Payments firms may have advanced fraud models; small advisers use lighter software.

Watchpoint: Match controls to regulated activity and sensitive data.

Group 662 | Auxiliary insurance and pension services

6621 | Risk and damage evaluation

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: insurance loss adjusters; claims assessors; risk surveyors; damage appraisal firms.

Software: Claims files, field inspection, photos, estimating, document management and billing.

AI applications / candidates: Report extraction and image-assisted damage review.

Adoption context: AI can extract reports and assist image review; site-specific judgment remains important.

Watchpoint: AI estimates are not proof of cause, coverage or fraud.

6622 | Insurance agents and brokers

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: independent insurance agents; commercial insurance brokers; insurance comparison intermediaries.

Software: Customer records, policy comparisons, submissions, renewals and commissions.

AI applications / candidates: Policy comparison and application-document assistance.

Adoption context: Larger brokers integrate document tools; small agencies often use carrier portals.

Watchpoint: Validate policy terms and client suitability.

6629 | Other insurance and pension support

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: insurance administration providers; pension administration firms; actuarial support businesses.

Software: Administration platforms, member or policy records, calculations and document workflows.

AI applications / candidates: Document extraction and administration exception checks.

Adoption context: Repetitive administration supports AI assistance, subject to auditability.

Watchpoint: Check calculations and retain source-document trails.

Group 663 | Fund management

6630 | Fund management

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: asset managers; mutual fund managers; private-equity managers; investment portfolio managers.

Software: Portfolio and order management, research, risk, compliance and investor reporting.

AI applications / candidates: Research search, portfolio analytics and compliance review.

Adoption context: AI can support research and surveillance; it does not guarantee investment outperformance.

Watchpoint: Control conflicts, confidential data and model risk.

COMPLETE ACTIVITY CATALOGUE / SECTION M

M | Real estate

1 divisions / 2 groups / 3 classes / 12 business examples

Property ownership, brokerage and fee-based management are different businesses. Large portfolios have recurring data; a single landlord may use only spreadsheets and messaging.

Evidence scope: RICS commercial-property respondents and Eurostat enterprises provide limited measured anchors; small residential operators may look very different. [S14] [S04] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 68 | Real estate

Group 681 | Owned or leased real estate

6810 | Owned or leased real estate

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: residential landlords; commercial property owners; self-owned property lessors; real-estate development for sale.

Software: Lease or tenancy records, property accounting, work orders, portfolios and document management.

AI applications / candidates: Lease extraction, maintenance triage and pricing support.

Adoption context: Large portfolios integrate building and tenant data; small landlords use basic tools.

Watchpoint: Review legal terms and do not automate discriminatory decisions.

Group 682 | Fee-based or contract real-estate services

6821 | Real-estate intermediation

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: estate agents; property brokers; real-estate marketplaces; property referral agencies.

Software: Listings, customer records, lead management, appointments, contracts and transactions.

AI applications / candidates: Listing drafts, lead matching and valuation support.

Adoption context: Larger brokers use more analytics; small agents often rely on portals and messaging.

Watchpoint: Verify listings and avoid discriminatory matching.

6829 | Other fee-based real-estate activities

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: property managers; real-estate appraisers; building management firms; real-estate service contractors.

Software: Lease or tenancy records, property accounting, work orders, portfolios and document management.

AI applications / candidates: Lease abstraction, work-order triage and valuation support.

Adoption context: Large portfolios integrate building and tenant data; small landlords use basic tools.

Watchpoint: Review legal terms and do not automate discriminatory decisions.

COMPLETE ACTIVITY CATALOGUE / SECTION N

N | Professional, scientific and technical services

7 divisions / 16 groups / 17 classes / 64 business examples

Law, accounting, consulting, engineering, research, advertising and veterinary care share knowledge work but not the same obligations. Technical expertise remains necessary even when documents become easier to produce.

Evidence scope: Professional-survey results cover selected occupations. Broader engineering, creative and scientific applications use additional sector evidence and explicit analyst judgment. [S22] [S11] [S18] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 69 | Legal and accounting services

Group 691 | Legal services

6910 | Legal services

D2-D3 / A1-A3 / Estimate confidence: medium

Businesses: law firms; solo lawyers; notaries; legal-aid practices.

Software: Practice management, legal research, document systems, matter billing and e-signatures.

AI applications / candidates: Research assistance, document review and drafting.

Adoption context: AI adoption differs by firm and task; research and document review are frequent entry points.

Watchpoint: Check every legal authority and preserve confidentiality.

Group 692 | Accounting, bookkeeping, audit and tax

6920 | Accounting, bookkeeping, audit and tax

D2-D3 / A1-A3 / Estimate confidence: medium

Businesses: accounting firms; bookkeeping bureaus; audit practices; tax consultancies.

Software: Accounting ledgers, tax, audit, practice management, documents and client portals.

AI applications / candidates: Transaction classification and anomaly-led review.

Adoption context: Structured records support assisted classification; judgment-heavy audit work needs review.

Watchpoint: Reconcile outputs to evidence and applicable rules.

Division 70 | Head offices and management consultancy

Group 701 | Head-office activities

7010 | Head-office activities

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: corporate headquarters; regional group headquarters; business group administrative centres.

Software: Enterprise planning, finance, HR, reporting, collaboration and governance systems.

AI applications / candidates: Management-report synthesis and internal knowledge search.

Adoption context: Large groups deploy enterprise assistants; fragmented subsidiaries limit integration.

Watchpoint: Confidential strategy and employee data need access controls.

Group 702 | Management consultancy

7020 | Management consultancy

D2-D3 / A1-A3 / Estimate confidence: medium

Businesses: strategy consultancies; operations consultants; HR management consultants; business process advisers.

Software: Project delivery, research, analysis, customer records, knowledge bases and billing.

AI applications / candidates: Research synthesis and analysis assistance.

Adoption context: Knowledge work suits AI assistance, but adoption does not prove advice quality.

Watchpoint: Validate evidence and disclose assumptions.

Division 71 | Architecture, engineering and technical testing

Group 711 | Architecture, engineering and technical consultancy

7110 | Architecture, engineering and technical consultancy

D2-D4 / A0-A3 / Estimate confidence: medium

Businesses: architecture practices; engineering consultancies; surveying firms; urban planning consultancies.

Software: Design, simulation, building models, geospatial tools, projects and technical documents.

AI applications / candidates: Drawing/document review and design-option assistance.

Adoption context: Digital design can be mature while AI remains experimental or narrowly deployed.

Watchpoint: Professional engineers must approve safety-critical outputs.

Group 712 | Technical testing and analysis

7120 | Technical testing and analysis

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: materials testing labs; environmental testing labs; product certification labs; vehicle inspection businesses.

Software: Laboratory information, instrument data, calibration, sample tracking and reporting.

AI applications / candidates: Test anomaly detection and report assistance.

Adoption context: AI can identify anomalies where validated test datasets exist.

Watchpoint: Accreditation and traceability require verified methods.

Division 72 | Scientific research and development

Group 721 | Natural-science and engineering R&D

7210 | Natural-science and engineering R&D

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: biotechnology research firms; engineering research labs; agricultural research institutes; physical-science R&D businesses.

Software: Scientific computing, laboratory records, instruments, literature and experiment data systems.

AI applications / candidates: Literature search, experiment analysis and modeling support.

Adoption context: Computational fields use more AI than data-poor or physical experimental niches.

Watchpoint: Reproduce results and guard against fabricated evidence.

Group 722 | Social-science and humanities R&D

7220 | Social-science and humanities R&D

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: economic research institutes; social research organizations; humanities research centres.

Software: Survey tools, qualitative analysis, statistical software, literature and research records.

AI applications / candidates: Text coding, literature search and survey analysis.

Adoption context: AI can assist coding and discovery, but evidence quality depends on research design.

Watchpoint: Protect participants and check coding bias.

Division 73 | Advertising, market research and public relations

Group 731 | Advertising

7310 | Advertising

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: advertising agencies; media-buying agencies; outdoor advertising firms; performance marketing agencies.

Software: Campaign management, ad platforms, analytics, design tools and customer records.

AI applications / candidates: Creative variants, targeting analysis and campaign monitoring.

Adoption context: AI is often embedded in platforms as well as agency workflows.

Watchpoint: Avoid deceptive claims and unlawful personal-data use.

Group 732 | Market research and polling

7320 | Market research and polling

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: market research firms; polling organizations; consumer insights agencies; survey businesses.

Software: Survey platforms, panels, analytics, interview records and insight reporting.

AI applications / candidates: Interview coding, sentiment analysis and quality checks.

Adoption context: Text and survey workflows support assistance; synthetic responses are not real evidence.

Watchpoint: Separate observed respondents from model-generated material.

Group 733 | Public relations

7330 | Public relations

D2-D3 / A1-A3 / Estimate confidence: medium

Businesses: PR agencies; corporate communications consultancies; media relations firms.

Software: Media monitoring, journalist databases, content calendars and client reporting.

AI applications / candidates: Media monitoring and communication-draft assistance.

Adoption context: AI can support monitoring and drafting; relationship work remains human-led.

Watchpoint: Verify claims and avoid impersonation or fabricated coverage.

Division 74 | Other professional, scientific and technical services

Group 741 | Specialized design

7410 | Specialized design

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: graphic design studios; industrial design firms; interior designers; fashion design studios.

Software: Design software, asset libraries, prototypes, collaboration and project systems.

AI applications / candidates: Concept variations, visualization and asset preparation.

Adoption context: AI capabilities vary by discipline; physical design still needs testing.

Watchpoint: Check copyright, feasibility and client permissions.

Group 742 | Photography

7420 | Photography

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: portrait studios; commercial photographers; event photographers; specialist photography businesses.

Software: Image editing, digital asset management, booking, online galleries and invoicing.

AI applications / candidates: Image selection, retouching and catalogue tagging.

Adoption context: Editing tools may include AI even in small studios; capture remains a physical service.

Watchpoint: Disclose consequential edits and protect likeness rights.

Group 743 | Translation and interpretation

7430 | Translation and interpretation

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: translation agencies; freelance translators; interpreting businesses; localization firms.

Software: Translation memory, terminology, localization, project and quality-assurance systems.

AI applications / candidates: Machine translation and transcript assistance.

Adoption context: Machine translation is established in some language pairs; expert review remains important.

Watchpoint: Medical and legal translations need specialist verification.

Group 749 | Other professional, scientific and technical services

7491 | Patent brokering and marketing

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: patent brokers; intellectual-property marketing agencies; invention licensing intermediaries.

Software: Patent databases, customer records, licensing pipelines and contract documents.

AI applications / candidates: Patent search and buyer-licensee matching.

Adoption context: AI can improve search, but commercial and legal assessments remain specialist tasks.

Watchpoint: Validate ownership, validity and permitted disclosures.

7499 | Other professional and technical services

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: environmental consultants; non-real-estate valuation specialists; talent agents; other specialist technical advisers.

Software: Specialist analysis, project records, client documents, knowledge search and invoicing.

AI applications / candidates: Document search, reporting and analysis assistance.

Adoption context: This residual class is heterogeneous; its range is a broad proxy, not a measured average.

Watchpoint: Check the exact service and appropriate professional oversight.

Division 75 | Veterinary services

Group 750 | Veterinary services

7500 | Veterinary services

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: companion-animal clinics; livestock veterinary practices; veterinary hospitals; equine veterinary practices.

Software: Practice management, appointments, animal records, imaging, pharmacy and billing tools.

AI applications / candidates: Clinical note drafting and reviewed diagnostic support.

Adoption context: Larger hospitals have more digital data; AI is usually task-specific assistance.

Watchpoint: Veterinarians retain clinical responsibility.

COMPLETE ACTIVITY CATALOGUE / SECTION O

O | Administrative and support services

6 divisions / 18 groups / 25 classes / 93 business examples

Rental, recruitment, travel, facilities, cleaning, call centres and business outsourcing range from manual local services to fully digital platforms. The digital base should be assessed before AI ambition.

Evidence scope: Research on SMEs and document-heavy services supports workflow analysis; many individual classes lack direct measured adoption evidence. [S07] [S16] [S22] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 77 | Rental and leasing

Group 771 | Motor-vehicle rental and leasing

7710 | Motor-vehicle rental and leasing

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: car-rental firms; van rental businesses; truck lessors without drivers; long-term vehicle rental firms.

Software: Asset inventory, reservations, contracts, inspections, maintenance and billing.

AI applications / candidates: Fleet demand, damage review and maintenance forecasting.

Adoption context: Fleet-scale operators can use predictive tools; small lessors may use simple booking software.

Watchpoint: Verify damage findings and contract terms.

Group 772 | Personal and household-goods rental

7721 | Recreational and sports-goods rental

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: bicycle rental shops; ski hire businesses; boat hire without crew; sports-equipment rental firms.

Software: Bookings, stock availability, deposits, maintenance records and payments.

AI applications / candidates: Demand forecasts and booking assistance.

Adoption context: Most AI comes through general booking or communication tools.

Watchpoint: Confirm availability, condition and liability terms.

7729 | Other personal and household-goods rental

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: furniture rental firms; costume hire shops; tool hire for households; household equipment rental businesses.

Software: Bookings, stock availability, deposits, maintenance records and payments.

AI applications / candidates: Stock allocation and booking assistance.

Adoption context: Most AI comes through general booking or communication tools.

Watchpoint: Confirm availability, condition and liability terms.

Group 773 | Other machinery and tangible-goods rental

7730 | Other machinery and tangible-goods rental

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: construction equipment lessors; industrial equipment rental firms; aircraft leasing without crew; agricultural machinery lessors.

Software: Asset inventory, reservations, contracts, inspections, maintenance and billing.

AI applications / candidates: Utilization prediction and maintenance alerts.

Adoption context: Fleet-scale operators can use predictive tools; small lessors may use simple booking software.

Watchpoint: Verify damage findings and contract terms.

Group 774 | Intellectual-property leasing other than copyright

7740 | Intellectual-property leasing other than copyright

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: patent licensors; trademark licensing businesses; franchise brand licensors; non-copyright IP licensors.

Software: License registers, contracts, royalties, partner records and compliance systems.

AI applications / candidates: Contract search and royalty-report checking.

Adoption context: Larger licensors can add document analytics; small owners may outsource administration.

Watchpoint: Rights scope and royalty obligations need contract review.

Group 775 | Rental and leasing intermediation

7751 | Car, motorhome and trailer rental intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: car-rental comparison platforms; motorhome rental marketplaces; trailer rental brokers.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Fleet matching and booking-risk analysis.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

7752 | Other rental and leasing intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: equipment rental marketplaces; household-goods hire platforms; IP licensing intermediaries.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Asset matching and listing verification support.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

Division 78 | Employment services

Group 781 | Employment placement

7810 | Employment placement

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: recruitment agencies; executive search firms; employment placement services; recruiting marketplaces.

Software: Applicant tracking, candidate databases, job postings, interviews and customer records.

AI applications / candidates: Candidate matching and application summarization.

Adoption context: Digital agencies use matching tools; automated screening quality varies widely.

Watchpoint: Test bias and retain meaningful human review.

Group 782 | Temporary staffing and other HR provision

7820 | Temporary staffing and other HR provision

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: temporary staffing agencies; contract staff suppliers; employer-of-record providers; workforce leasing firms.

Software: Workforce scheduling, payroll, onboarding, time records and client billing.

AI applications / candidates: Staff matching and roster-demand forecasting.

Adoption context: Repeat staffing operations have more usable data than occasional placements.

Watchpoint: Respect worker rights, privacy and employment rules.

Division 79 | Travel agencies, tour operators and related services

Group 791 | Travel agencies and tour operators

7911 | Travel agencies

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: retail travel agents; corporate travel agencies; online travel agencies.

Software: Booking systems, supplier inventories, itineraries, customer records and payments.

AI applications / candidates: Itinerary drafting and booking-policy assistance.

Adoption context: Online agencies are more integrated than small specialists; AI often assists planning and support.

Watchpoint: Recheck prices, availability, visas and cancellation terms.

7912 | Tour operators

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: package-tour operators; adventure tour companies; inbound tour operators; specialist holiday organizers.

Software: Booking systems, supplier inventories, itineraries, customer records and payments.

AI applications / candidates: Itinerary planning and demand forecasting.

Adoption context: Online agencies are more integrated than small specialists; AI often assists planning and support.

Watchpoint: Recheck prices, availability, visas and cancellation terms.

Group 799 | Other travel-related services

7990 | Other travel-related services

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: tourist guides; tourism information services; travel reservation support firms; destination assistance services.

Software: Booking systems, supplier inventories, itineraries, customer records and payments.

AI applications / candidates: Multilingual visitor assistance and content drafting.

Adoption context: Online agencies are more integrated than small specialists; AI often assists planning and support.

Watchpoint: Recheck prices, availability, visas and cancellation terms.

Division 80 | Investigation and security

Group 801 | Investigation and security

8011 | Investigation and private security

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: private investigation firms; guarding companies; cash-in-transit firms; personal security providers.

Software: Incident records, patrol schedules, alarm or access systems, dispatch and customer management.

AI applications / candidates: Incident-document review and supervised alert triage.

Adoption context: AI is generally alert or document assistance; surveillance deployment varies and is sensitive.

Watchpoint: Require lawful use, privacy safeguards and human review.

8019 | Other security activities

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: alarm monitoring firms; security-system monitoring services; other specialist security service providers.

Software: Incident records, patrol schedules, alarm or access systems, dispatch and customer management.

AI applications / candidates: Alarm prioritization and equipment-event analysis.

Adoption context: AI is generally alert or document assistance; surveillance deployment varies and is sensitive.

Watchpoint: Require lawful use, privacy safeguards and human review.

Division 81 | Building and landscape services

Group 811 | Combined facilities support

8110 | Combined facilities support

D2-D4 / A0-A2 / Estimate confidence: medium

Businesses: integrated facilities managers; bundled building-service contractors; workplace facilities support firms.

Software: Asset registers, work orders, building controls, space records and supplier management.

AI applications / candidates: Maintenance prioritization and energy anomaly alerts.

Adoption context: Large estates can connect sensors and maintenance models; smaller sites may not.

Watchpoint: Protect building control networks and occupant data.

Group 812 | Cleaning

8121 | General building cleaning

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: office cleaners; residential cleaning companies; contract janitorial firms; small cleaning businesses.

Software: Job scheduling, workforce time, quotes, customer records and billing.

AI applications / candidates: Quote assistance and staff-route planning.

Adoption context: Small firms often use basic apps; AI is mainly administrative or supplied within equipment.

Watchpoint: Check chemical, hygiene and worker-safety requirements.

8129 | Other cleaning

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: industrial cleaners; window cleaners; pest-control services; specialized sanitation contractors.

Software: Job scheduling, workforce time, quotes, customer records and billing.

AI applications / candidates: Service scheduling and inspection-image review.

Adoption context: Small firms often use basic apps; AI is mainly administrative or supplied within equipment.

Watchpoint: Check chemical, hygiene and worker-safety requirements.

Group 813 | Landscape services

8130 | Landscape services

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: landscaping companies; garden maintenance firms; tree-care contractors; grounds maintenance businesses.

Software: Scheduling, site plans, estimating, equipment records and invoicing.

AI applications / candidates: Plant-health image assistance and job estimates.

Adoption context: AI is more likely in quoting and identification than autonomous outdoor work.

Watchpoint: Confirm plant identification and local environmental conditions.

Division 82 | Office administration and business support

Group 821 | Office administration and support

8210 | Office administration and support

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: virtual assistant businesses; administrative outsourcing firms; document preparation services; shared administrative offices.

Software: Documents, spreadsheets, workflow, customer records, accounting and collaboration tools.

AI applications / candidates: Document processing and correspondence drafting.

Adoption context: Repeated digital tasks are practical AI targets; manual exceptions remain common.

Watchpoint: Protect client information and verify completed actions.

Group 822 | Call centres

8220 | Call centres

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: customer support centres; outsourced contact centres; inbound service centres; telemarketing call centres.

Software: Telephony, customer records, workforce scheduling, knowledge bases and quality monitoring.

AI applications / candidates: Agent assistance, summaries and quality review.

Adoption context: AI can be embedded in established platforms; fully automated service has narrower safe scope.

Watchpoint: Provide human escalation and prevent unsupported promises.

Group 823 | Conventions and trade shows

8230 | Conventions and trade shows

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: conference organizers; exhibition organizers; trade-fair operators; business event planners.

Software: Registration, ticketing, event websites, supplier plans, schedules and attendee records.

AI applications / candidates: Attendee matching, content drafting and demand forecasts.

Adoption context: Larger recurring events have better data; small organizers often use general tools.

Watchpoint: Protect attendee data and verify event logistics.

Group 824 | Other business-support intermediation

8240 | Other business-support intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: business-service marketplaces; administrative service brokers; specialist B2B matching agents.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Supplier matching and request classification.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

Group 829 | Other business support

8291 | Collection agencies and credit bureaus

D2-D3 / A0-A2 / Estimate confidence: low

Businesses: debt collection agencies; credit reporting bureaus; receivables recovery firms.

Software: Case management, credit records, payments, communications and dispute systems.

AI applications / candidates: Case prioritization and supervised dispute analysis.

Adoption context: AI may prioritize cases, but errors can cause serious individual harm.

Watchpoint: Keep dispute rights, fairness and human escalation.

8292 | Packaging services

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: contract packers; fulfillment packing services; labeling contractors; co-packing businesses.

Software: Customer orders, packaging specifications, inventory, labor scheduling and quality records.

AI applications / candidates: Packing-defect inspection and volume forecasting.

Adoption context: Larger lines can use inspection tools; small contract packers stay partly manual.

Watchpoint: Check labels, traceability and customer specifications.

8299 | Other business support

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: business document support firms; loyalty programme administrators; business service bureaus; other outsourced support providers.

Software: Documents, spreadsheets, workflow, customer records, accounting and collaboration tools.

AI applications / candidates: Workflow triage and document extraction.

Adoption context: Repeated digital tasks are practical AI targets; manual exceptions remain common.

Watchpoint: Protect client information and verify completed actions.

COMPLETE ACTIVITY CATALOGUE / SECTION P

P | Public administration, defence and social security

1 divisions / 3 groups / 8 classes / 28 business examples

These are public functions, not ordinary private businesses. Administration, regulation, defence, public safety and benefits have different accountability and confidentiality requirements.

Evidence scope: OECD documents government use cases. This atlas intentionally limits defence and security AI examples to administrative, documentary and support functions. [S23] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 84 | Public administration, defence and social security

Group 841 | State administration and economic, social and environmental policy

8411 | General public administration

D1-D4 / A0-A2 / Estimate confidence: medium

Businesses: national administrative departments; municipal governments; public treasury offices; general government agencies.

Software: Case management, public finance, records, identity, portals and service workflows.

AI applications / candidates: Document retrieval and citizen-service assistance.

Adoption context: Digital maturity varies substantially by agency and country; AI is often limited to selected services.

Watchpoint: Provide accountable decisions, access controls and appeal routes.

8412 | Social-sector regulation and administration

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: education regulators; health regulators; cultural service regulators; social service administration agencies.

Software: Licensing, case records, inspections, reporting portals and policy documents.

AI applications / candidates: Submission triage and policy-document search.

Adoption context: AI can assist triage and search; binding decisions should remain accountable.

Watchpoint: Check legal basis, evidence and procedural fairness.

8413 | Environmental regulation

D2-D4 / A0-A2 / Estimate confidence: medium

Businesses: environment agencies; pollution-control authorities; conservation regulators.

Software: Monitoring networks, maps, permits, inspections, laboratory results and case systems.

AI applications / candidates: Monitoring-data analysis and compliance case triage.

Adoption context: AI can support remote sensing and anomaly review, not replace enforcement evidence.

Watchpoint: Verify signals through approved investigation processes.

8414 | Economic and business regulation

D2-D3 / A0-A2 / Estimate confidence: medium

Businesses: commerce departments; trade promotion authorities; industry regulators; business licensing agencies.

Software: Licensing, case records, inspections, reporting portals and policy documents.

AI applications / candidates: Application checks and regulatory information assistance.

Adoption context: AI can assist triage and search; binding decisions should remain accountable.

Watchpoint: Check legal basis, evidence and procedural fairness.

Group 842 | Services to the community as a whole

8421 | Foreign affairs

D1-D4 / A0-A2 / Estimate confidence: medium

Businesses: foreign ministries; diplomatic missions; consular services.

Software: Case management, public finance, records, identity, portals and service workflows.

AI applications / candidates: Document translation and consular enquiry assistance.

Adoption context: Digital maturity varies substantially by agency and country; AI is often limited to selected services.

Watchpoint: Provide accountable decisions, access controls and appeal routes.

8422 | Defence administration and services

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: armed forces administration; defence ministries; military support establishments.

Software: Secure records, logistics, procurement, staffing and controlled case or asset systems.

AI applications / candidates: Administrative records, procurement review and logistics documentation only.

Adoption context: This report covers administrative AI only; operational deployment is highly sensitive and uneven.

Watchpoint: Human rights, security and accountable oversight are essential.

8423 | Public order and safety

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: courts; fire services; police administration; correctional institutions.

Software: Secure records, logistics, procurement, staffing and controlled case or asset systems.

AI applications / candidates: Records retrieval and administrative triage with oversight.

Adoption context: This report covers administrative AI only; operational deployment is highly sensitive and uneven.

Watchpoint: Human rights, security and accountable oversight are essential.

Group 843 | Compulsory social security

8430 | Compulsory social security

D2-D4 / A0-A2 / Estimate confidence: medium

Businesses: mandatory social insurance agencies; statutory pension agencies; compulsory benefit administrators.

Software: Benefit cases, identity, contributions, payments, appeals and reporting systems.

AI applications / candidates: Case-document extraction and supervised error checks.

Adoption context: AI may assist document processing; erroneous decisions affect livelihoods.

Watchpoint: Never make unreviewable automated benefit denials.

COMPLETE ACTIVITY CATALOGUE / SECTION Q

Q | Education

1 divisions / 6 groups / 12 classes / 42 business examples

Education levels and educational support are separated. Teacher or student use of a chatbot is not equivalent to an institution operating an approved AI learning system.

Evidence scope: UNESCO provides guidance rather than prevalence estimates. Funding, connectivity, language and learner age produce substantial variation. [S24] [S06] [S07]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 85 | Education

Group 851 | Pre-primary education

8510 | Pre-primary education

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: preschools; kindergartens; nursery schools; early-learning centres.

Software: Enrollment, attendance, parent communication, staff records and billing.

AI applications / candidates: Teacher preparation and administrative assistance.

Adoption context: AI is more appropriate for staff preparation than unsupervised use by young children.

Watchpoint: Protect children, consent and developmental needs.

Group 852 | Primary education

8520 | Primary education

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: primary schools; elementary schools; independent primary schools.

Software: Student records, learning platforms, attendance, timetables and parent communication.

AI applications / candidates: Lesson preparation and teacher-reviewed learning support.

Adoption context: Resources differ widely; teacher AI use is not the same as school-wide deployment.

Watchpoint: Use age-appropriate tools and teacher oversight.

Group 853 | Secondary and post-secondary non-tertiary education

8531 | General secondary education

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: secondary schools; high schools; general academic schools.

Software: Student records, learning platforms, attendance, timetables and parent communication.

AI applications / candidates: Teacher assistance and supervised learning support.

Adoption context: Resources differ widely; teacher AI use is not the same as school-wide deployment.

Watchpoint: Use age-appropriate tools and teacher oversight.

8532 | Vocational secondary education

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: vocational high schools; technical secondary schools; trade-focused secondary schools.

Software: Learning platforms, course records, scheduling, simulations and assessment tools.

AI applications / candidates: Teaching-material support and training analytics.

Adoption context: Digital theory teaching may coexist with manual workshop training.

Watchpoint: Verify technical instruction and practical competence.

8533 | Post-secondary non-tertiary education

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: post-secondary vocational institutes; non-tertiary technical programmes; bridging education providers.

Software: Learning platforms, course records, scheduling, simulations and assessment tools.

AI applications / candidates: Course preparation and learner-support assistance.

Adoption context: Digital theory teaching may coexist with manual workshop training.

Watchpoint: Verify technical instruction and practical competence.

Group 854 | Tertiary education

8540 | Tertiary education

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: universities; degree colleges; higher professional institutes; tertiary technical institutions.

Software: Student administration, learning platforms, research computing, libraries and finance systems.

AI applications / candidates: Teaching assistance, research support and administration.

Adoption context: AI use varies across staff, students, research fields and institutional policies.

Watchpoint: Protect integrity, privacy and independent learning.

Group 855 | Other education

8551 | Sports and recreation education

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: sports coaching academies; swimming schools; yoga schools; martial arts schools.

Software: Bookings, learning content, payments, attendance and learner communications.

AI applications / candidates: Scheduling support and reviewed movement feedback.

Adoption context: Online providers are more digital than individual tutors or coaches.

Watchpoint: Human instruction and valid assessment remain important.

8552 | Cultural education

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: music schools; dance schools; art schools; drama training studios.

Software: Bookings, learning content, payments, attendance and learner communications.

AI applications / candidates: Practice-material preparation and feedback assistance.

Adoption context: Online providers are more digital than individual tutors or coaches.

Watchpoint: Human instruction and valid assessment remain important.

8553 | Driving schools

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: car driving schools; commercial driver training schools; motorcycle riding schools.

Software: Bookings, learning content, payments, attendance and learner communications.

AI applications / candidates: Theory tutoring and lesson scheduling.

Adoption context: Online providers are more digital than individual tutors or coaches.

Watchpoint: Human instruction and valid assessment remain important.

8559 | Other education

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: language schools; tutoring centres; professional training firms; online course providers.

Software: Bookings, learning content, payments, attendance and learner communications.

AI applications / candidates: Tutoring assistance and course-content production.

Adoption context: Online providers are more digital than individual tutors or coaches.

Watchpoint: Human instruction and valid assessment remain important.

Group 856 | Educational support

8561 | Course and tutor intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: tutor marketplaces; course comparison services; learning-provider matching platforms.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Learner-tutor matching and search assistance.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

8569 | Other educational support

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: education consultants; student guidance services; testing support providers; educational advisory firms.

Software: Student cases, course information, application records, assessments and communications.

AI applications / candidates: Guidance-document search and administrative triage.

Adoption context: AI usually supports information retrieval and administration.

Watchpoint: Check advice and protect learner records.

COMPLETE ACTIVITY CATALOGUE / SECTION R

R | Health and social work

3 divisions / 9 groups / 11 classes / 38 business examples

Hospitals, physician practices, allied health, residential care and social support have different systems and risks. A laboratory is not a useful maturity proxy for a small physiotherapy clinic.

Evidence scope: US hospital predictive-AI evidence is strong within its scope. It must not be extrapolated to all countries, clinics or social-care providers. [S25] [S26] [S27]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 86 | Human health

Group 861 | Hospitals

8610 | Hospitals

D2-D4 / A1-A3 / Estimate confidence: medium

Businesses: general hospitals; specialist hospitals; psychiatric hospitals; surgical hospitals.

Software: Electronic health records, imaging, laboratory, pharmacy, billing and hospital operations systems.

AI applications / candidates: Imaging support, note drafting and bed-demand forecasts.

Adoption context: Digitally mature hospitals can deploy several AI tools; resource-limited hospitals may lack the foundations.

Watchpoint: Validate clinical performance and retain clinician accountability.

Group 862 | Medical and dental practices

8620 | Medical and dental practices

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: general practices; specialist clinics; dental practices; outpatient physician clinics.

Software: Patient records, appointments, billing, prescriptions and clinical documents.

AI applications / candidates: Note drafting, scheduling and reviewed clinical assistance.

Adoption context: Large groups use more integrated tools than solo clinics; AI is often limited to documentation.

Watchpoint: Protect patient data and review clinical outputs.

Group 869 | Other human health services

8691 | Health-service intermediation

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: doctor-booking marketplaces; diagnostic-service booking agents; health-provider matching services.

Software: Provider directories, availability, booking, payments and patient communications.

AI applications / candidates: Provider matching and administrative enquiry triage.

Adoption context: Platforms can use matching and support tools without delivering clinical care themselves.

Watchpoint: Do not confuse matching with diagnosis or quality guarantees.

8699 | Other human health services

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: physiotherapy clinics; diagnostic laboratories; ambulance services; allied-health practices.

Software: Appointments, patient or sample records, billing and discipline-specific clinical systems.

AI applications / candidates: Documentation, scheduling and specialist decision support.

Adoption context: Laboratories and imaging centres can be more automated than small therapy practices.

Watchpoint: Use discipline-specific validation and qualified clinical review.

Division 87 | Residential care

Group 871 | Residential nursing care

8710 | Residential nursing care

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: nursing homes; skilled nursing facilities; residential rehabilitation nursing facilities.

Software: Care plans, medication records, staffing, occupancy, billing and family communications.

AI applications / candidates: Care documentation and staffing forecasts.

Adoption context: Larger care groups have better data; AI is usually administrative or supervised monitoring.

Watchpoint: Respect dignity, consent and clinical responsibility.

Group 872 | Residential mental-health and substance-use care

8720 | Residential mental-health and substance-use care

D1-D3 / A0-A1 / Estimate confidence: low

Businesses: residential mental-health centres; substance-use rehabilitation residences; therapeutic residential communities.

Software: Restricted care records, care plans, staffing, incidents and billing systems.

AI applications / candidates: Administrative documentation with strict clinical oversight.

Adoption context: Sensitive populations call for cautious, narrow AI use rather than autonomous care.

Watchpoint: Strong consent, privacy and safeguarding are essential.

Group 873 | Residential elder and disability care

8730 | Residential elder and disability care

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: assisted-living residences; elder-care homes; residential physical-disability care facilities.

Software: Care plans, medication records, staffing, occupancy, billing and family communications.

AI applications / candidates: Staff planning and reviewed care-record summaries.

Adoption context: Larger care groups have better data; AI is usually administrative or supervised monitoring.

Watchpoint: Respect dignity, consent and clinical responsibility.

Group 879 | Other residential care

8791 | Residential-care intermediation

D2-D4 / A0-A2 / Estimate confidence: low

Businesses: care-home placement services; residential-care comparison platforms; care accommodation brokers.

Software: Provider directories, availability, booking, payments and patient communications.

AI applications / candidates: Care-provider matching and information assistance.

Adoption context: Platforms can use matching and support tools without delivering clinical care themselves.

Watchpoint: Do not confuse matching with diagnosis or quality guarantees.

8799 | Other residential care

D1-D3 / A0-A1 / Estimate confidence: low

Businesses: children's residential homes; residential shelters; other supported residential care facilities.

Software: Restricted care records, care plans, staffing, incidents and billing systems.

AI applications / candidates: Records assistance and staff scheduling.

Adoption context: Sensitive populations call for cautious, narrow AI use rather than autonomous care.

Watchpoint: Strong consent, privacy and safeguarding are essential.

Division 88 | Non-residential social work

Group 881 | Non-residential elder and disability social work

8810 | Non-residential elder and disability social work

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: home-help agencies; adult day-care centres; disability support services; elder social support organizations.

Software: Case records, visit schedules, referrals, safeguarding and grant or service billing.

AI applications / candidates: Visit scheduling and reviewed case summaries.

Adoption context: Small providers may be partly paper-based; AI is usually limited assistance.

Watchpoint: Never replace case-worker judgment or safeguarding duties.

Group 889 | Other non-residential social work

8890 | Other non-residential social work

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: child day-care providers; family support services; community welfare organizations; refugee social support providers.

Software: Case records, visit schedules, referrals, safeguarding and grant or service billing.

AI applications / candidates: Case-document organization and service navigation.

Adoption context: Small providers may be partly paper-based; AI is usually limited assistance.

Watchpoint: Never replace case-worker judgment or safeguarding duties.

COMPLETE ACTIVITY CATALOGUE / SECTION S

S | Arts, sports and recreation

4 divisions / 10 groups / 19 classes / 72 business examples

Independent creators, cultural institutions, elite sports and local recreation occupy very different digital environments. AI should not erase distinctions between original evidence, creative work and synthetic output.

Evidence scope: UNESCO-convened experts and the IOC provide selected sector context. Much local recreation and gambling-class maturity remains an estimate. [S18] [S28] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 90 | Arts creation and performing arts

Group 901 | Arts creation

9011 | Literary creation and musical composition

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: independent writers; composers; lyricists; freelance scriptwriters.

Software: Writing or art tools, portfolios, rights records, online sales and project administration.

AI applications / candidates: Draft ideation and research assistance.

Adoption context: Some creators embrace AI; others deliberately avoid it. Handmade practice remains valid.

Watchpoint: Respect creative ownership and client disclosure expectations.

9012 | Visual arts creation

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: painters; sculptors; independent illustrators; visual artists.

Software: Writing or art tools, portfolios, rights records, online sales and project administration.

AI applications / candidates: Concept development and digital image assistance.

Adoption context: Some creators embrace AI; others deliberately avoid it. Handmade practice remains valid.

Watchpoint: Respect creative ownership and client disclosure expectations.

9013 | Other arts creation

D0-D3 / A0-A2 / Estimate confidence: medium

Businesses: independent multidisciplinary artists; multimedia creators; other artistic creators.

Software: Writing or art tools, portfolios, rights records, online sales and project administration.

AI applications / candidates: Ideation and production-document assistance.

Adoption context: Some creators embrace AI; others deliberately avoid it. Handmade practice remains valid.

Watchpoint: Respect creative ownership and client disclosure expectations.

Group 902 | Performing arts

9020 | Performing arts

D0-D3 / A0-A2 / Estimate confidence: low

Businesses: theatre companies; musicians; dance companies; independent live performers.
Software: Bookings, promotion, rehearsals, ticketing interfaces and rights records.
AI applications / candidates: Promotion, rehearsal administration and captioning.
Adoption context: Administrative AI is easier to adopt than replacing live performance.
Watchpoint: Protect performer consent and creative rights.

Group 903 | Support to arts creation and performing arts

9031 | Arts facilities and sites

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: theatres; concert halls; performing arts venues; arts-centre operators.
Software: Ticketing or reservations, payments, membership, staffing and event schedules.
AI applications / candidates: Audience forecasts and venue-service assistance.
Adoption context: Chains and online-first venues integrate more data than independent sites.
Watchpoint: Keep customer and event information accurate.

9039 | Other support to arts creation and performance

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: stage production services; performing arts production firms; live show technical support businesses.
Software: Production schedules, stage plans, crew rosters, equipment and supplier systems.
AI applications / candidates: Production schedules and technical-document assistance.
Adoption context: Technical production is increasingly digital; AI is usually a supporting tool.
Watchpoint: On-site safety must remain independently supervised.

Division 91 | Libraries, archives, museums and cultural activities

Group 911 | Libraries and archives

9111 | Libraries

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: public libraries; academic libraries; specialist libraries; subscription libraries.
Software: Catalogues, circulation, digital repositories, discovery and user-service systems.
AI applications / candidates: Catalogue enrichment and discovery assistance.
Adoption context: Large institutions can improve discovery with AI; small libraries may lack digitized collections.
Watchpoint: Show source provenance and protect reader privacy.

9112 | Archives

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: public archives; business archives; historical document archives; specialist record archives.
Software: Archival catalogues, scanning, preservation storage, metadata and access systems.
AI applications / candidates: Handwriting recognition, metadata and document discovery.
Adoption context: AI is useful only where records can be lawfully digitized and checked.
Watchpoint: Retain original records and label uncertain transcription.

Group 912 | Museums, collections, historical sites and monuments

9121 | Museums and collections

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: art museums; science museums; private collections open to visitors; specialist museums.

Software: Collection management, ticketing, digital exhibits, archives and conservation records.

AI applications / candidates: Collection tagging and visitor information support.

Adoption context: Digitized collections support discovery; smaller sites may use minimal software.

Watchpoint: Protect provenance, community rights and historical accuracy.

9122 | Historical sites and monuments

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: historic house operators; heritage-site operators; monument visitor services.

Software: Collection management, ticketing, digital exhibits, archives and conservation records.

AI applications / candidates: Visitor assistance and heritage-document search.

Adoption context: Digitized collections support discovery; smaller sites may use minimal software.

Watchpoint: Protect provenance, community rights and historical accuracy.

Group 913 | Cultural heritage conservation and restoration

9130 | Cultural heritage conservation and restoration

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: art conservators; heritage restoration studios; conservation laboratories; collection-preservation businesses.

Software: Condition records, images, laboratory analysis, projects and collection documents.

AI applications / candidates: Condition-image comparison and documentation.

Adoption context: AI can compare images and organize records; physical restoration remains specialist work.

Watchpoint: Do not treat generated reconstruction as historical evidence.

Group 914 | Botanical and zoological gardens and nature reserves

9141 | Botanical and zoological gardens

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: zoos; botanical gardens; aquariums; wildlife display parks.

Software: Animal or plant records, care schedules, ticketing, research and maintenance tools.

AI applications / candidates: Animal or plant monitoring and visitor forecasts.

Adoption context: AI monitoring is selective and depends on suitable sensors and labeled data.

Watchpoint: Keep animal welfare and expert review central.

9142 | Nature reserves

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: nature reserve operators; conservation park managers; protected-area visitor operators.

Software: Habitat maps, ranger records, visitor permits, ecological databases and remote sensing.

AI applications / candidates: Habitat imagery and biodiversity monitoring.

Adoption context: Larger programmes can use image analysis; local sites may rely on basic tools.

Watchpoint: Protect sensitive species locations and community rights.

Division 92 | Gambling and betting

Group 920 | Gambling and betting

9200 | Gambling and betting

D2-D4 / A1-A3 / Estimate confidence: low

Businesses: licensed casinos; licensed betting operators; lottery operators; regulated online gambling businesses.

Software: Account or venue systems, payments, risk, compliance and game operations.

AI applications / candidates: Fraud detection and responsible-gambling risk review.

Adoption context: Digital operators may use sophisticated models; small licensed venues vary.

Watchpoint: Prioritize legal compliance and harm prevention, not exploitation.

Division 93 | Sports, amusement and recreation

Group 931 | Sports

9311 | Sports facilities

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: gyms; sports stadium operators; swimming pool operators; golf-course operators.

Software: Membership, bookings, access, payments, coaching and facility maintenance tools.

AI applications / candidates: Demand forecasts and facility maintenance alerts.

Adoption context: Chains are more integrated than independent facilities; AI often enters through vendor apps.

Watchpoint: Health and injury advice requires qualified supervision.

9312 | Sports clubs

D1-D4 / A0-A3 / Estimate confidence: medium

Businesses: professional sports clubs; amateur member clubs; team sports organizations; competitive sports clubs.

Software: Membership, ticketing, coaching records, video analysis and athlete management.

AI applications / candidates: Performance analysis and member-service assistance.

Adoption context: Elite clubs can use advanced analytics; community clubs often use spreadsheets.

Watchpoint: Respect athlete privacy and limits of performance prediction.

9319 | Other sports activities

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: sports event promoters; independent sports officials; sports league services; other sports support businesses.

Software: Event systems, video records, scheduling, scoring support and commercial administration.

AI applications / candidates: Video tagging and event-operation assistance.

Adoption context: AI differs sharply between elite competition and small local events.

Watchpoint: Maintain review and appeal for consequential judgments.

Group 932 | Amusement and recreation

9321 | Amusement and theme parks

D2-D4 / A0-A3 / Estimate confidence: low

Businesses: theme parks; amusement parks; water parks; attraction parks.

Software: Ticketing, queue systems, rides maintenance, staffing, retail and visitor analytics.

AI applications / candidates: Queue forecasts and ride-maintenance alerts.

Adoption context: Large parks have connected assets; smaller attractions have simpler systems.

Watchpoint: Keep ride safety independent of unvalidated AI.

9329 | Other amusement and recreation

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: escape rooms; recreation centres; leisure activity operators; other entertainment venues.

Software: Ticketing or reservations, payments, membership, staffing and event schedules.

AI applications / candidates: Booking forecasts and customer enquiry assistance.

Adoption context: Chains and online-first venues integrate more data than independent sites.

Watchpoint: Keep customer and event information accurate.

COMPLETE ACTIVITY CATALOGUE / SECTION T

T | Membership, repair and personal services

3 divisions / 12 groups / 22 classes / 77 business examples

This sector includes associations, device and vehicle repair, laundry, hair and beauty, funerals and other personal services. Many are microbusinesses where simple scheduling matters more than sophisticated AI.

Evidence scope: SME research supports the scale caveat; detailed personal-service bands rely mainly on transparent workflow inference, not global survey evidence. [S07] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 94 | Membership organizations

Group 941 | Business, employer and professional organizations

9411 | Business and employer organizations

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: chambers of commerce; employer associations; industry trade associations.

Software: Membership, dues, events, communications, case files and financial records.

AI applications / candidates: Member-service assistance and policy-document summaries.

Adoption context: Large associations may deploy assistants; volunteer groups often use basic tools.

Watchpoint: Protect sensitive membership data and verify communications.

9412 | Professional membership organizations

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: professional institutes; specialist professional associations; chartered professional bodies.

Software: Membership, dues, events, communications, case files and financial records.

AI applications / candidates: Member support and knowledge-library search.

Adoption context: Large associations may deploy assistants; volunteer groups often use basic tools.

Watchpoint: Protect sensitive membership data and verify communications.

Group 942 | Trade unions

9420 | Trade unions

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: labor unions; worker federations; sectoral trade unions.

Software: Membership, dues, events, communications, case files and financial records.

AI applications / candidates: Case-document search and member-service assistance.

Adoption context: Large associations may deploy assistants; volunteer groups often use basic tools.

Watchpoint: Protect sensitive membership data and verify communications.

Group 949 | Other membership organizations

9491 | Religious organizations

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: congregations; religious associations; faith-based organizations with religious primary activity.
Software: Membership, dues, events, communications, case files and financial records.
AI applications / candidates: Administrative support and archive search.
Adoption context: Large associations may deploy assistants; volunteer groups often use basic tools.
Watchpoint: Protect sensitive membership data and verify communications.

9492 | Political organizations

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: political parties; political membership organizations; party administrative bodies.
Software: Membership administration, events, public information and financial records.
AI applications / candidates: Administrative records and public-information assistance.
Adoption context: This profile concerns routine administration, not persuasion or targeting.
Watchpoint: Protect political privacy and avoid deceptive content.

9499 | Other membership organizations

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: community associations; civic clubs; hobby associations; membership-based advocacy groups.
Software: Membership, dues, events, communications, case files and financial records.
AI applications / candidates: Member communication and document search.
Adoption context: Large associations may deploy assistants; volunteer groups often use basic tools.
Watchpoint: Protect sensitive membership data and verify communications.

Division 95 | Repair of devices, household goods and motor vehicles

Group 951 | Computer and communications-equipment repair

9510 | Computer and communications-equipment repair

D1-D3 / A0-A2 / Estimate confidence: low

Businesses: computer repair shops; phone repair businesses; network equipment repairers; device refurbishment services.
Software: Repair tickets, diagnostics, parts inventory, service manuals and billing.
AI applications / candidates: Fault lookup and parts matching.
Adoption context: AI assists knowledge search; technicians still perform diagnosis and physical repairs.
Watchpoint: Secure customer devices and never expose personal files.

Group 952 | Personal and household-goods repair

9521 | Consumer electronics repair

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: television repairers; audio equipment repair shops; consumer video-equipment service firms.
Software: Work orders, manuals, parts, scheduling, customer records and invoices.
AI applications / candidates: Fault-record search and quotation assistance.
Adoption context: Small repair shops may rely on messaging; AI is mainly occasional lookup assistance.
Watchpoint: Verify safe procedures and compatible parts.

9522 | Home appliances and garden-equipment repair

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: appliance repair businesses; lawnmower repair shops; household equipment service firms.

Software: Work orders, manuals, parts, scheduling, customer records and invoices.

AI applications / candidates: Fault lookup and job triage.

Adoption context: Small repair shops may rely on messaging; AI is mainly occasional lookup assistance.

Watchpoint: Verify safe procedures and compatible parts.

9523 | Footwear and leather-goods repair

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: cobblers; shoe repair shops; luggage repairers; leather-goods repair workshops.

Software: Job records, customer messages, payments and simple inventory or design tools.

AI applications / candidates: Quote drafting and customer-message assistance.

Adoption context: Hands-on skill dominates; dedicated AI deployment is usually limited.

Watchpoint: Preserve the original object and customer instructions.

9524 | Furniture and home-furnishing repair

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: furniture repairers; upholstery restoration businesses; home furnishing repair workshops.

Software: Job records, customer messages, payments and simple inventory or design tools.

AI applications / candidates: Design visualization and job estimation.

Adoption context: Hands-on skill dominates; dedicated AI deployment is usually limited.

Watchpoint: Preserve the original object and customer instructions.

9529 | Other personal and household-goods repair

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: watch repairers; bicycle repair shops; clothing alteration shops; musical-instrument repairers.

Software: Job records, customer messages, payments and simple inventory or design tools.

AI applications / candidates: Fault-reference search and quotation assistance.

Adoption context: Hands-on skill dominates; dedicated AI deployment is usually limited.

Watchpoint: Preserve the original object and customer instructions.

Group 953 | Motor-vehicle and motorcycle repair

9531 | Motor-vehicle repair and maintenance

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: auto repair garages; body shops; vehicle detailing and washing businesses; specialist vehicle service centres.

Software: Workshop management, vehicle diagnostics, parts catalogues, estimating and service records.

AI applications / candidates: Diagnostic assistance and damage-estimate support.

Adoption context: Dealer workshops are more digital than small garages; AI supports rather than proves diagnosis.

Watchpoint: Qualified technicians must verify safety-related repairs.

9532 | Motorcycle repair and maintenance

D1-D3 / A0-A2 / Estimate confidence: medium

Businesses: motorcycle workshops; scooter repairers; motorcycle service centres.

Software: Workshop management, vehicle diagnostics, parts catalogues, estimating and service records.

AI applications / candidates: Fault lookup and service scheduling.

Adoption context: Dealer workshops are more digital than small garages; AI supports rather than proves diagnosis.

Watchpoint: Qualified technicians must verify safety-related repairs.

Group 954 | Repair and maintenance intermediation

9540 | Repair and maintenance intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: repair booking platforms; garage comparison services; device-repair marketplaces.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Provider matching and service-request triage.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

Division 96 | Personal services

Group 961 | Textile and fur washing and cleaning

9610 | Textile and fur washing and cleaning

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: laundries; dry cleaners; linen rental and laundering services; laundromats.

Software: Order tracking, garment tags, machine or route records, payments and customer communications.

AI applications / candidates: Demand forecasts and garment recognition.

Adoption context: Industrial laundries use more process software than small laundromats.

Watchpoint: Check garment care, hygiene and handling instructions.

Group 962 | Hairdressing, beauty care and day spas

9621 | Hairdressing and barbering

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: hair salons; barber shops; mobile hairdressers; hair-styling studios.

Software: Appointment calendars, payments, client notes, inventory and marketing tools.

AI applications / candidates: Booking assistance and style visualization.

Adoption context: Chains and app-based salons are more digital than independent workers.

Watchpoint: Protect client images and distinguish cosmetic from medical claims.

9622 | Beauty care and treatments

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: beauty salons; nail salons; non-medical skincare studios; makeup artists.

Software: Appointment calendars, payments, client notes, inventory and marketing tools.

AI applications / candidates: Appointment support and cosmetic visualization.

Adoption context: Chains and app-based salons are more digital than independent workers.

Watchpoint: Protect client images and distinguish cosmetic from medical claims.

9623 | Day spas, saunas and steam baths

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: day spas; sauna businesses; steam baths; non-medical wellness spas.

Software: Appointment calendars, payments, client notes, inventory and marketing tools.

AI applications / candidates: Booking support and staff-demand forecasts.

Adoption context: Chains and app-based salons are more digital than independent workers.

Watchpoint: Protect client images and distinguish cosmetic from medical claims.

Group 963 | Funeral and related services

9630 | Funeral and related services

D1-D3 / A0-A1 / Estimate confidence: low

Businesses: funeral directors; crematoria; cemetery operators; mortuary service businesses.

Software: Case records, scheduling, permits, inventories, family communications and billing.

AI applications / candidates: Paperwork assistance and scheduling.

Adoption context: Software mainly handles coordination; AI should remain a carefully reviewed administrative aid.

Watchpoint: Respect grief, cultural practices and sensitive records.

Group 964 | Personal-service intermediation

9640 | Personal-service intermediation

D3-D4 / A1-A3 / Estimate confidence: medium

Businesses: salon booking platforms; personal-service marketplaces; beauty-service booking agents.

Software: Search, listings, matching, booking, payments, customer support and seller-management systems.

AI applications / candidates: Provider matching and booking assistance.

Adoption context: Digital platforms may embed ranking and fraud models; offline agents can be much less digital.

Watchpoint: Explain rankings; protect payments, identity and sensitive data.

Group 969 | Other personal services

9690 | Other personal services

D0-D3 / A0-A1 / Estimate confidence: low

Businesses: pet grooming businesses; pet boarding businesses; tattoo studios; personal concierge services.

Software: Bookings, customer records, payments, messaging and service-specific forms.

AI applications / candidates: Scheduling, customer messages and record organization.

Adoption context: This broad class includes many hands-on microbusinesses; AI is usually general assistance.

Watchpoint: Apply service-specific safety, welfare and privacy checks.

COMPLETE ACTIVITY CATALOGUE / SECTION U

U | Household employment and own-use production

2 divisions / 3 groups / 3 classes / 11 business examples

Included for classification completeness. Households employing domestic workers and households producing for their own use are not ordinary commercial businesses.

Evidence scope: These entries describe optional household tools. Their scores must not be included in an enterprise AI-adoption calculation. [S01] [S06]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 97 | Households employing domestic personnel

Group 970 | Households employing domestic personnel

9700 | Households employing domestic personnel

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: households employing cleaners; households employing nannies; households employing cooks; households employing gardeners.

Software: Optional payroll, payment, calendar and household task tools.

AI applications / candidates: Optional household administration assistance.

Adoption context: A private household is not a business; tool use may be informal or absent.

Watchpoint: Respect domestic workers' rights and personal privacy.

Division 98 | Undifferentiated household production for own use

Group 981 | Household goods production for own use

9810 | Household goods production for own use

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: mixed subsistence household production; undifferentiated household goods-making; own-use production households.

Software: Optional household budgeting, messaging, calendars and information apps.

AI applications / candidates: Occasional information assistance, not a business deployment.

Adoption context: These are own-use activities, not commercial businesses; no business adoption rate is implied.

Watchpoint: Do not count personal AI use as enterprise adoption.

Group 982 | Household services production for own use

9820 | Household services production for own use

D0-D2 / A0-A1 / Estimate confidence: low

Businesses: own-household cooking; own-household cleaning; own-household care; undifferentiated household services.

Software: Optional household budgeting, messaging, calendars and information apps.

AI applications / candidates: Optional planning assistance, not a business deployment.

Adoption context: These are own-use activities, not commercial businesses; no business adoption rate is implied.

Watchpoint: Do not count personal AI use as enterprise adoption.

COMPLETE ACTIVITY CATALOGUE / SECTION V

V | Extraterritorial organizations and bodies

1 divisions / 1 groups / 1 classes / 4 business examples

International bodies carry out programmes, finance, research and coordination across jurisdictions. Institutional and field-office technology can differ substantially.

Evidence scope: UNHCR provides a primary humanitarian example. It is not representative evidence for every international organization. [S29] [S23] [S30]

All maturity bands below are analyst estimates. Business examples inherit their parent profile unless a focused subtype entry gives a refinement. AI lines identify applications or candidates, not universal deployment.

Division 99 | Extraterritorial organizations and bodies

Group 990 | Extraterritorial organizations and bodies

9900 | Extraterritorial organizations and bodies

D2-D4 / A0-A2 / Estimate confidence: medium

Businesses: United Nations bodies; international financial institutions; regional intergovernmental organizations; qualifying international organizations.

Software: Programme management, grants, procurement, records, logistics and multilingual information systems.

AI applications / candidates: Multilingual document search and programme-data analysis.

Adoption context: Large bodies may pilot or deploy assistants; field-office capability varies.

Watchpoint: Human rights, sensitive data and humanitarian accountability matter.

APPENDIX A / FIND THE LABEL YOU KNOW

Modern and cross-cutting ecosystem finder

Many familiar "industries" are ecosystems rather than a single classification branch. The routes below are illustrative, not exhaustive legal crosswalks. Classify what the business actually does: research, manufacture, software publishing, intermediation or direct service delivery. [S01] [S34]

Ecosystem	Starting class codes	How to read the route
Artificial intelligence	5829, 6219, 6310, 7210	AI software publishers, custom AI developers, computing infrastructure and research.
Robotics and automation	2819, 2829, 3320, 6219, 7210	Equipment, specialist machinery, industrial installation, software and R&D; exact machine purpose matters.
Cybersecurity	5829, 6220, 6290	Security software, managed technical services and related IT activity; non-IT security is separate.
Semiconductors and chips	2619, 2829, 7210	Electronic components, specialized manufacturing equipment and research.
Quantum technology	7210, 2619, 2620, 6219	Research, components, computing hardware and custom software according to output.
Cloud and data infrastructure	6310, 6110, 6220	Hosting and computing resources, telecom connectivity and managed IT.
Fintech and payments	5829, 6219, 6419, 6499, 6619	A software vendor is different from a regulated financial provider or auxiliary service.
Insurance technology	5829, 6511, 6512, 6621, 6622, 6629	Insurance software, insurers, loss assessment, broking and auxiliary insurance work.
Property technology	5829, 6810, 6821, 6829	Property software, ownership, brokerage and fee-based property activity.
Construction technology	5829, 7110, 4100, 4390	Software, architecture/engineering, construction and specialist execution; classify each separately.
Agricultural technology	5829, 2821, 0161, 7210	Farm software, machinery, support services and research; growers remain in division 01.
Food technology	7210, 5829, 1075, 5610	Research, software, prepared-food production and foodservice are separate activities.
Health technology	5829, 2660, 3250, 8699	Software, medical electronics, supplies and care delivery; medical-device status is a separate issue.
Biotechnology and life sciences	7210, 2100, 8699	R&D, pharmaceutical manufacture and selected clinical/laboratory services.
Education technology	5829, 6219, 8559, 8569	Software and development differ from teaching and educational support.
Climate technology	7210, 3512, 3830, 7499	Research, renewable power, material recovery and environmental advice; not one industry.

Ecosystem	Starting class codes	How to read the route
Renewables and energy storage	3512, 2611, 2720, 2710, 4321	Power generation, batteries, electrical equipment and electrical installation.
Electric mobility	2910, 2920, 2930, 3513, 9531	Vehicles, bodies, parts, charging operation and repair.
Circular economy	3830, 4774, 9510, 9529, 9531	Material recovery, second-hand retail and repair; reuse is broader than recycling.
Water technology	2813, 3600, 3700, 7110	Pumps, water supply, sewerage and engineering services.
Space economy	3030, 5120, 6110, 7210	Spacecraft, commercial cargo space transport, satellite telecom and R&D.
Ocean economy	0311, 0321, 3011, 5012, 5222	Marine fishing, aquaculture, shipbuilding, shipping and ports; many more coastal services connect.
Travel and tourism	5510, 5520, 5110, 7911, 7912, 7990	Accommodation, transport, agencies, tour operators and reservation services.
Creator economy	5911, 5920, 6039, 7310, 9011	Video, audio, distribution platforms, advertising and creative writing/composition.
Video games and esports	5821, 6211, 6020, 9319	Publishing, game development, broadcasting and sporting/event activity as applicable.
Beauty and wellness	2023, 4772, 9621, 9622, 9623	Product manufacture/retail, hair, beauty and physical-wellness services; clinical care is separate.
Pet economy	1080, 4773, 7500, 9690	Pet food, retail, veterinary care and non-veterinary pet services.
Senior and care economy	8710, 8730, 8810, 8699	Nursing facilities, residential support, non-residential social care and health services.
Sports technology	5829, 3230, 9311, 9312, 9319	Software, sporting goods, facilities, clubs and other sports services.
Defence and security economy	2520, 3040, 8422, 8011	Regulated manufacturing, public defence and private security; only support AI is profiled here.
Geospatial and mapping services	7110, 5829, 6310, 7210	Surveying/engineering, software, infrastructure and R&D; product and service boundaries matter.
Advertising and marketing technology	7310, 7320, 7330, 5829	Advertising, market research, public relations and software publishing.
Legal and regulatory technology	6910, 6920, 5829	Professional legal/accounting services versus the software tools they use.
Human-resources technology	5829, 7810, 7820	Workforce software, employment placement and temporary employment provision.
On-demand and sharing platforms	4790, 5232, 5540, 5640, 7751, 7752	The category follows the intermediated service; an asset-owning operator may classify elsewhere.

Ecosystem	Starting class codes	How to read the route
Nonprofits and social enterprises	9499, 8890, 8559, 8699	Legal purpose does not determine activity: membership, social support, teaching and care differ.
Events, conventions and experiences	8230, 5621, 9039, 9329	Event organization, catering, arts support and recreational services.
Green hydrogen and industrial gases	2011, 3520, 4730, 7210	Manufacture, utility supply, vehicle-fuel retail and research depend on the actual operation.

Examples of boundaries checked against official notes: EV charging operation is in 3513 rather than automotive fuel retail 4730; spacecraft manufacture and cargo space transport are separate; environmental advisory work is not automatically engineering; commercial art galleries sell art rather than operate museums. Confirm the exact primary activity before using any code for registration or reporting. [S01] [S03]

APPENDIX B / 72 FINER BUSINESS LENSES

Focused business subtype profiles

These entries expand broad official classes into recognizable niches. They are editorial subtypes, not 72 new ISIC classes, and some overlap with the main examples. All bands here are **low-confidence analyst refinements**, not measured adoption rates. The linked parent class supplies context and sector evidence.

01 | Accounting and ERP software publishers

Parent class 5829 / D4 / A1-A3 / Estimate confidence: low

Software: Ledgers, purchasing, inventory, reporting, implementation and subscription operations.

AI applications / candidates: Invoice extraction, reconciliation suggestions and business-document assistants.

Boundary / watchpoint: Software publishing is distinct from operating an accounting practice.

02 | Human-resources and payroll software publishers

Parent class 5829 / D4 / A1-A3 / Estimate confidence: low

Software: Employee records, payroll rules, attendance, identity controls and application operations.

AI applications / candidates: Employee support, document processing and workforce analysis.

Boundary / watchpoint: Do not treat model output as authoritative payroll or employment law.

03 | Customer-service and CRM software publishers

Parent class 5829 / D4 / A1-A4 / Estimate confidence: low

Software: Customer databases, communication channels, workflow rules and support operations.

AI applications / candidates: Conversation assistance, ticket routing and controlled task completion.

Boundary / watchpoint: A vendor may offer AI even when customers do not enable it.

04 | Cybersecurity software publishers

Parent class 5829 / D4 / A1-A4 / Estimate confidence: low

Software: Security telemetry, detection rules, threat analysis, cloud delivery and incident tools.

AI applications / candidates: Alert correlation, anomaly detection and analyst assistance.

Boundary / watchpoint: False alerts and privileged automated actions require controls.

05 | Education software publishers

Parent class 5829 / D4 / A1-A4 / Estimate confidence: low

Software: Learning content, assessments, student integrations, subscriptions and product operations.

AI applications / candidates: Tutoring support, lesson generation and reviewed feedback.

Boundary / watchpoint: The publisher is not necessarily an education provider under section Q.

06 | Clinical and practice-management software publishers

Parent class 5829 / D4 / A1-A3 / Estimate confidence: low

Software: Patient workflows, interoperability, appointments, billing and controlled software delivery.

AI applications / candidates: Documentation, scheduling assistance and validated clinical modules.

Boundary / watchpoint: A software vendor is distinct from the clinical service operator.

07 | Supply-chain and logistics software publishers

Parent class 5829 / D4 / A1-A3 / Estimate confidence: low

Software: Shipment, inventory and planning products, customer integrations and application operations.

AI applications / candidates: Demand forecasts, exception detection and planning assistance.

Boundary / watchpoint: Traditional optimization and rules are not automatically AI.

08 | Vertical business SaaS publishers

Parent class 5829 / D4 / A1-A3 / Estimate confidence: low

Software: Industry-specific records, workflow, billing, integrations and cloud application delivery.

AI applications / candidates: Domain-document search and controlled workflow assistance.

Boundary / watchpoint: Examples include software for salons, farms, clinics and property managers.

09 | Design, engineering and simulation software publishers

Parent class 5829 / D4 / A1-A3 / Estimate confidence: low

Software: Design engines, simulation tools, customer licensing, rendering and collaboration platforms.

AI applications / candidates: Design assistance, surrogate modeling and document checking.

Boundary / watchpoint: Engineering outputs still require qualified validation.

10 | AI assistant and agent software publishers

Parent class 5829 / D4 / A3-A4 / Estimate confidence: low

Software: Model access, orchestration, evaluations, permissions, usage billing and observability.

AI applications / candidates: AI is part of the product being sold rather than merely an internal aid.

Boundary / watchpoint: Agent permissions and verifiable task completion are central.

11 | Custom AI application developers

Parent class 6219 / D3-D4 / A2-A3 / Estimate confidence: low

Software: Client code, model integration, retrieval systems, testing, deployment and documentation.

AI applications / candidates: Coding assistance and building client-specific AI workflows.

Boundary / watchpoint: Use separate controls for the developer's own AI and the delivered product.

12 | Web and mobile application agencies

Parent class 6219 / D3-D4 / A1-A3 / Estimate confidence: low

Software: Design, source control, build pipelines, testing, deployment and client project management.

AI applications / candidates: Code generation, tests, content drafts and debugging.

Boundary / watchpoint: Human review is needed for security, accessibility and acceptance criteria.

13 | Business automation and integration developers

Parent class 6219 / D3-D4 / A1-A3 / Estimate confidence: low

Software: APIs, workflow engines, connectors, event logs, testing and deployment tools.

AI applications / candidates: Document interpretation and bounded agentic workflow steps.

Boundary / watchpoint: Conventional integration scripts and robotic process automation are not necessarily AI.

14 | Cloud and infrastructure managed-service providers

Parent class 6220 / D3-D4 / A1-A3 / Estimate confidence: low

Software: Monitoring, service desks, configuration, security, billing and customer access controls.

AI applications / candidates: Incident summarization and operational knowledge retrieval.

Boundary / watchpoint: Do not allow unapproved infrastructure changes.

15 | Cybersecurity consulting and managed IT security

Parent class 6220 / D3-D4 / A1-A3 / Estimate confidence: low

Software: Security monitoring, incident response, customer systems and case records.

AI applications / candidates: Detection support, alert triage and investigation assistance.

Boundary / watchpoint: Classification follows the actual IT service; non-IT guarding belongs elsewhere.

16 | Cloud infrastructure and compute providers

Parent class 6310 / D4 / A1-A3 / Estimate confidence: low

Software: Compute, storage, networking, orchestration, metering, security and customer operations.

AI applications / candidates: Capacity forecasts and infrastructure anomaly detection.

Boundary / watchpoint: Providing compute for AI does not prove AI runs the provider's own operations.

17 | Data-centre operators and colocation providers

Parent class 6310 / D3-D4 / A0-A2 / Estimate confidence: low

Software: Power and cooling management, racks, access controls, maintenance and customer billing.

AI applications / candidates: Cooling-demand forecasts and equipment failure alerts.

Boundary / watchpoint: Digital building controls alone are not AI.

18 | Hosting and data-processing bureaus

Parent class 6310 / D3-D4 / A0-A2 / Estimate confidence: low

Software: Hosting platforms, job processing, data storage, customer support and billing.

AI applications / candidates: Document processing and service-ticket assistance.

Boundary / watchpoint: Software publishing and hosting are different principal activities.

19 | Search and information-discovery businesses

Parent class 6390 / D4 / A2-A4 / Estimate confidence: low

Software: Indexing, retrieval, crawling, databases, information rights and query serving.

AI applications / candidates: Semantic retrieval and source-linked answer generation.

Boundary / watchpoint: Generated answers require provenance and evaluation.

20 | AI and machine-learning research organizations

Parent class 7210 / D3-D4 / A2-A4 / Estimate confidence: low

Software: Scientific computing, datasets, experiments, evaluation and research publication workflows.

AI applications / candidates: AI methods are part of the research activity itself.

Boundary / watchpoint: Commercial software publishing is a different activity from R&D.

21 | Biotechnology R&D organizations

Parent class 7210 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Laboratory records, sequencing or assay data, scientific computing and research governance.

AI applications / candidates: Molecule or biological-data analysis and experiment-support models.

Boundary / watchpoint: Research hypotheses are not clinical evidence or approved products.

22 | Materials and advanced-manufacturing R&D

Parent class 7210 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Materials databases, simulations, instruments and experiment tracking.

AI applications / candidates: Property prediction and experiment prioritization.

Boundary / watchpoint: Experimental validation is still required.

23 | Quantum and physical-science R&D

Parent class 7210 / D3-D4 / A1-A3 / Estimate confidence: low

Software: Scientific instruments, simulations, laboratory records and research code.

AI applications / candidates: Data analysis, control calibration support and research assistance.

Boundary / watchpoint: Quantum technology is not a separate universal industry code.

24 | Physiotherapy and rehabilitation practices

Parent class 8699 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Patient records, recurring appointments, treatment plans, billing and home-exercise systems.

AI applications / candidates: Clinical note drafting and reviewed movement-analysis support.

Boundary / watchpoint: Most clinics need sound scheduling and records before advanced AI.

25 | Occupational therapy practices

Parent class 8699 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Care plans, functional assessments, appointments and patient documentation.

AI applications / candidates: Report drafts and reviewed activity-assessment support.

Boundary / watchpoint: Clinical judgment and individualized care remain central.

26 | Speech and language therapy practices

Parent class 8699 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Assessment records, therapy sessions, audio exercises and billing.

AI applications / candidates: Speech-analysis assistance and practice-material preparation.

Boundary / watchpoint: Validate for the patient's language, age and condition.

27 | Psychology and non-physician therapy practices

Parent class 8699 / D1-D3 / A0-A1 / Estimate confidence: low

Software: Confidential notes, appointments, consent and billing systems.

AI applications / candidates: Administrative support and carefully reviewed note assistance.

Boundary / watchpoint: Psychiatrist medical practice instead belongs under 8620; privacy is critical.

28 | Clinical diagnostic laboratories

Parent class 8699 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Laboratory information, sample tracking, instruments, quality controls and reporting.

AI applications / candidates: Anomaly checks and validated analytical interpretation support.

Boundary / watchpoint: Do not release unvalidated test interpretations.

29 | Independent diagnostic imaging services

Parent class 8699 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Imaging archives, radiology workflow, bookings and diagnostic reporting.

AI applications / candidates: Validated image-analysis assistance and reporting support.

Boundary / watchpoint: The operator's principal activity and professional structure affect classification.

30 | Ambulance and patient transport services with medical care

Parent class 8699 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Dispatch, fleet tracking, clinical handover records and billing.

AI applications / candidates: Demand forecasts and dispatch decision support.

Boundary / watchpoint: Emergency care decisions require qualified personnel.

31 | Independent nursing and midwifery services

Parent class 8699 / D1-D3 / A0-A1 / Estimate confidence: low

Software: Care records, visit schedules, medication documentation and billing.

AI applications / candidates: Visit planning and reviewed documentation.

Boundary / watchpoint: Pure social support without clinical care belongs under social work.

32 | Dietetic and nutrition care practices

Parent class 8699 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Consultation records, assessments, meal-planning tools and appointments.

AI applications / candidates: Documentation and individualized educational-content assistance.

Boundary / watchpoint: Do not automate clinical nutrition advice without qualified review.

33 | Dental clinics and specialist dentistry

Parent class 8620 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Dental records, radiographs, appointments, treatment plans and billing.

AI applications / candidates: Image-review assistance and documentation.

Boundary / watchpoint: Validated clinical performance is different from a convincing visualization.

34 | General and specialist physician practices

Parent class 8620 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Clinical records, appointments, prescriptions, results and billing.

AI applications / candidates: Note drafting and reviewed decision support.

Boundary / watchpoint: The responsible clinician must verify medical outputs.

35 | Small-animal veterinary clinics

Parent class 7500 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Animal records, appointments, imaging, pharmacy and billing.

AI applications / candidates: Note preparation and reviewed image-analysis assistance.

Boundary / watchpoint: Do not substitute human medical models without veterinary validation.

36 | Farm and mobile veterinary practices

Parent class 7500 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Mobile clinical records, herd histories, visits and medicine inventories.

AI applications / candidates: Record summaries and herd-health alert interpretation.

Boundary / watchpoint: Offline access and field conditions may limit deployment.

37 | Florists and plant retailers

Parent class 4773 / D0-D3 / A0-A1 / Estimate confidence: low

Software: Orders, event bookings, perishables stock, payments and delivery records.

AI applications / candidates: Demand forecasts and arrangement visualization.

Boundary / watchpoint: Retail differs from flower growing and plant propagation.

38 | Pet-product retailers

Parent class 4773 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Product catalogues, stock, subscriptions, payments and online selling.

AI applications / candidates: Product search and replenishment suggestions.

Boundary / watchpoint: Veterinary advice is not a normal retail recommendation task.

39 | Jewellery retailers

Parent class 4773 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Item inventories, valuations, certificates, customer records and sales.

AI applications / candidates: Visual search and listing assistance.

Boundary / watchpoint: Verify authenticity and ownership using reliable evidence.

40 | Commercial art galleries as retailers

Parent class 4769 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Artwork inventories, provenance, consignments, clients and sales.

AI applications / candidates: Catalogue enrichment and discovery assistance.

Boundary / watchpoint: A commercial art retailer is not automatically a museum.

41 | Antique and collectible dealers

Parent class 4774 / D0-D3 / A0-A2 / Estimate confidence: low

Software: Acquisition records, provenance, valuations, listings and inventory.

AI applications / candidates: Object identification and comparable-item search.

Boundary / watchpoint: Model identification is not proof of authenticity.

42 | General e-commerce marketplaces

Parent class 4790 / D4 / A2-A4 / Estimate confidence: low

Software: Seller listings, search, payments, fulfillment interfaces, returns and trust systems.

AI applications / candidates: Recommendations, fraud alerts and seller assistance.

Boundary / watchpoint: A marketplace intermediary differs from a retailer taking ownership of stock.

43 | Specialist B2C resale marketplaces

Parent class 4790 / D3-D4 / A1-A3 / Estimate confidence: low

Software: Listings, seller identity, payments, item condition and dispute workflows.

AI applications / candidates: Listing creation and counterfeit-risk triage.

Boundary / watchpoint: Human dispute handling and authentic evidence remain necessary.

44 | No-code and low-code software publishers

Parent class 5829 / D4 / A1-A4 / Estimate confidence: low

Software: Application builders, connectors, deployment, customer support and subscriptions.

AI applications / candidates: Natural-language application creation and workflow assistance.

Boundary / watchpoint: Generated applications need testing and access control.

45 | Virtual assistants and remote administrative services

Parent class 8210 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Calendars, documents, spreadsheets, communications and client workflows.

AI applications / candidates: Email drafting, document extraction and meeting summaries.

Boundary / watchpoint: Confirm external actions before sending, buying or changing records.

46 | Outsourced customer-support operations

Parent class 8220 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Customer records, telephony or chat, knowledge bases and quality monitoring.

AI applications / candidates: Agent suggestions, summaries and quality review.

Boundary / watchpoint: Customer promises must be grounded in current company policy.

47 | AI data-labeling and business-support operations

Parent class 8299 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Task queues, annotation interfaces, quality sampling and workforce tools.

AI applications / candidates: Pre-labeling and quality-review assistance.

Boundary / watchpoint: Classification depends on whether work is routine support, data processing or technical development.

48 | Corporate and commercial law firms

Parent class 6910 / D2-D3 / A1-A3 / Estimate confidence: low

Software: Matters, contracts, legal research, documents and billing.

AI applications / candidates: Contract analysis and reviewed drafting.

Boundary / watchpoint: Verify authorities, obligations and confidentiality.

49 | Litigation and local legal practices

Parent class 6910 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Case files, deadlines, evidence, court documents and client communications.

AI applications / candidates: Chronologies, document search and drafting assistance.

Boundary / watchpoint: Never use invented precedents or unverified factual allegations.

50 | Small bookkeeping and tax practices

Parent class 6920 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Ledgers, tax forms, document collection, bank feeds and client portals.

AI applications / candidates: Receipt extraction and transaction classification.

Boundary / watchpoint: Applicable rules and final submissions require verification.

51 | Audit and assurance firms

Parent class 6920 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Audit workpapers, sampling, financial data, evidence and review systems.

AI applications / candidates: Anomaly screening and evidence-document search.

Boundary / watchpoint: An AI score is not sufficient audit evidence by itself.

52 | Architecture and building-design studios

Parent class 7110 / D2-D4 / A0-A2 / Estimate confidence: low

Software: Building models, drawings, visualization, specifications and project records.

AI applications / candidates: Design-option exploration and document checks.

Boundary / watchpoint: Building-code and engineering compliance require professional verification.

53 | Civil and industrial engineering consultancies

Parent class 7110 / D2-D4 / A0-A3 / Estimate confidence: low

Software: Simulation, mapping, design, technical documentation and project controls.

AI applications / candidates: Modeling assistance and reviewed design analysis.

Boundary / watchpoint: Do not rely on unvalidated generated calculations.

54 | Interior design studios

Parent class 7410 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Space plans, visualizations, material schedules and client projects.

AI applications / candidates: Concept renders and product matching.

Boundary / watchpoint: Verify dimensions, accessibility and construction feasibility.

55 | Graphic and digital product-design studios

Parent class 7410 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Design assets, prototypes, collaboration and client delivery tools.

AI applications / candidates: Concept variants, image editing and layout support.

Boundary / watchpoint: Check licenses and accessibility; user research remains evidence-based.

56 | Specialist medical and legal translators

Parent class 7430 / D2-D3 / A1-A2 / Estimate confidence: low

Software: Terminology, translation memory, secure documents and review workflows.

AI applications / candidates: Draft translation and consistency checking.

Boundary / watchpoint: Expert linguistic and domain review is essential.

57 | Pet grooming salons and mobile groomers

Parent class 9690 / D0-D3 / A0-A1 / Estimate confidence: low

Software: Appointments, pet notes, payments, inventory and client messages.

AI applications / candidates: Booking support and communication drafts.

Boundary / watchpoint: Animal welfare and handling remain hands-on responsibilities.

58 | Pet boarding, walking and training services

Parent class 9690 / D0-D3 / A0-A1 / Estimate confidence: low

Software: Bookings, animal care instructions, rosters, payments and client records.

AI applications / candidates: Scheduling and record summaries.

Boundary / watchpoint: Do not replace professional behavioral or veterinary judgment.

59 | Tattoo and body-piercing studios

Parent class 9690 / D0-D3 / A0-A1 / Estimate confidence: low

Software: Appointments, consent forms, portfolios, hygiene records and payments.

AI applications / candidates: Design exploration and administrative support.

Boundary / watchpoint: Verify original designs, consent and infection-control practices.

60 | Personal concierge and domestic personal services

Parent class 9690 / D0-D3 / A0-A1 / Estimate confidence: low

Software: Tasks, bookings, customer preferences, expense records and communications.

AI applications / candidates: Planning and message drafting.

Boundary / watchpoint: Separate independent service businesses from household employment.

61 | Matchmaking and dating-service businesses

Parent class 9690 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Member profiles, permissions, appointments, communications and subscriptions.

AI applications / candidates: Profile assistance and reviewed matching.

Boundary / watchpoint: Sensitive personal data and non-discrimination need strong safeguards.

62 | Independent barber shops and hair salons

Parent class 9621 / D0-D2 / A0-A1 / Estimate confidence: low

Software: Appointments, payments, client notes and basic stock.

AI applications / candidates: Booking replies and style visualization.

Boundary / watchpoint: A social-media account is not an integrated salon system.

63 | Nail salons and beauty studios

Parent class 9622 / D0-D3 / A0-A1 / Estimate confidence: low

Software: Appointments, inventory, client preferences, consent and payments.

AI applications / candidates: Scheduling and promotional content assistance.

Boundary / watchpoint: Avoid unverified medical or treatment claims.

64 | Industrial linen and laundry services

Parent class 9610 / D2-D3 / A0-A2 / Estimate confidence: low

Software: Garment tracking, wash-cycle records, machine controls, routes and contracts.

AI applications / candidates: Volume forecasts and item-recognition support.

Boundary / watchpoint: Hygiene controls and traceability remain primary.

65 | Independent motor-repair garages

Parent class 9531 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Work orders, diagnostic tools, service data, parts catalogues and billing.

AI applications / candidates: Fault-reference search and service estimate assistance.

Boundary / watchpoint: A suggested fault code is not a confirmed diagnosis.

66 | Vehicle body shops and claims repairers

Parent class 9531 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Damage photos, repair estimating, parts, claims coordination and job tracking.

AI applications / candidates: Image-assisted damage estimation and document processing.

Boundary / watchpoint: Physical inspection and insurer or customer approval remain necessary.

67 | Electric-vehicle charging operators

Parent class 3513 / D2-D4 / A0-A2 / Estimate confidence: low

Software: Charge-point control, metering, payments, availability and maintenance systems.

AI applications / candidates: Demand forecasts and equipment anomaly alerts.

Boundary / watchpoint: ISIC Rev. 5 places charging-station operation here, not in fuel retail 4730.

68 | Electronic-waste recovery businesses

Parent class 3830 / D1-D4 / A0-A3 / Estimate confidence: low

Software: Asset intake, material tracking, disassembly records, sorting and sales.

AI applications / candidates: Object recognition and material-sorting assistance.

Boundary / watchpoint: Data destruction, hazardous components and recovery rules require controls.

69 | Climate and environmental technology R&D

Parent class 7210 / D2-D4 / A1-A3 / Estimate confidence: low

Software: Scientific data, experiments, models, field observations and research governance.

AI applications / candidates: Climate-data analysis and research prioritization.

Boundary / watchpoint: Commercial installation or operation belongs in the relevant operating industry.

70 | Environmental advisory practices

Parent class 7499 / D1-D3 / A0-A2 / Estimate confidence: low

Software: Site data, impact assessments, mapping, projects and reporting.

AI applications / candidates: Evidence search and report drafting.

Boundary / watchpoint: Engineering design and laboratory testing have separate classifications.

71 | Commercial cargo space-transport operators

Parent class 5120 / D3-D4 / A0-A2 / Estimate confidence: low

Software: Mission planning, vehicle operations, tracking, payload records and contracts.

AI applications / candidates: Operations-document search and anomaly review.

Boundary / watchpoint: Spacecraft manufacture belongs under 3030; operational decisions need specialist validation.

72 | Satellite and spacecraft manufacturers

Parent class 3030 / D3-D4 / A1-A3 / Estimate confidence: low

Software: Engineering models, parts traceability, assembly, simulation and testing.

AI applications / candidates: Test-data analysis and manufacturing inspection.

Boundary / watchpoint: AI-generated designs are not mission-qualified hardware.

APPENDIX C / AVOID AN INFINITE, DUPLICATED TREE

Business models: another axis, not another industry

Two businesses can share one activity class and still be commercially very different. Describe these dimensions alongside the activity code rather than duplicating the entire economy for each model. This is an editorial framework for extending the atlas.

Dimension	Common variants	Why the software/AI profile changes
Customer	Consumer, business, government, institution; mixed customer base.	Changes sales, procurement, contracting and support workflows.
Revenue	One-off sale, subscription, usage-based charge, commission, licence, rental, sponsorship, donation.	Changes billing, revenue records, customer retention and collection systems.
Market role	Producer, wholesaler, retailer, service provider, intermediary, asset owner, operator.	Changes inventory ownership, work delivery and regulatory obligations.
Production model	Own manufacture, contract production, private label, made-to-order, standardized service.	Changes planning, traceability, quality and supplier systems.
Channel	Store, field service, web, mobile app, platform, hybrid.	Changes ordering, scheduling, fulfilment and data capture.
Ownership / governance	Sole proprietor, partnership, company, cooperative, public entity, nonprofit.	Changes reporting and governance, not necessarily the underlying activity.
Network	Independent operator, branch network, chain, franchisee, franchisor, marketplace.	Changes integration needs, permissions, standards and local autonomy.
Size / resources	Solo, microbusiness, small/medium enterprise, large enterprise.	Changes affordability, dedicated staff, data volume and implementation capacity.
Geographic reach	Local, regional, national, exporter, multinational.	Changes currencies, languages, tax, logistics and compliance requirements.
Assets / delivery	Asset-heavy or asset-light; remote or on-site; labour-intensive or automated.	Changes maintenance, workforce, monitoring and delivery systems.
Formality / regulation	Formal or informal operation; licensed or unlicensed; regulated or unregulated activity.	No code proves legality. Classification coverage does not endorse unlawful conduct.

Worked example: a physiotherapy business

The activity may sit in allied health class 8699. It can then be a solo clinic or a chain; sell individual sessions or care packages; deliver in person or remotely; and use paper, a scheduling app or an integrated clinical system. These variants change the D/A profile without creating a completely new top-level industry. The focused subtype appendix distinguishes physiotherapy from laboratories and other businesses sharing the broad parent class.

APPENDIX D / TURN THE MAP INTO A VERIFIED PROFILE

Assessing a specific business

A class-level atlas is a starting point. To claim how much technology a particular business actually uses, collect direct evidence. The following record is designed to replace assumptions with observations.

Record	Questions to answer
Activity and scope	What does it sell or deliver? Which establishment, country, size and customer group are being assessed? What activities are outsourced?
Software in operation	List the systems actually used for customers, scheduling/orders, delivery, assets, money, people and records. Distinguish a licence bought from a system used routinely.
Workflow integration	Where is information retyped? Which systems exchange data? Which critical processes stop when software is unavailable?
AI purpose and stage	For each AI use, record: personal experiment, approved pilot, bounded production, integrated production, or core product. Record the deployment date.
Evidence of use	Look for live workflows, documented users, transaction volumes, usage logs, standard operating procedures and accountable owners. Vendor marketing alone is insufficient.
Human review and controls	Identify who checks results, which decisions AI cannot make, how errors are reported, how access is controlled and which sensitive data enters the tool.
Results and limitations	Track task completion, quality, error rate, time saved, rework and operational cost using a defined baseline. Record what remains unknown.
Final maturity statement	Assign the observed D/A level for the defined business unit. State evidence date and confidence. Do not copy the class range as a verified company fact.

Use appropriate oversight in high-impact workflows. The NIST framework supplies a general risk-management reference; sector rules and professional obligations require additional local checks. [S30]

APPENDIX E / EXTEND WITHOUT BREAKING THE HIERARCHY

Going deeper: countries, products and occupations

ISIC supplies global comparability. Country classifications, product taxonomies and occupation systems answer different questions and can add detail without pretending they are interchangeable.

Route	System	Use and caution
North America	NAICS 2022	Use the official detailed North American categories and concordances. Proposed 2027 revisions are not treated as the current classification here. [S31]
European Union	NACE Rev. 2.1	Use the 2025 manual for European activity detail. The cited Eurostat 2025 enterprise ICT results still identify their sectors using NACE Rev. 2. [S32]
India	NIC 2025	Use the official 2025 code finder and detailed national subclasses. Its six-digit detail extends the global hierarchy; those extra national subclasses are not all separately profiled here. [S33]
Products and services	Central Product Classification	Use a product taxonomy to enumerate outputs such as specific goods and services. Do not make every product a separate industry branch. [S34]
Occupations	ISCO	Use occupation categories for jobs and skill groups. Accountants, drivers, designers and programmers can work in many industries. [S35]
AI adoption updates	US BTOS and other repeated surveys	Preserve wording and reference dates. The US AI question changed in November 2025; incompatible measures should not be spliced into a continuous trend. [S36]

NIC 2025 was released on 18 November 2025. It is included as a current route for finer India-specific analysis rather than defaulting to NIC 2008. [S38]

What remains outside this edition

The atlas does not individually enumerate all country-level subclasses, products, occupations or business-model permutations. Its examples are not a directory of named firms, a market-size dataset, a software-vendor comparison or a list of startup recommendations. Those are separate expansions. The main hierarchy is complete at the stated global class resolution; the finer examples deliberately have a bounded scope.

APPENDIX F / WHAT WAS ACTUALLY INCLUDED

Coverage audit

Sec.	Activity family	Div.	Groups	Classes	Examples
A	Agriculture, forestry and fishing	3	14	39	144
B	Mining and quarrying	5	10	14	47
C	Manufacturing	24	71	141	494
D	Electricity, gas, steam and cooling	1	4	6	21
E	Water, sewerage, waste and remediation	4	6	8	27
F	Construction	3	9	12	46
G	Wholesale and retail	2	17	41	154
H	Transportation and storage	5	13	23	82
I	Accommodation and food services	2	9	10	37
J	Publishing, broadcasting and content	3	7	15	56
K	Telecom, computing and information services	3	8	9	33
L	Finance and insurance	3	10	24	81
M	Real estate	1	2	3	12
N	Professional, scientific and technical services	7	16	17	64
O	Administrative and support services	6	18	25	93
P	Public administration, defence and social security	1	3	8	28
Q	Education	1	6	12	42
R	Health and social work	3	9	11	38
S	Arts, sports and recreation	4	10	19	72
T	Membership, repair and personal services	3	12	22	77
U	Household employment and own-use production	2	3	3	11
V	Extraterritorial organizations and bodies	1	1	1	4
Total	**Complete global backbone**	**87**	**258**	**463**	**1,663**

Checks performed: all 463 class codes are unique and connected to a division and group; all classes have a software/AI profile; all 72 focused subtypes point to an included parent class; all ecosystem links point to included classes; source citations use the 38-entry bibliography. The 1,663 count is example entries, not an independently deduplicated count of every business niche in existence.

Coverage is based on the UN structure. The presence of a class does not establish how many firms operate in it, whether an activity is lawful in a particular jurisdiction, or how much AI it uses. [S01] [S02]

APPENDIX G / 38 LINKED RESEARCH REFERENCES

Sources and evidence notes

All links were researched for this edition with a cut-off of 11 October 2026. Reference years are retained below. Source titles are clickable. Official classifications establish coverage; surveys establish only the quantities within their own scope; operator cases demonstrate feasibility, not universal adoption.

S01 | United Nations Statistics Division

ISIC Revision 5: white-cover publication

2026 / Official global activity classification

Backbone for the hierarchy and activity boundaries; short labels and business examples in this atlas are editorial interpretations, not copied explanatory notes.

S02 | United Nations Statistics Division

ISIC Rev. 5 classification registry entry

Accessed 11 October 2026 / Official classification metadata

Confirms 22 sections, 87 divisions, 258 groups and 463 classes. A classification is not a census of every distinct business model.

S03 | United Nations Statistics Division

Errata for ISIC Rev. 5 explanatory notes

17 August 2026 / Official corrections

Corrections concern explanatory content, not the classification's structure or codes. The atlas uses the revised structure and checks selected boundary cases.

S04 | Eurostat

Use of artificial intelligence in enterprises

Reference year 2025 / Official EU enterprise survey

Covered enterprises have at least 10 employees or self-employed persons. Sector coverage excludes important parts of the economy; these are not global rates.

S05 | Eurostat

E-business integration

Reference year 2025; extracted May 2026 / Official EU enterprise survey

Evidence for websites, ERP, CRM and business intelligence in covered enterprises. These selected applications do not represent all possible software.

S06 | World Bank

Digital Progress and Trends Report 2025: Strengthening AI Foundations

2025 report / Global development analysis

Highlights connectivity, compute, data context and skills as foundations; country-level digital gaps prevent simple global extrapolation.

S07 | OECD

AI adoption by small and medium-sized enterprises

9 December 2025 / Research and policy synthesis

Supports separate consideration of firm size, digital maturity and implementation capability; not a 463-class adoption dataset.

S08 | USDA National Agricultural Statistics Service

Technology Use: Farm Computer Usage and Ownership

August 2025 / Official US farm survey

Farm connectivity and technology evidence. The internet-access measure is not an AI-adoption measure.

S09 | Food and Agriculture Organization

Digital agriculture and AI innovation

Accessed 11 October 2026 / Sector programme and use cases

Agricultural application context, including farm decisions and value chains. Programme examples do not establish global prevalence.

S10 | Rio Tinto

Smart mining

Accessed 11 October 2026 / Primary operator case evidence

Illustrates data-driven mining and equipment monitoring at a large operator; not representative of artisanal or all commercial mines.

S11 | US National Institute of Standards and Technology

2026 Roadmap for Artificial Intelligence and Machine Learning in Smart Manufacturing

2026 / Technical research roadmap

Supports manufacturing workflow and integration analysis, not global adoption percentages by product category.

S12 | International Energy Agency

Energy and AI

10 April 2025 / Energy-sector analysis

Energy applications and system constraints. AI potential and selected deployments are distinct from sector-wide adoption.

S13 | US Environmental Protection Agency

Smart sewers

Accessed 11 October 2026 / Primary utility-technology guidance

Connected monitoring and analytical control examples for sewer infrastructure; not all water utilities use AI.

S14 | Royal Institution of Chartered Surveyors

AI in commercial property and construction report 2026

August 2026; Q1 survey / Multi-region professional survey

1,883 construction and 1,265 commercial-property responses. Professional samples and weighting limit generalization to all businesses.

S15 | National Retail Federation

AI focus area

Accessed 11 October 2026 / Retail industry research and cases

Retail use cases and governance context. Selected retail AI leaders are not a representative sample of all shops.

S16 | DHL

AI in Action

7 May 2025 / Primary logistics operator presentation

Examples of logistics vision, operational analysis and back-office applications; not a global adoption survey.

S17 | UN Tourism

Digital transformation

Accessed 11 October 2026 / Tourism-sector programme

Provides tourism digitalization context; tourism itself spans several activity classes.

S18 | UNESCO-convened independent expert group

Report on Artificial Intelligence and Culture

2025 / Expert sector analysis

Creative and cultural applications, rights and inclusion. Expert findings are not necessarily UNESCO or member-state positions.

S19 | GSMA Intelligence

Telco AI: Q4 2025

Q4 2025; 2026 publication / Telecommunications industry research

Network, customer-service and operational AI context; examples and deployments are not equivalent to operator adoption rates.

S20 | Stack Overflow

2026 Developer Survey: AI data

2026 / Voluntary respondent survey

AI-tool question has 17,464 respondents. Reports people using tools at work, not the percentage of software companies using AI.

S21 | Bank of England and Financial Conduct Authority

Artificial intelligence in UK financial services 2024

21 November 2024 / Survey of regulated financial firms

118 respondents; useful financial-sector evidence with UK, sample and reference-year limits.

S22 | Thomson Reuters Institute

2026 AI in Professional Services Report

2026 / Professional respondent survey

More than 1,500 professionals across selected legal, tax, accounting, risk, fraud and government fields. Not every professional-service industry.

S23 | OECD

Governing with Artificial Intelligence

18 September 2025 / Government use-case research

Evidence across government functions and deployment constraints; not an agency-level global prevalence measure.

S24 | UNESCO

Guidance for generative AI in education and research

2023; web update January 2026 / Education guidance

Human agency, age-appropriate use, data protection and institutional validation; not an adoption survey.

S25 | US Assistant Secretary for Technology Policy

Hospital Trends in the Use, Evaluation and Governance of Predictive AI, 2023-2024

September 2025 / US hospital survey analysis

2024 adoption denominator: 2,080 informative non-federal acute-care hospital responses. Missing and do-not-know responses excluded; predictive models linked to EHRs only.

S26 | World Health Organization

Ethics and governance of AI for health: large multi-modal models

2024 / Health-sector guidance

Clinical safety, privacy, bias and accountable deployment. This atlas is not clinical advice.

S27 | US Food and Drug Administration

Artificial intelligence-enabled medical devices

Accessed 11 October 2026 / Regulatory device information

Authorization listings demonstrate regulated products, not actual use at every hospital or clinic; the list is not necessarily exhaustive.

S28 | International Olympic Committee

Olympic AI Agenda

2024; resources updated 2026 / Primary sports strategy and governance

Sports applications and governance context. Elite sport is not representative of all community clubs.

S29 | UNHCR

UNHCR AI Approach

2025 / Primary humanitarian strategy and implementation

Administrative and humanitarian applications with human oversight and rights-based safeguards.

S30 | US National Institute of Standards and Technology

AI Risk Management Framework: Generative AI Profile, NIST AI 600-1

July 2024 / Cross-sector risk-management guidance

Used for general verification, governance and risk-control principles; not a legal compliance guarantee.

S31 | US Census Bureau

North American Industry Classification System

2022 classification; accessed October 2026 / Official regional classification

A finer North American route for industry research. Proposed 2027 revisions are not treated as the current 2022 classification.

S32 | Eurostat

NACE Rev. 2.1: statistical classification of economic activities

2025 edition / Official European classification

European activity detail. Do not confuse this taxonomy with the older NACE Rev. 2 grouping still used in the cited 2025 ICT survey.

S33 | India Ministry of Statistics and Programme Implementation

NIC 2025 code finder

Accessed 11 October 2026 / Official national classification

India-specific extension for more detailed business research; preserve the edition when mapping to ISIC.

S34 | United Nations Statistics Division

Central Product Classification

Accessed 11 October 2026 / Official product classification

Use to enumerate products and services rather than treating every product as a separate industry.

S35 | International Labour Organization

International Standard Classification of Occupations

Accessed 11 October 2026 / Official occupation classification

Jobs and occupations cut across industries and should not be mixed into the activity hierarchy.

S36 | US Census Bureau

Business Trends and Outlook Survey

Accessed 11 October 2026 / Official US business survey series

Useful for updates, but the AI question changed in November 2025. This atlas does not splice incompatible pre-change and post-change measures.

S37 | OECD, BCG and INSEAD

The adoption of artificial intelligence in firms

2025 / Selected AI-adopter research

Research includes firms already adopting AI; useful for implementation constraints, not the prevalence of AI in all firms.

S38 | India Press Information Bureau / MoSPI

MoSPI releases the National Industrial Classification 2025

18 November 2025 / Official release announcement

Confirms NIC 2025's release and alignment with newer economic activities; national classifications add local detail beyond the global backbone.